

3. Basic measurement in Epidemiology

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Content

- ▣ Measures of occurrence of events frequency (counts, ratios, proportions, rates)
- ▣ Measures of morbidity (prevalence and incidence rates)
- ▣ Measures of disease burden (DaLY, QUALY)
- ▣ Measures of mortality
- ▣ Standardization of rates
- ▣ Measures of disease occurrence

Measuring Disease Occurrence/frequency

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- Epidemiology is mainly a **quantitative** discipline, so we should quantify health and health related events.
- Epidemiologists use a variety of methods to summarize data.
- One fundamental method is the frequency distribution.
- Comparing mortality and morbidity rates in two or more different geographic areas is important for the evaluation of community health status.

Disease in population...

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- ✂ If membership of the population changes during the observation period, it is called an **Open Population (or “dynamic population”)**.
- ✂ People may join – e.g., births, in-migration.
- ✂ People may leave – e.g., deaths, out-migrants.
- Examples:
 - Population of AA town during 2023.
 - Employees of the Federal government during 2015-2019.

Measurement...

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- **Occurrence of disease** is the fundamental outcome measurement of epidemiology
- **Occurrence of disease** is typically a **binary** (yes /no) outcome
- **Occurrence of disease** involves *time*
- The Three Elements in Measures of Disease Incidence
 - ❖ E = an event = a binary outcome (occurrence of disease)
 - ❖ N = number of at-risk persons in the population under study
 - ❖ T = time period during which the events are observed

Measurement...

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- Epidemiologists always consider three factors when they measure how commonly a disease occurs in a group of people:
- (1) the number of people that are affected by the disease (cases)
- (2) the size of the population from which the cases of disease arise (population)
- (3) The length of time that the population is followed (time).

How do we measure diseases?

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- Four quantitative descriptors
 - ▣ Numbers , Ratios , Proportions , Rates
- Counting case, events, phenomena (*numbers*)
 - ▣ *Numbers*: Use of actual number of events
- E.g. 100 cases of TB in community A

Measurement...

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- The number of cases in a given community can give more epidemiologic sense if they are related to the size of the population
- Types of calculations:
- Three types of calculations are used to describe and compare measures of disease occurrence:
 - Ratios
 - Proportions , and
 - Rates
- ◇ Epidemiology of science and Epidemiology in science

How do diseases evolve in individuals over time?

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Disease states:

- Many diseases afflict an individual over time.
- Some are **recurrent** – person is diseased for a while, then recovers, but may experience the disease again

Examples:

- ✓ Malaria
- ✓ Tuberculosis
- ✓ Urinary tract infection
- Some are **non-recurrent** – an individual can develop the disease at most once
E.g. Measles
- Diseases may be **irreversible E.g HIV**

How do diseases evolve ...

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- At any given time, an individual can be in one of three states:
 1. **Diseased:** Currently afflicted.
 2. **Susceptible:** not currently afflicted, but capable of developing the disease in the next moment of time.
 3. **Immune:** not currently afflicted, and incapable of developing the disease in the next moment

How disease evolve...

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- Over time, individuals may move from one state to another:



- Moving from immune to diseased is always impossible, by definition.
- Depending on the disease, some other transitions shown may also be impossible, leading to a simpler model.

How disease evolve....

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- General Model:
Examples:
 - Uterine cancer – men are immune.
 - Benign prostatic hypertrophy – women are immune.
 - Measles – those who become ill (or immunized) develop protective antibodies.
 - Polio – oral vaccines confer immunity.
- Susceptible individuals are termed “at-risk” (there may be wide variability in risk among them, however).
- Immune and diseased “not at-risk” – they have no

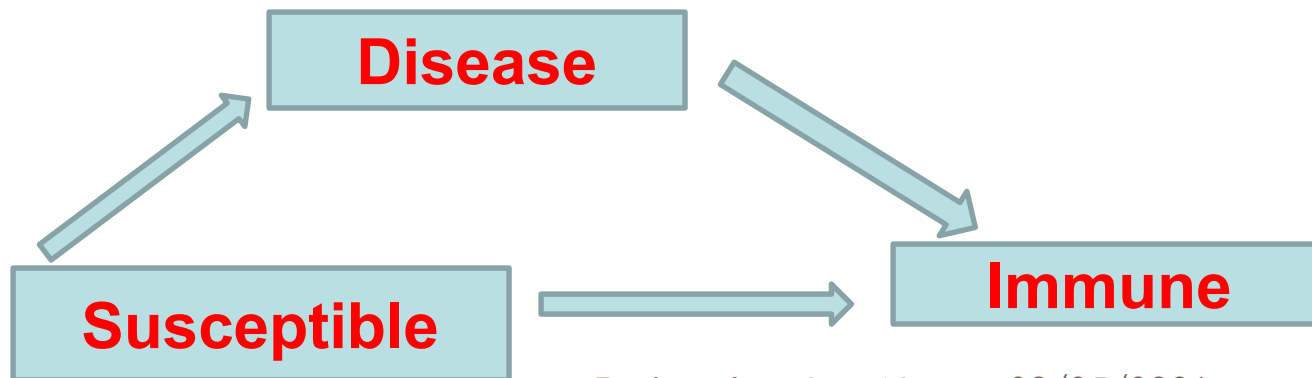
How disease evolve...

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Example #1: HIV – nobody is immune, and recovery does not occur.

Susceptible  Disease

Example #2: Measles – afflicted individuals recover with lifelong immunity, or susceptible individuals may be immunized with vaccine.

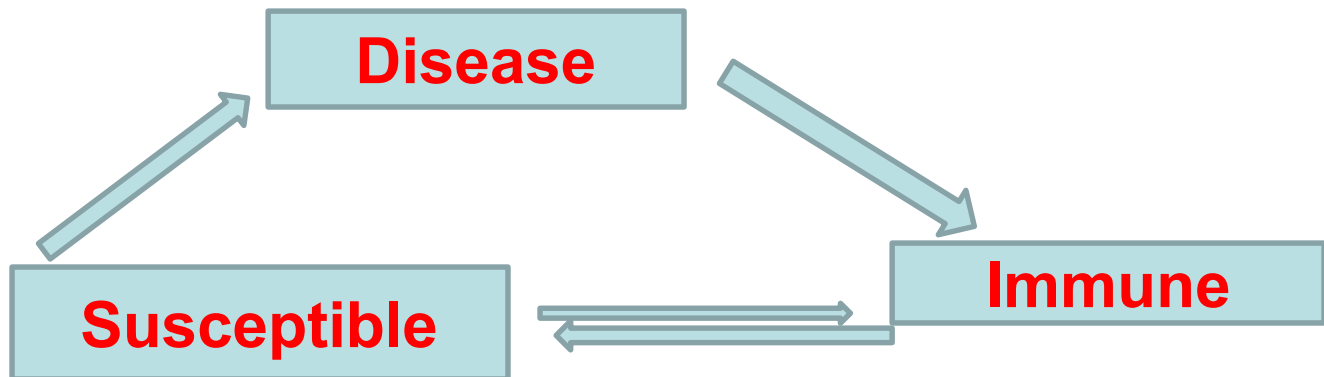


How dx evolve...

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Example #3:

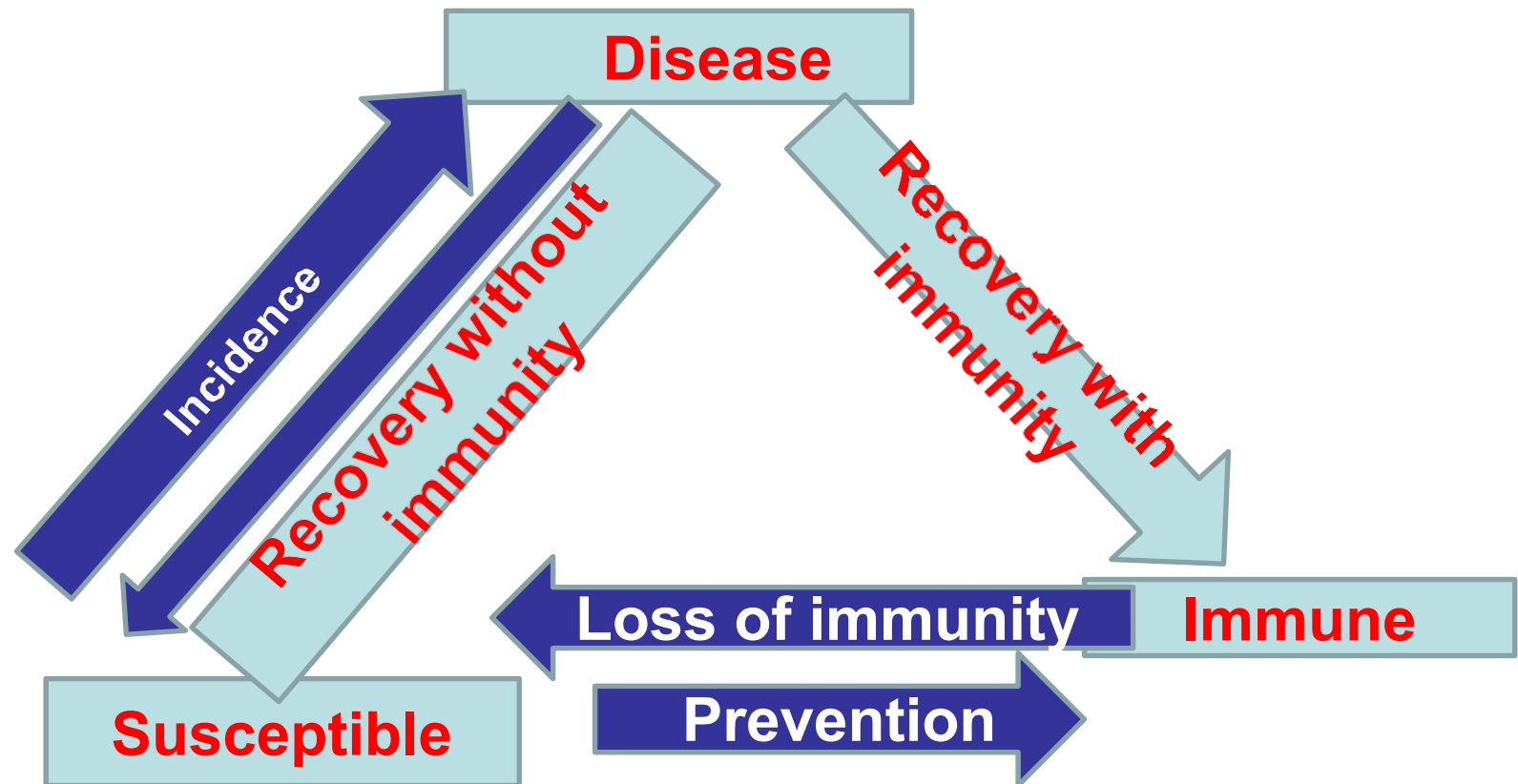
- Diphtheria – afflicted individuals develop immunity, but it wears off over time. Vaccination is also possible.



How dx evolve..

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- In general, the transitions can be interpreted and labeled as follows:



How disease evolve...

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Disease events:

- Some diseases are most naturally regarded as instantaneous occurrences at a particular point in time.
- Unlike states, they have negligible durations.

Examples:

- ✓ Rupture of uterus
- ✓ Spontaneous abortion.
- ✓ Fall in a man.

Disease in population

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- In a population that is observed over time, membership in the population may or may not change.
- If no changes in membership occur, it is called a **Closed Population (or “fixed cohort”)**.

Examples:

- ✓ Persons involved in a food borne illness outbreak that occurs over a short period of time.
- ✓ Clinical trials participants whose mortality experience is followed for the duration of the trial, with no losses to follow-up.

Disease in population...

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- If membership of the population changes during the observation period, it is called an **Open Population (or “dynamic population”)**.
- People may join – e.g., births, in-migration.
- People may leave – e.g., deaths, out-migrants.

Examples:

- Population of Debre Markos town during 2019.
- Employees of the Federal government during 2015-2019.

Why we measure disease frequency ?

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It is fundamental to epidemiology

- To study distribution of disease
- To make comparisons.
- For monitoring the health status of the population
- For planning health services

Measure disease frequency

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Epidemiologists must always consider three factors when they measure how commonly a disease occurs in a group of people:

- 1) the number of people that are affected by the disease,
 - 2) the size of the population from which the cases of disease arise, and
 - 3) the length of time that the population is followed.
- Failure to consider all three components will give a false impression about the impact of the disease on a population.

Measurement

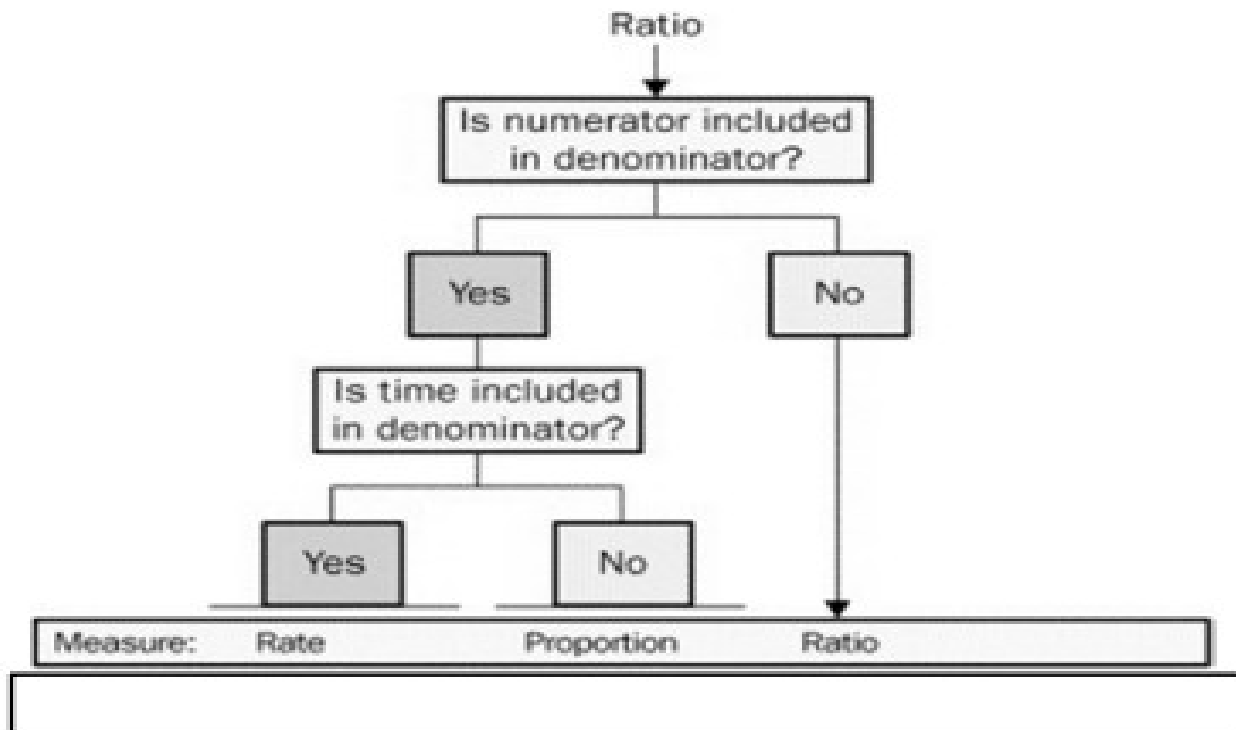
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Four quantitative Tools in Measures of frequency

- I. Counts (absolute numbers)
- II. Ratio
- III. Proportion
- IV. Rate

Algorithm for distinguishing rates, proportions, and ratios

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Simple Count(absolute numbers)

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- Number of cases in a defined population
- May be sufficient by itself for some purposes e.g. to determine the quantity of health care resources required to meet needs for people who currently have the disease.
 - Can be used to compare disease frequency between populations if and only if the population are known to be of similar size.
 - Often hard to interpret if populations being compared *differ in size*
 - hence counts are sometimes termed “numerator data”.

Measurement

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Ratio

- The value obtained by dividing one quantity by another.
- Used to compare two quantities both descriptive measures and as analytic tools
- The entities represented by the two numbers are not required to be related to one another.
- In other words, the individuals in the numerator can be different from those in the denominator.
- For example, the “sex ratio” is a ratio of two unrelated numbers
- Usually expressed as the number of males per 100 females.
- Child-women ratio
- e.g. Ratio of TB cases in community A to B is 1:10
- Sex ratio
- **Fetal death *ratio*: Fetal deaths/live births.** **Ratio= N/D**
- # PG students per lect.

Measurement...

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Proportion

- The ratio of a part to the whole
 - Is also one number divided by another, but the entities represented by
 - ▣ These numbers are related to one another.
 - In fact, the numerator of a proportion is always a subset of the denominator.
 - Proportions, also known as decimal /common fractions, are often expressed as percentages and range from 0 to 1 or 0% to 100%.
 - e.g. proportion of diabetics in a population
 - Fetal death *rate*: Fetal deaths/all births
- All births includes both live births and fetal deaths.
- =N/D (prevalence, case fatality of an event)

Measurement...

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□ Rate

- Is also one number divided by another, but **time** is an integral part of the denominator.
- We are familiar with rates in our daily travels since a rate is a measure of how fast we travel.
- For example, the maximum speed or rate at which cars are permitted to travel may be 65 miles per hour.
- This rate can also be written as 65miles/one hour. The measure of time in the denominator is what makes this number a rate.
- It has five components(numerator, denominator, time, population, and multiplier)
- Measles cases in under five children in 2000 x k
Under five children in 2000

$$\text{Rate} = \frac{\text{Number of events in a specfied population during a specfied period}}{\text{Average popluation during the period}} * 10^n$$

Characteristics of simple frequency measures

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- ◇ All rates are proportions.
- ◇ All rates are ratios too.
- ◇ All proportions are ratios.
- ◇ Not all proportions are rates.
- ◇ But not all ratios are proportions.

	Incidence	Prevalence
Numerator	Number of new cases of disease during a specified period of time	Number of existing cases of disease at a given point of time
Denominator	Population at risk	Population at risk
Focus	Whether the event is a new case Time of onset of the disease	Presence or absence of a disease Time period is arbitrary; rather a "snapshot" in time
Uses	Expresses the risk of becoming ill The main measure of acute diseases or conditions, but also used for chronic diseases More useful for studies of causation	Estimates the probability of the population being ill at the period of time being studied. Useful in the study of the burden of chronic diseases and implication for health services

Note: If incident cases are not resolved, but continue over time, then they become existing (prevalent) cases. In this sense, $\text{prevalence} = \text{incidence} \times \text{duration}$.

Measurement...

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Types of rates

- A. Crude rates
- B. Specific rates
- C. Adjusted (standardized) rates: It is a...
 - ◇ **Classic Epidemiological Method for the Comparison of Rates.**

Crude rate

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- Accounts the total burden of health outcomes in all ages and cause.
- Are overall measure that does not take explicit account of the composition of the population.
- Its value will be an average of the values for the individual subgroups, weighted by their relative sizes [Democracy].
- **Is the simplest and most straightforward summary.**
- **$CR = \frac{E}{P} * K$ (E= Number of events, p=mid-year total population, k=constant, may be 100, 1000, etc.**
- Are summary rates based on the actual number of events (births, deaths, diseases) in the total population over a given time period
- Crude birth rate (CBR) and the crude death rate (CDR) are widely used
- CDR depend on two factors
 - The probability of dying for individuals
 - The age distribution of the population

Crude...

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➤ **Advantages**

- Actual summary rates
- Calculable from minimum information
- Widely used despite limitations

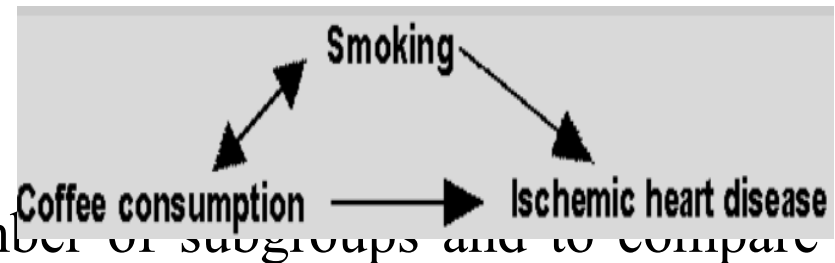
➤ **Disadvantages**

- Difficult to interpret due to variation in composition (e.g. age)
- Obscure significant differences in risk between subgroups

Specific rates

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- Apply to specific subgroups in the population.
- **SR** are calculated based on the specific characteristic(s).
- Denominator should be the population in that specific group except for **cause-specific rates**
- **Advantages**
- The rates apply to homogenous subgroups
- The rates are detailed and useful for epidemiological and public health purposes.
- Important to control third variables
- **Disadvantages**
- Impractical to work with a large number of subgroups and to compare many of two or more populations
- Furthermore, if the subgroups consist of small populations, the specific rates can be very imprecise.



Category of Specific Rates

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❑ Calculation of factor specific disease frequency measures such as:

- ✓ Cause specific rates
- ✓ Age specific incidence rates
- ✓ Age specific death rate
- ✓ Sex-specific death rates
- ✓ Race-specific rates
- ✓ Place-specific rates

Adjusted/standardization rates

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- Epidemiologists are always mindful of population diversity.
- Is a way to make fairer comparison between groups with different age distribution.
- Different FD in different populations, a comparison between CR would be misleading since CRs are not very informative about the health status of a population.
- Standardization for the characteristic(s) responsible for the differences in comparison is necessary.
- Age and sex are two of the most common variables used for standardization and they are called standardized rates.
- The difference between CR and SR is that the former are calculated based on the population under study as a whole whereas the latter are based on particular characteristic(s) as standard.

Adjusted/standardization rates...

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- A principal role in epidemiology is to compare the incidence of disease or mortality between two or more populations.
- However, the comparison of crude mortality or morbidity rates is often misleading because the populations being compared may differ significantly with respect to certain underlying characteristics, such as age or sex, that will affect the overall rate of morbidity or mortality
- For example, age is an important determinant of mortality.
- An older population will have a higher overall mortality rate than a younger population.
- As a result, variations in age will complicate any comparison between two or more populations that have different age structures.
- To understand how a comparison of crude rates can be affected by differing population distributions, it should be recognized that a crude overall rate is simply a weighted average of the individual category specific rates, with the weights being the proportion of the population in each category

Adjusted/standardization rates...

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Category specific rates

- One method of overcoming the effects of confounding variables such as age is to simply present and compare the age specific rates.
- While this allows for a more comprehensive comparison of mortality or morbidity rates between two or more populations, as the number of stratum specific rates being compared increases, the volume of data being examined may become unmanageable.
- It is, therefore, more useful to combine category specific rates into a single summary rate that has been adjusted to take into account its age structure or other confounding factor.
- **This is achieved by using the methods of standardisation**

Adjusted/standardization rates...

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- Virtually every large population is heterogeneous in regard to
 - ▣ Socio-demographic (e.g., age, gender, education, religion), geographic, genetic, occupational, dietary, medical history, and
 - ▣ Numerous other personal attributes and
 - ▣ Environmental factors related to health.
- The adjusted or “standardized” rate is obtained by dividing the total of expected cases by the standard population

Adjustment...

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- In many circumstances, it is useful to have single summary rate for each population that takes into account any differences in the structure of the population
- This is done through **adjustment** or **standardization**
- Adjusted rates are summary rates that have undergone statistical transformation, to permit fair comparison between groups differing in some characteristics that may affect risk of disease.

Adjusted...

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- Adjustment (standardization) procedures are any of a variety of procedures performed during data analysis to attempt to remove the effect of one or more extraneous sources of variation that could affect (or are believed to affect) a particular result (Meinert, 1996).
- When comparing rates adjusted for a particular factor, any remaining observed differences between the groups can not be attributed to confounding by that variable.

Measurement...

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- When comparing the crude death rates of two or more places, it is impossible to know whether the difference is due to age composition, age specific death rate or both.
- In age adjusted rates the difference is exclusively attributed to differences in age specific mortality rates, since the effect of age composition is artificially removed.

Goals of Adjustment Procedures

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- ❖ Appropriately **combine** data
- ❖ Make appropriate **comparisons** among groups
- ❖ Reach appropriate **conclusions** and inferences
- ❖ There are two methods for calculating standardized rates, namely
- ❖ **Direct and indirect standardization**
- ❖ The ratio of the observed number of deaths to the expected number of deaths is called: “Standardized mortality ratio” or SMR.

$$\text{Directly standardized rate} = \frac{\sum (\text{stratum-specific rates} \times \text{standard weights})}{\sum (\text{standard weights})}$$

$$\text{Directly standardized rate} = \frac{(r_1 N_1 + r_2 N_2 + r_3 N_3 + \dots + r_n N_n)}{(N_1 + N_2 + N_3 + \dots + N_n)}$$

$$R_s = \frac{\sum (r_k \times N_k)}{\sum (N_k)} = \sum \left(r_k \times \frac{N_k}{\sum (N_k)} \right) = \sum \left(r_k \times \frac{N_k}{N} \right)$$

$$R_s = \sum (r_k W_k)$$

where:

r_k = rate in k-th stratum of the *study* population

N_k = number of persons in k-th stratum of the *standard* population

N = total number of persons in the *standard* population ($\sum N_k$)

W_k = weight for each stratum (equal to N_k/N)

$\sum$ means summation over the k strata.

Standardization

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- The 3's (Standardization, simplification, specifications.
- Applicable for factors other than age(socioeconomic status, income, race, are of residence, etc.)
- The adjusted or “standardized” rate is **obtained by dividing the total of expected cases by the standard population.**

Direct standardization

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- Used when specific mortality rates for two or more populations are known such as age(ASMR).
- Steps
 - A. Calculate the age-specific mortality rates for each age group in each population
 - B. Then choose the standard (reference) population from one of the populations like national mortality
 - C. Multiply the age-specific mortality rates of the other population under study to the number of persons in each age group of the standard population

Direct standardization...

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- D. By this way, you will get the expected deaths for each age group of each population(c)
- D. Add the number of expected deaths from all age groups.
- D. Finally to get the age-adjusted mortality rates, divide the total number of expected deaths by the standard population
- D. Now you can conclude by comparing the age-standardized mortality rates of two populations

Adjusted rate...

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➤ Advantages

- Are summary rates
- Permit unbiased comparison
- Easy to interpret

➤ Disadvantages

- Ineffectual rates
- Absolute magnitude depends on standard population
- Opposing trends in subgroups masked

Direct standardization: steps WITH example

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- Standardized rates are **a statistical measure of any rates in a population.**
- These are adjusted rates that take into account the vital differences between populations that may affect their birthrates or death rates.
- Traditionally, there are two methods of standardization: **direct and indirect standardization.**
- Population of interest ASMR^k Standard population
→ compare Age Adjusted MR

Methods of Standardisation

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- There are two methods of standardisation commonly used in epidemiological studies, and these are characterized by whether the standard used is a population distribution (direct method) or a set of specific rates (indirect method).
- Both direct and indirect standardisation involves the calculation of numbers of expected events (e.g. deaths), which are compared to the number of observed events.
- Age is a factor that is frequently adjusted for in epidemiological investigations, particularly in comparative mortality studies, since the age structure of a population will greatly affect the population's overall mortality.
- To illustrate the methods of both direct and indirect standardisation, the age specific mortality rates for two hypothetical populations are compared below in Ethiopia.

Direct method of standardisation

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	Country A			Country B		
Age - group	No. of deaths	Population	Rate per 1,000 pyrs	No. of deaths	Population	Rate per 1,000 pyrs
0-29	7,000	6,000,000	1.2	6,300	1,500,000	4.2
30-59	20,000	5,500,000	3.6	3,000	550,000	5.5
60+	120,000	2,500,000	48	6,000	120,000	50
Total	147,000	14,000,000	10.5	15,300	2,170,000	7

presents CMR
The overall CMR is higher for country A (10.5/1,000 person years) but the ASMR higher among all age-groups in country B.

For Standard population of the above three age groups 100 thu; 65 thu; and 20 thu, the Expected Deaths as

	Country A	Country B
	Expected deaths	Expected deaths
0-29	$0.0012 \times 100,000 = 120$	$0.0042 \times 100,000 = 420$
30-59	$0.0036 \times 65,000 = 234$	$0.0055 \times 65,000 = 357.5$
60+	$0.048 \times 20,000 = 960$	$0.05 \times 20,000 = 1,000$
Total expected deaths	1,314	1,777.5
Age adjusted rate	$1,314 / 185,000 = 7.1$ per 1,000 pyrs	$1,777.5 / 185,000 = 9.6$ per 1,000 pyrs
Age standard rate ratio (B:A) = $9.6 / 7.1 = 1.35$		

Indirect method of standardisation

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- The indirect method of standardisation is commonly used when age-specific rates are unavailable.
- For example if we did not know the age specific mortality rates for country B
- In this method, instead of taking one population structure as standard and applying sets of rates to it to estimate expected events, a set of rates from a standard population (country A) is applied to each of the populations being compared to calculate standardized morbidity/mortality ratios.
- Indirect method of standardisation: Number of expected deaths if the population had the same age-specific mortality rates as Country A.

Indirect method of standardization...

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	Country A	Country B
	Expected deaths	Expected deaths
0-29	$0.0012 \times 6,000,000 = 7,200$	$0.0012 \times 1,500,000 = 1,800$
30-59	$0.0036 \times 5,500,000 = 19,800$	$0.0036 \times 550,000 = 1,980$
60+	$0.048 \times 2,500,000 = 120,000$	$0.048 \times 120,000 = 5,760$
Total expected deaths (E)	147,000	9,540
Total observed deaths (O)	147,000	15,300
Standardised Mortality Ratio O/E x 100	100	160

- The IMS is used to calculate how many deaths would be expected in Country B if it had the same age-specific mortality rates as Country A.
- The expected deaths in Country B are calculated by multiplying the age specific rate for Country A by the population of Country B in the corresponding age group.
- The sum of the age categories gives the total number of deaths that would be experienced in country B if it had the same mortality experience as country A.
- An overall summary measure can then be calculated, that is, the standardized mortality ratio (SMR), which is the ratio of the observed number of deaths to the expected number of deaths.
- $SMR = \frac{\text{Observed number of deaths (O)} \times 100\%}{\text{Expected number of deaths (E)}}$
- $SMR = \frac{160}{100} = 1.6 \times 100 = 160$, which means that the number of observed deaths in Country B is 60% higher than Expected

The crude mortality rate for all ages in the United States for 1995-1997 is 8.7 per 1,000 population. In Mexico it is much lower: 4.7 per 1,000 population.

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Calculate

- The adjusted rate of the two countries(DSR)

	Standard population (1)	Age-specific mortality rate per 100,000 population, 1995-1997		Expected number of deaths	
		Mexico (2)	United States (3)	Mexico (4)=(1)x(2)/100,000	United States (5)=(1)x(3)/100,000
<1	2,400	1693.2	737.8	41	18
1-4	9,600	112.5	38.5	11	4
5-14	19,000	36.2	21.7	7	4
15-24	17,000	102.9	90.3	17	15
25-44	26,000	209.6	176.4	55	46
45-64	19,000	841.1	702.3	160	133
65+	7,000	4,967.4	5,062.6	348	354
	100,000			639	574

- When eliminating the effect of the difference in the age structure in both countries, we obtain a rate that is higher in ---than in the ---.
- Is the conclusion of the comparison of the rates is inversed or not when using adjusted rates rather than crude rates.

Table 1
Population and Deaths by Age in 1970 for White Females in
Miami, Alaska, and the U.S.

Age	Miami			Alaska			U.S.		
	Pop.	Deaths		Pop.	Deaths		Pop. ⁺	Deaths ⁺	
< 15	114,350	136		37,164	59		23,961	32	
15-24	80,259	57		20,036	18		15,420	9	
25-44	133,440	208		32,693	37		21,353	30	
45-64	142,670	1,016		14,947	90		19,609	140	
65+	92,168	3,605		2,077	81		10,685	529	

EX1. Home take exercise

Calculate the CDR and SAR Of Each State?

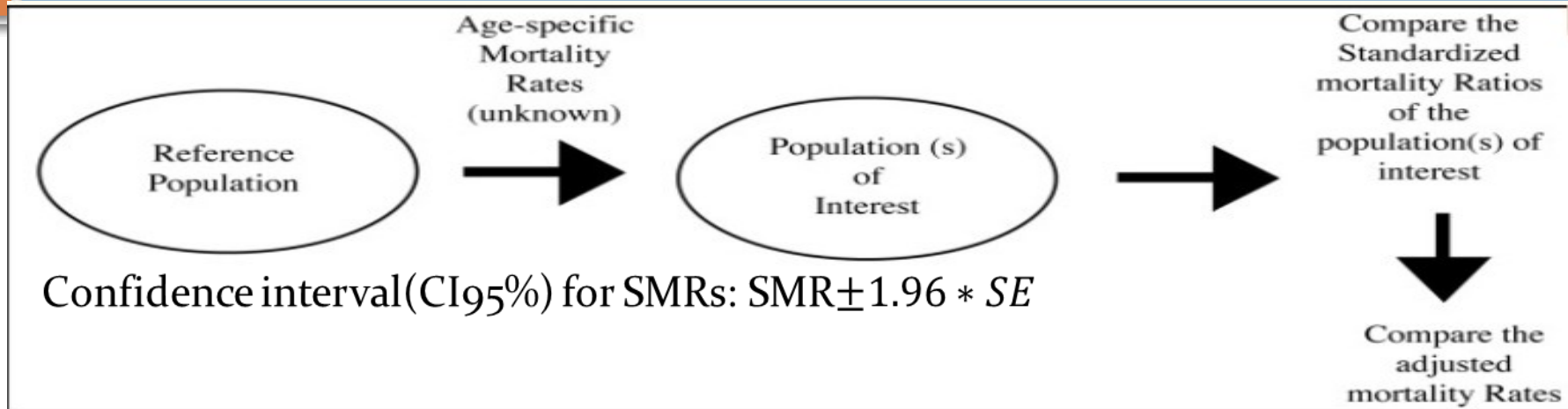
Indirect standardization (IS)

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- Common practice is to compare specific mortality rate indirectly
 - A. Choose a reference or standard population.
 - B. Calculate the observed number of deaths in the population (s) of interest
 - C. Apply the age-specific mortality rates from the chosen reference population to the population(s) of interest.
 - D. Multiply the number of people in each age group of the population(s) of interest by the age-specific mortality rate in the comparable age group of the reference population.
 - E. Sum the total number of expected deaths for each population of interest.
 - F. Divide the total number of observed deaths of the population(s) of interest by the expected deaths

IS: is different in both method-needs specific rates, comparison group and interpretation

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- A conclusion can be reached by simply calculating and looking at the SMR (obs/exp).
- A SMR higher than one indicates that the risk of dying in the observed population is higher than what would be expected if it had the same experience or risk than the standard population. On the other hand, a SMR lower than one indicates that the risk of dying is lower in the observed population than expected if its distribution were the same as the reference population.
- The actual adjusted rates can also be calculated using the indirect method by multiplying the crude rate of every population by its SMR
- In order to use the indirect method you need:
 - The age-specific mortality rates by age group in reference population
 - The population of the state of specific region stratified by age
 - The total number of deaths observed in the region
- How to construct z distribution at 95%CI?

Note

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- If the age-specific mortality rates of the reference population is applied, CDR must be calculated from that reference population.
- Standardization methods are not difficult but sometimes the health personnel have some confusion about selecting which method and how to calculate and apply the particular method.
- **Standardization: A Classic Epidemiological Method for the Comparison of Rates**
- One of the fundamentals of health situation analysis (HSA) is the comparison of basic health indicators.
- HAS allows to identify risk areas, define needs, and document inequalities in health, in two or more populations, in subgroups of a population, or else in a single population at different points in time.
- Crude rates, whether they represent mortality, morbidity or other health events, are summary measures of the experience of populations that facilitate this comparative analysis

Note...

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- However, the comparison of crude rates can sometimes be inadequate, particularly when the population structures are not comparable for factors such as age, sex or socioeconomic level.
- The direct method is most often used.
- However, it requires rates specific to population strata corresponding to the variable of interest in all the populations we wish to compare, which are sometimes not available.
- Even when these specific rates are available for all the subgroups, they are sometimes calculated from very small numbers and can be very imprecise. In this case, the indirect standardization method is recommended
- **Adjusted mortality rates = Standardized mortality ratio × crude death rate**
- In *direct standardization*, the **study population** provides the **rates** and the **standard population** provides the **weights**. In *indirect standardization*, the **standard population** provides the **rates** and the **study population** provides the **weights**.

Note...

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	Study population	Standard population
Directly-standardized rate	Rates	Weights
Indirectly-standardized rate	Weights	Rates

$$\begin{aligned} \text{Expected number of deaths} &= \sum \left(\begin{array}{c} \text{[Stratum-specific rates from} \\ \text{the standard population]} \end{array} \times \begin{array}{c} \text{[stratum sizes from} \\ \text{the study population]} \end{array} \right) \\ &= \sum (R_k n_k) \end{aligned}$$

and the observed number of deaths is

$$\sum d_k$$

so that

$$\text{SMR} = \frac{\text{Observed deaths}}{\text{Expected deaths}} = \frac{\sum d_k}{\sum (R_k n_k)}$$

where d_k = number of deaths in the k -th stratum of the *study* population ("observed deaths")

n_k = size of the k -th stratum of the *study* population

R_k = death rate in the k -th stratum of the *standard* population

The number of observed deaths can also be expressed as the sum of stratum-specific death rates multiplied by stratum sizes:

$$\begin{aligned} \text{Observed number of deaths} &= \sum \left(\begin{array}{c} \text{[Stratum-specific rates from} \\ \text{the study population]} \end{array} \times \begin{array}{c} \text{[stratum sizes from} \\ \text{the study population]} \end{array} \right) \\ &= \sum (r_k n_k) \end{aligned}$$

where: r_k = death rate in the k -th stratum,

Example

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Death rates by age in two occupations and a standard population

Age	Occupation A			Occupation B			Standard population		
	Persons	Deaths	Rate	Persons	Deaths	Rate	Persons	Deaths	Rate
40-49	1,000	2	0.002	5,000	10	0.002	30,000	30	0.001
50-59	5,000	20	0.004	1,000	4	0.004	40,000	120	0.003
Total	6,000	22		6,000	14		70,000	150	

Cont.

$$\text{Indirectly-standardized rate} = \text{SMR} \times \left(\frac{\text{Crude death rate in the standard Population}}{\text{Crude death rate in the standard Population}} \right)$$

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SMR	22	14
	$\frac{22}{(0.001)(1,000) + (0.003)(5,000)}$	$\frac{14}{(0.001)(5,000) + (0.003)(1,000)}$
	1.38	1.75

Though both occupations have exactly the same stratum-specific rates, their SMR's differ, due to the substantially different age distributions for the two occupations. However, the directly standardized rates for both occupations are, reassuringly, the same:

$$\text{Directly standardized rate for A} = (0.002 \times 30,000 + 0.004 \times 40,000) / 70,000 = 0.0031$$

$$\text{Directly standardized rate for B} = (0.002 \times 30,000 + 0.004 \times 40,000) / 70,000 = 0.0031$$

Similarly, the SRR for each occupation relative to the standard population is $0.0031/0.0021 = 1.48$, indicating a 48% higher age-standardized rate of death in each occupational population compared to the standard population. However, the apparent equivalence of the directly standardized rates is misleading. With so few deaths in the younger age stratum in Occupation A and in the older age stratum in Occupation B, the rate estimates are very unstable. In other words, we cannot really estimate some of the rates, so direct standardization is a dubious procedure. Given the substantial uncertainty about what the stratum-specific rates really are, the only conclusion we can be confident of is that both occupations have elevated mortality rates compared to the standard, or reference population. Without assumptions or additional information, we have no evidence from standardization to conclude that one of the occupations is more hazardous (or is not more hazardous) than the other.

Issues in the use of standardisation

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- Standardised rates are used for the comparison of two or more populations; they represent a weighted average of the age specific rates taken from a 'standard population' and are not actual rates.
- The direct method of standardisation requires that the age-specific rates for all populations being studied are available and that a standard population is defined.
- The indirect method of standardisation requires the total number of cases
- The ratio of two directly standardised rates is called the Comparative Incidence Ratio or Comparative Mortality Ratio.
- The ratio of two indirectly standardised rates is called the Standardized Incidence Ratio or the Standardized Mortality Ratio.
- Indirect standardisation is more appropriate for use in studies with small numbers or when the rates are unstable.
- As the choice of a standard population will affect the comparison between populations, it should always be stated clearly which standard population has been applied.
- Standardisation may be used to adjust for the effects of a variety of confounding factors including age, sex, race or socio-economic status.
- Standardisation of rates can be difficult to understand and is explained in several different ways depending on the literature source. It is recommended that as well as using this resource text you use one of the sources in the related links, choosing the text which gives you the appropriate amount of detail.

Measures of morbidity rate

61

- Morbidity rates are rates that are used to quantify the magnitude/frequency of **diseases/illness**.
- The frequency of health related events are measured by risk, prevalence and incidence rate.
- Two common morbidity rates
 - ❖ **Incidence rates (Cumulative incidence, incidence density)**
 - ❖ **Prevalence (Period prevalence, point prevalence)**

Incidence {Risk}

62

- The occurrence of new cases of disease that develop in a candidate population **over a specified time** period (measure of absolute risk).
- Measures the newly diagnosed patients develop over time in [Week, month, years so on]
- If the IR of a certain disease is high in one area, then the risk of acquiring that disease by other healthy individuals will be high.

THREE KEY IDEAS

- Incidence measures new disease events.
- New cases of disease are measured in a candidate/risk/ population
- Account the specific amount of time (transition from health to disease).
- Example. In Sept 2019 there were 50 new cases of relapsing fever in “Kebele X”. The average total population of “Kebele X” was 5000. Calculate the incidence rate of relapsing fever in “Kebele X” in Ginbot 2019.
- The types of incidence rate is attack rate (used during outbreak)
- Used for etiological study

Risk(cumulative incidence-CI)

63

- Likelihood that an individual will contract a disease.
- The proportion of a candidate **population{closed}** that becomes diseased over a specified period of time.

Mathematically

$$\text{Risk} = \frac{\text{New cases occurring during a given time period} * k}{\text{Population at risk during the same period}}$$

- Numerator (new cases of disease) is subset of denominator (candidate population).

CI...

64

- The average risk of getting a disease over a certain period of time.
- A risk is the probability of getting a disease.
- CI is mainly used in fixed populations when there are no or small losses to follow-up.
- Possible value ranges from 0 to 1 or 0% to 100%
- Time is not integral part but expressed by words that accompany by the CI number.
- Thus CI is dimensionless
- **For example: -**
- If we count all new cases of influenza occurring in X undergraduates from September 1, 2018 – October 28, 2019, and we take as the denominator all undergraduates enrolled in September 1, 2018, we would be describing the **cumulative incidence rate** of influenza.

Incidence density/person-time/PTY

65

- The risk/probability of an individual developing a disease overtime {closed or open Population}
 - The rapidity with which new cases of a disease develop overtime
 - Occurrence of new cases of disease that arise during *person time* of observation.
- Incidence Rate =
$$\frac{\text{Number of new cases of disease}}{\text{Person-time of observation in candidate population}}$$
- The difference between
 - The incidence rate's denominator integrates time (t), and so it is a true rate.
 - Incidence rate only up until he or she is diagnosed with the disease of interest.
 - Unlike for cumulative incidence, Person-time is accrued only while the candidate is being followed.

IR...

66

- Other things being equal, the number of disease events observed in a population over times varies directly with the **number of people at risk and the amount of time** over which they are observed.
- Is a fundamental tool for etiologic studies of diseases since it is a direct measure of risk.
- If the IR is high in a specific area , the event must be systematically searched.
- Person-time rates are often used in cohort (follow-up) studies of diseases with long incubation or latency periods, such as occupationally related diseases, AIDS, and chronic diseases.
- **ATTACK RATE:** The proportion of a group that experiences the outcome under study over a given period (e.g., the period of an epidemic)
- **SAR:** The number of cases of an infection that occur among contacts within the incubation period following exposure to a primary case in relation to the total number of exposed contacts
- The SAR is a measure of contagiousness and is useful in evaluating control measures.

Why PTY/ID?

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- ❖ Impossible to follow everyone for entire time period.
- ❖ In a dynamic population, individuals enter population **over different times**, become lost, etc.
- ❖ Length of follow-up is not uniform for all .
- ❖ Incidence rates do not make assumption of complete follow-up.
- ❖ **Since the number of cases is divided by a measure of time of observation, rather than people, this helps address the problem of competing risks.**
- ❖ **CR:** An event that removes a subject from being at risk for the outcome under investigation.
- ❖ For example, in a study of smoking and cancer of the lung, a subject who dies of CHD is no longer at risk of lung cancer; in this situation, CHD is a competing risk.

Example

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- Investigators enrolled 2,100 men in a study and followed them over 4 years to determine the rate of CHD.
 - We assume that persons diagnosed with disease and those lost to follow-up were disease-free for half of the year, and thus contribute $\frac{1}{2}$ year to the denominator.
 - Initial enrolment: 2,100 men free of disease
 - After 1 year: 2,000 disease-free, 0 with disease, 100 lost to follow-up
 - After 2 years: 1,900 disease-free, 1 with disease, 99 lost to follow-up
 - After 3 years: 1,100 disease-free, 7 with disease, 793 lost to follow-up
 - After 4 years: 700 disease-free, 8 with disease, 392 lost to follow-up
1. Identify x : $x = \text{cases diagnosed} = 1 + 7 + 8 = 16$
 2. Calculate y , the person-years of observation:

Results

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$$(2,000 + \frac{1}{2} \times 100) + (1,900 + \frac{1}{2} \times 1 + \frac{1}{2} \times 99) + (1,100 + \frac{1}{2} \times 7 + \frac{1}{2} \times 793) + (700 + \frac{1}{2} \times 8 + \frac{1}{2} \times 392) = 6,400 \text{ person-years of observation.}$$

A second way to calculate the person-years of observation is to turn the data around to reflect how many people were followed for how many years, as follows:

700 men x 4.0 years = 2,800 person-years

8 + 392 = 400 men x 3.5 years = 1,400 person-years

7 + 793 = 800 men x 2.5 years = 2,000 person-years

1 + 99 = 100 men x 1.5 years = 150 person-years

$$\begin{aligned} 3. \text{ Person-time rate} &= \frac{\text{number of cases during 4-year study}}{\text{time each person was observed, totaled for all persons}} \times 10^n \\ &= \frac{16}{6,400} \times 10^n = .0025 \times 10^n \end{aligned}$$

or, if 10^n is set at 1,000, there were 2.5 cases per 1,000 person-years of observation. This quantity is also commonly expressed as 2.5 cases per 1,000 persons per year.

Relation Between CI and IR

70

- It is possible to obtain cumulative incidence from an incidence rate.
- in a fixed population with a constant incidence rate and small cumulative incidence (less than 10%).

$$CI = IR_i \times t_i$$

- When the incidence rate is not constant, it is necessary to take into account the different rates that prevail during each time period:

$$CI = \Sigma (IR_i \times t_i)$$

- where CI is cumulative incidence, IR is incidence rate, and t_i is the specified period of time.

Practical challenges in measuring incidence

71

- I. Identification of population at risk
- II. Population is not static/it fluctuates/as a result of births, deaths and migration
- III. People are at risk only until they get the disease and then no more at risk

I. Use the total population as a denominator

- This gives an estimate of the incidence rate and not the actual incidence rate

II. Use person-time at risk

- Incidence density = number of new cases of a disease over a specified period / person-time at risk

Prevalence {Burden}

73

- ▣ Measures the existence of current disease.
 - Didn't provide strong evidence of causality.
 - Has three components numerator, denominator, and multiplier.
 - Proportion of the total population that is diseased during a specified period or at a point in time

$$\text{Prevalence} = \frac{\text{Number of people with the disease/conditions at specified time}}{\text{Number of people in the population at risk at the specified period}} * 100$$

- ▣ **Types**
 - ▣ **Point prevalence rate**
 - ▣ **Period prevalence rate**
 - ▣ **Lifetime prevalence**

Point PR

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- Measures the **proportion** of a population with a disease at a point in time
- Determined by conducting cross-sectional/survey study.
- Point prevalence rate=All persons with a disease at a point in time/Total population
- proportion of the population that is diseased at a single point in time and can be thought of as a single snapshot of the population.
- It is not a rate, but a true proportion
- e.g. December 1, 2010 or a point in someone's life such as college graduation.
- **Point prevalence** =
$$\frac{\text{Number of existing cases of disease}}{\text{Number in total population}}$$
 At a point in time

Period PR

75

- Measures the proportion of a population with a disease in a specified time period
- Period prevalence rate = $\frac{\text{All persons with a disease over time period}}{\text{Average (mid-year) population in the same period}}$
- e.g. during the year 2010.
- The period prevalence includes the number of cases that were present at the start of the period.
- e.g. January 1, 2010 as well as the number that developed during the period (for example, January 2 through December 31, 2011).
- E.g. annual prevalence rate/ lifetime prevalence rate.
- A series of snapshots $\frac{\text{Number of existing cases of disease}}{\text{Number in total population}}$ During a period of time
- **Period prevalence** =

Example

76

Two surveys were done in the same community 12 months apart. Of 5,000 people surveyed the first time, 25 had antibodies to histoplasmosis. Twelve months later, 35 had antibodies, including the original 25. Calculate the prevalence at the second survey, and compare the prevalence with the 1-year incidence.

1. Prevalence at the second survey:

x = antibody positive at second survey = 35

y = population = 5,000

$$x/y \times 10^n = 35/5,000 \times 1,000 = 7 \text{ per } 1,000$$

2. Incidence during the 12-month period:

x = number of new positives during the 12-month period = $35 - 25 = 10$

y = population at risk = $5,000 - 25 = 4,975$

$$x/y \times 10^n = 10/4,975 \times 1,000 = 2 \text{ per } 1,000$$

Lifetime prevalence

77

- ❖ Is the proportion of a population who, at some point in life has ever had the characteristic or
- ❖ The proportion of a population that might become afflicted with a given disease at any point in their lifetime(up to the time of assessment) have experinaced as a “case”.

Characteristics of Prevalence

78

- Cause and effect measured simultaneously
 - ▣ Impossible to infer causation
- Useful for planning (e.g. beds, clinics, workforce needs)
- High prevalence $\neq$ high risk
 - ▣ could reflect increased survival(improved care, behavior change - long duration)
- Low prevalence $\Rightarrow$ could reflect rapid fatal or cure process - short duration)
- Easy to obtain $\Rightarrow$ need only 1 measurement

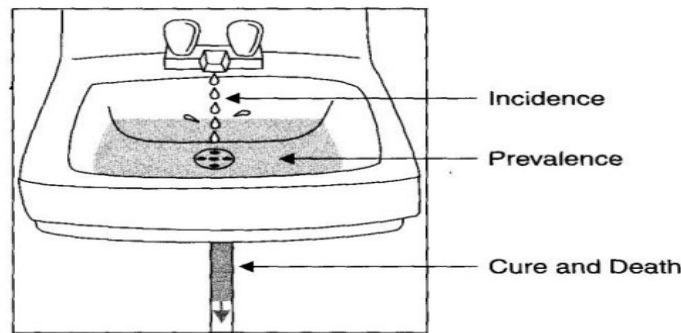
Relationship between Prevalence and Incidence

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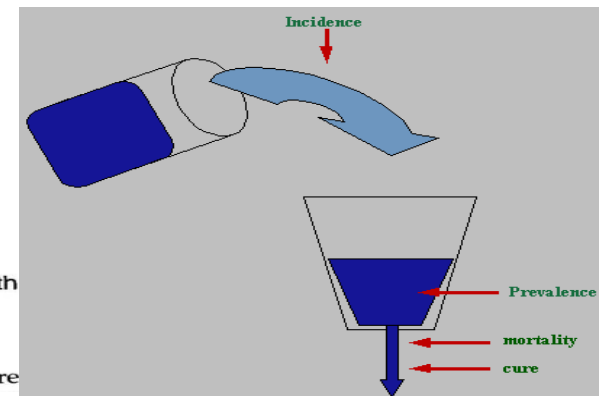
- Prevalence depends on the rate at which
- new cases of disease develop (the incidence rate)
- duration or length of time that individuals have the disease.
- The duration of a disease starts at the time of diagnosis and ends when the person either is cured or dies.

◇ Features of P and I

- $P = IR \times D$



Relationship between Incidence, Prevalence, Mortality and Cure



Measure of mortality/death rates

80

- These rates measures magnitude of **deaths** in a community
 - ▣ Some are crude like the crude death rate
 - ▣ Others are cause-specific mortality rate
 - ▣ Some others are adjusted like standardized mortality ration
- Rates are calculated using mid-interval/average population due to population fluctuation over time.
- **Below are given some formulas for the commonly used mortality rates and ratios.**

Common Mortality rates

81

- Crude death rate
- Age-specific mortality rate
- Sex-specific mortality rate
- Cause-specific mortality rate
- Proportionate mortality ratio
- Case fatality rate
- Fetal death rate
- Perinatal mortality rate
- Neonatal mortality rate
- Infant mortality rate
- Child mortality rate
- Under-five mortality rate
- Maternal mortality ratio

Measure of mortality...

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$$\text{CDR} = \frac{\text{Total no. of deaths reported during a given time interval}}{\text{Estimated mid interval population}} \times 1000$$

It reflects the risk of death in that community or country.

$$\text{Age-specific mortality rate} = \frac{\text{No. of deaths in a specific age group during a given time}}{\text{Estimated mid interval population of specific age group}} \times 1000$$

E.G infant Mortality Rate.

$$\text{Sex-specific mortality rate} = \frac{\text{No. of deaths in a specific sex during a given time}}{\text{Estimated mid interval population of same sex}} \times 1000$$

Mortality...

83

$$\text{Proportionate mortality ratio} = \frac{\text{No. of deaths from a specific cause during a given time}}{\text{Total no. of deaths from all causes in the same time}} \times 100$$

The proportionate mortality ratio asks the question: What proportion of deaths are due to a certain cause? For example when we say the proportionate mortality ratio for HIV/AIDS is 30 %, this means out of 100 total (of all) deaths 30 of them died from HIV/AIDS.

$$\text{Case Fatality Rate (CFR)} = \frac{\text{No. of deaths from a specific disease during a given time}}{\text{No. of cases of that disease during the same time}} \times 100$$

Represents the probability of death among diagnosed cases or the killing power of a disease.

Mortality...

84

Neonatal Mortality Rate = No. of deaths under 28 days of age reported
during a given time X 1000
No. of live births reported during the same time

Infant Mortality Rate (IMR) = No. of deaths under 1 year of age
during a given time X 1000
No. of live births reported during the
same time interval

Mortality...

85

Under- five mortality rate = No. of deaths of 0-4 years of age
_____ during a given time _____ X 1000
Average (mid-year) population of
the same age at the same time

Maternal Mortality Rate = No. of pregnancy associated deaths of
_____ mothers in a given time _____ X 100,000
No. of live births in the same time

Frequently used measures of mortality

Measure	Numerator (x)	Denominator (y)	Expressed per number at risk (10 ⁿ)
Crude Death Rate	total number of deaths reported during a given time interval	Estimated mid-interval population	1,000 or 100,000
Cause-specific Death Rate	# deaths assigned to a specific cause during a given time interval	Estimated mid-interval population	100,000
Proportional Mortality	# deaths assigned to a specific cause during a given time interval	Total number of deaths from all causes during the same interval	100 or 1,000
Death-to-Case Ratio	# deaths assigned to a specific disease during a given time interval	# new cases of that disease reported during the same time interval	100
Neonatal Mortality Rate	# deaths under 28 days of age during a given time interval	# live births during the same time interval	1,000
Postneonatal Mortality Rate	# deaths from 28 days to, but not including, 1 year of age, during a given time interval	# live births during the same time interval	1,000
Infant Mortality Rate	# deaths under 1 year of age during a given time interval	# live births reported during the same time interval	1,000
Maternal Mortality Rate	# deaths assigned to pregnancy-related causes during a given time interval	# live births during the same time interval	100,000

...Cont

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2. Age-specific mortality rate =
$$\frac{\text{No. of deaths in a specific age group}}{\text{during a given time}} \times 1000$$

 Estimated mid interval population of specific age group

One example of age specific mortality rate is Infant Mortality Rate.

3. Sex-specific mortality rate =
$$\frac{\text{No. of deaths in a specific sex}}{\text{during a given time}} \times 1000$$

 Estimated mid interval population of same sex

5. Neonatal Mortality Rate =
$$\frac{\text{No. of deaths under 28 days of age reported}}{\text{during a given time}} \times 1000$$

 No. of live births reported during the same time

7. Under-five mortality rate =
$$\frac{\text{No. of deaths of 0-4 years of age}}{\text{during a given time}} \times 1000$$

 Average (mid-year) population of
 the same age at the same time

4. Case Fatality Rate (CFR) =
$$\frac{\text{No. of deaths from a specific disease}}{\text{during a given time}} \times 100$$

 No. of cases of that disease during the same time

6. Infant Mortality Rate (IMR) =
$$\frac{\text{No. of deaths under 1 year of age}}{\text{during a given time}} \times 1000$$

 No. of live births reported during the
 same time interval

8. Maternal Mortality Rate =
$$\frac{\text{No. of pregnancy associated deaths of}}{\text{mothers in a given time}} \times 100,000$$

 No. of live births in the same time

Table 1.1 Neonatal sepsis, Hospital A, Ethiopia, 2003

ID	Sex	Culture Date	Symptom Date	DOB	Delivery Type	Delivery Site	Outcome	Admitting Symptoms
CS	F	6/2	6/2	6/2	vaginal	Del rm	Lived	dyspnea
CT	M	6/8	6/2	6/2	c-section	Oper rm	Lived	fever
WG	F	6/15	6/15	6/8	vaginal	Emer rm	Died	dyspnea
PA	F	6/15	6/12	6/8	vaginal	Del rm	Lived	fever
SA	F	6/15	6/15	6/11	c-section	Oper rm	Lived	pneumonia
HP	F	6/22	6/20	6/14	c-section	Oper rm	Lived	fever
SS	M	6/22	6/21	6/14	vaginal	Del rm	Lived	fever
JB	F	6/22	6/18	6/15	c-section	Oper rm	Lived	fever
BS	M	6/22	6/20	6/15	c-section	Oper rm	Lived	pneumonia
JG	M	6/23	6/19	6/16	forceps	Del rm	Lived	fever
NC	M	7/21	7/21	7/21	vaginal	Del rm	Died	dyspnea

Source: 11

Abbreviations

vaginal = vaginal delivery

Del rm = delivery room

Oper rm = operating room

Emer rm = emergency room

Exercise

The following information is about kebele X which was collected for the year 1999.

- Total average population = 40,000
- Total number of live births = 4000
- Total number of deaths = 400
- Total number of deaths before the age of 28 days = 50
- Total number of infant deaths = 200
- Number of women who died from pregnancy related causes = 160
- New cases of tuberculosis = 100
- All cases of tuberculosis = 300
- Deaths from tuberculosis = 60

Based on the above information calculate the following.

Class work...

90

- The incidence rate of tuberculosis.
- The period prevalence rate of tuberculosis.
- The case fatality rate of tuberculosis
- The Neonatal mortality rate.
- The infant mortality rate.
- The maternal mortality ratio

Natality frequency measure

91

- Natalty measures are used in the area of maternal and child health and less so in other areas.
- **Reading assignment**

Frequently used measures of natality

Measure	Numerator (x)	Denominator (y)	Expressed per Number at Risk (10^n)
Crude Birth Rate	# live births reported during a given time interval	Estimated total population at mid interval	1,000
Crude Fertility Rate	# live births reported during a given time interval	Estimated number of women age 15-44 years mid-interval	1,000
Crude Rate of Natural Increase	# live births minus # deaths during a given time interval	estimated total population at mid-interval	1,000
Low Birth Weight Ratio	# live births under 2,500 grams during a given time interval	# live births reported during the same time interval	100

Measures of disease burden

92

- Disease burden is the **impact of a health problem as measured by financial cost, mortality, morbidity, or other indicators.**
- It is often quantified in terms of quality-adjusted life years (QALYs) or disability-adjusted life years (DALYs).
- BOD is a summary measures provide a full account of the health of the population because it estimates the effect of both morbidity and mortality.
- It is quantifies by summary measures of population health(SMPH).

Measures of disease burden

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- Mortality does not give a complete picture of the burden of disease borne by individuals in different populations.
- The overall burden of disease is assessed using the disability-adjusted life year (DALY), a time-based measure that combines years of life lost due to premature mortality (YLLs) and years of life lost due to time lived in states of less than full health, or years of healthy life lost due to disability (YLDs).
- One DALY represents the loss of the equivalent of one year of full health. Using DALYs, the burden of diseases that cause premature death but little disability (such as drowning or measles) can be compared to that of diseases that do not cause death but do cause disability (such as cataract causing blindness).

Measures of disease burden

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- One DALY represents the loss of the equivalent of one year of full health.
- DALYs for a disease or health condition are the sum of the years of life lost to due to premature mortality (YLLs) and the years lived with a disability (YLDs) due to prevalent cases of the disease or health condition in a population.
- DALYs for a specific cause are calculated as the sum of the years of life lost due to premature mortality (YLLs) from that cause and the years of years of healthy life lost due to disability (YLDs) for people living in states of less than good health resulting from the specific cause.
- The YLLs for a cause are calculated as the number of cause-specific deaths multiplied by a loss function specifying the years lost for deaths as a function of the age at which death occurs. The loss function is based on the frontier national life expectancy projected for the year 2050 by the World Population Prospects 2012 (UN Population Division, 2013), with a life expectancy at birth of 92 years.
Prevalence YLDs are used here. Prevalence YLDs are calculated as the prevalence of each non-fatal condition multiplied by its disability weight.

Global Burden of Disease (GBD)

95

- The Global Burden of Disease Study (GBD) is **the most comprehensive worldwide observational epidemiological study to date.**
- It describes mortality and morbidity from major diseases, injuries and risk factors to health at global, national and regional levels.
- Examining trends from 1990 to the present and making comparisons across populations enables understanding of the changing health challenges facing people across the world in the 21st century.
- WHO now regularly develops GBD estimates at regional and global level for a set of more than 135 causes of disease and injury.
-

Effects and Use of Disease Burden

96

- Diseases can affect people
 - Not only physically,
 - But also mentally,
 - **As contracting and**
 - **Living with a disease**
 - **Can alter the affected person's**
 - **Perspective on life.**
 - Death due to disease is called death by natural causes.



Use of DB

97

- To compare population health across communities and over time.
- To provide a full of which disease, injuries, and risk factors contribute the most to poor health in specific population, including identification of the most important health problems and whether they are getting/worse over time.
- To assess which information/source of information are missing, uncertain/of low quality.

Quantitative and Qualitative measure of DB

98

- We can measure the burden in quantity or quality
- Quantitative measurement of DB includes life expectancy(LE), Years of potential life lost(YPLL), and health adjusted life expectancy(HALYs).
- Qualitative measurements of DB includes quality adjusted life years(QALY), and Disability adjusted life years(DALY), which are two common approaches to measure HALYs.

Health Adjusted Life Expectancy/Years(HALE/Ys)

99

- HALY are the population health summary measures typically used in estimates of DB.
- HALY measure the combined effects of mortality and morbidity in populations, allowing for comparisons across illness or interventions as well as between populations.
- DALYs and QALYs are the two common measure of HALYs.

DALYs and QALYs

100

- Both are count on the latest available epidemiological data.
- Obtained from different sources once completeness and diagnostic accuracy assessed.
- To calculate all HALYs/DALYs/QALYs three steps involved.
 - I. The health state/disease conditions associated with a pathogen/disease analyzed are defined and described (morbidity)
 - II. Each health state described in step 1 given a weighted value, often called a health related quality of life (HRQL) value, and
 - III. The value of each health state is combined with estimates of life expectancy (mortality).

DALYs

101

- 📄 1 DALYs can be thought as one lost year of “**healthy**” life.
- The DALY measures health gaps as opposed to health expectancies. It measures the difference between a current situation and an ideal situation where everyone lives up to the age of the standard life expectancy, and in perfect health.
- It combines in one measure the time lived with disability and the time lost due to premature mortality
- Based on life tables, the standard life expectancy at birth is set at 80 years for men and 82.5 for women.
- The sum of these DALYs across h populations / the BOD, can be thought of as a measurement of the gap between current health status and an ideal health situations where the entire populations lives to an advanced age, free of disease and disability.
- **DALYs** = *Mortality (years of life lost due to prematur death + Morbidty* (The measure of all non fatal disease affects such as illness episodes or chronic disability).

DALYs...

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Mortality

- Life expectancy-Age at death= $YLL/LYL/LY$
- Disability weight= is the range of 0 or fully healthy to 1 or 1005 or disabled.
- Disability adjustment(DA)= Disability wt. $\times$ Duration of illness.
- HQOL= Age at death-DALY
- DALYs measures the overall burden of disease expressed as the cumulative of years lost due to ill-health, and calculated as
- $DALYs = YLD(\text{Years Lived with disability}) + YLL(\text{year life lost})$
- Healthy life Disease/Disability early death LE



DALYs

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- On the population basis the YLL for a given age basically corresponded to the number of deaths for that given age multiplied by the standard life expectancy (SLE) at the age which death occurs, given as
- $YLL = N(\text{No of deaths}) * L(\text{standard of life expectancy at the age death years})$, but for an individual level YLL corresponds to the **SLE** at the age at which death occurs.
- $YLD(\text{pop.}) = I(\text{No of incident cases}) * DW(\text{disability weight}) * L(\text{Ave. duration of cases until remission or death/years})$
- $YLD(\text{Indi.}) = DW * L$

QALYs

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- QALYs is a generic measure of DB, including both the quality and the quantity of life lived.
- It measured both the effectiveness and cost-effectiveness of an intervention.
- $QALYs = \text{additional no of years of life} * HRQL$ (Health related quality o life).

Differences

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DALYs

- It's an absolute measure of used to compare disease burden in populations
- The goal s to minimize the bad of gaps in health
- It use disability weights(0=perfect health, and 1= death).
- What are the critics of both **DALYs and QALYs?**

QALYs

- It's used to analyze clinical interventions
- The goal s to maximize the good of quality of life.
- It use utility weights(0=death, and 1= prefect health)
- What are **DALE and HALE?**

5. Epidemiological Study designs

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- Descriptive study designs
 - ▣ Ecological/correlation
 - ▣ Case reports/case series
 - ▣ Cross-sectional
- Analytical study designs:
 - ▣ Case-control
 - ▣ Cohort
 - ▣ Experimental

Epidemiological Study design...

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- **Study Design:** is an arrangement of conditions for the **collection and analysis** of data that leads to the most accurate answer to the research question and the most economical way.
- **Research Design** The “architecture” of a study: its structure, specific details of the studied population, time frame, method, and procedures, including ethical considerations, all of which should be explicitly stated in a research protocol.
- The basic design strategies used in epidemiologic research can be broadly categorized according to whether such investigations
 - ❖ Focus on **describing the distributions** of disease (**Descriptive**) or,
 - ❖ **Elucidating its determinants** (**Analytical**).
- Selection of study design counts on
 - Hypothesis being tested
 - Resources available
 - Current state of knowledge

Strategy of epidemiology

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- Epidemiologic surveillance
- Epidemiologic research
- ❖ The usual sequence of epidemiological study designs

Descriptive
study of the
disease



Case control, Cohort



Experimental
study

Purpose of Epidemiology

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1. To diagnosis community health problems
 2. To identify natural history and etiology of diseases.
 3. To evaluate and plan health services.
- To achieve these purposes, it is crucial to understand basic types of
- epidemiological study designs.**

Why Epidemiological Studies?

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- To answer questions like:
 - ▣ How big is the problem (magnitude)?
 - Prevalence, incidence, mortality
 - ▣ What, who and where of any health problem?
 - Person characteristic of affected population
 - Place characteristics (locality)
 - ▣ What factors are associated with certain disease
 - Specific factors related to causation
 - ▣ To evaluate interventions
 - Which drug is best for patients with X disease
 - To evaluate any program etc.

What is a research?

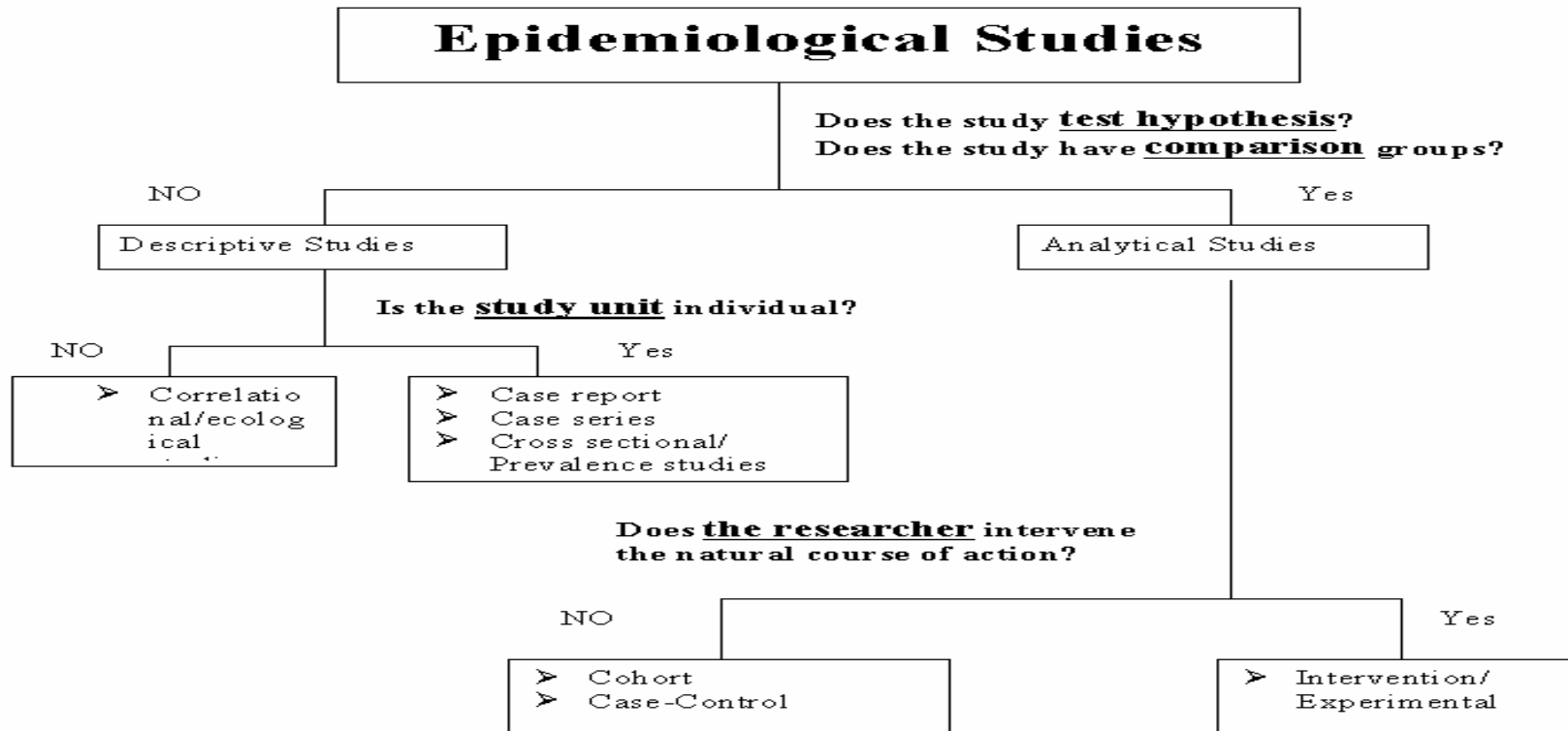
111

- It is a systematic approach to problem solving
- It Investigates a phenomenon to answer a burning question
- It tries to answer a vague study question

Taxonomy/hierarchy/typology/tree of Epidemiological study design

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Types of Epidemiological Study Design



□ **Descriptive epidemiology:**

- Generates idea(s)/ hypothesis to be tested using analytic study design

□ **Analytical epidemiology:**

- Uses comparison groups to establish an association between risk factors and illness in the two groups
- Tests a hypothesis

1. Descriptive study designs

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Contents

- Purpose of descriptive studies
- Types of descriptive study designs

Introduction

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Definition

- ❑ A study concerned with and designed only to describe the existing **distribution of variables** **without** much regard to **causal relationships** or other hypotheses.
- ❑ In a descriptive study, a parameter of disease occurrence is related to a **determinant** without concern for a **causal interpretation** of the relation.
- ❑ The major variables in descriptive Epidemiology are **person**, **place** and **time**, and it answered the questions **who**, **where** and **when**.

Descriptive study design

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Depend on the study subjects it can classified as

- Population as study subject
 - Correlational /ecological studies
- Individual as study subjects
 - Case report / Case series
 - Cross-sectional/ surveys/prevalence study
- A **study** concerned with and designed only to describe the existing distribution of variables without much regard to **causal relationships or other hypotheses**.

Descriptive...

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- In a descriptive study, a parameter of disease occurrence is related to a determinant without concern for a **causal interpretation of the relation**.
- **Examples**
 - A community health **survey** used to determine the health status of the people in a community **or**
 - Analyses of population registries to measure risks or trends in health indicators, generate hypotheses, monitor public health policies, etc.

Descriptive design...

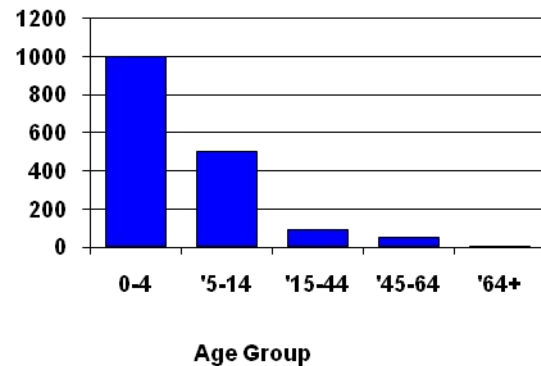
118

- Some studies simply describe occurrence of disease or health related problems
 - ▣ Prevalence of a disease,
 - ▣ Rate of certain behaviour
- When describing these factors, it does not link with anything
- However we can identify **unusual distributions** or correlations (e.g clusters)
- These insights can be used to generate interesting hypothesis (Case series, cross-sectional, ecological)

Descriptive ...

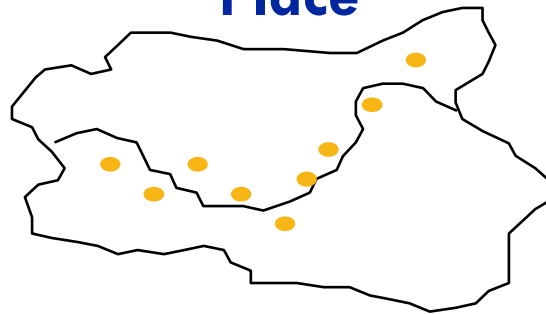
Cases

Person



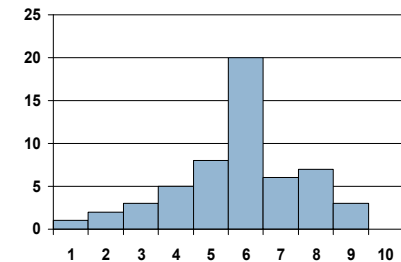
Who?

Place



Where?

Time



When?

1. Correlational/ Ecological study

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- Uses data from *entire population* (as a whole) to compare disease frequencies.
(ie it doesn't need data from individuals)
- Can be done quickly and inexpensively, often using already available data.
- The units of analysis are populations or groups of people rather than individuals.
- Ecological studies can reach valid causal inferences on causal relationships at the ecological level—i.e., on causal processes that occur at the group level or among groups.
- Ecological studies are necessary for decisions that affect entire groups (e.g., for public and private policies that are developed across an entire state, country, or region).

Rationale for ecological studies

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1. Low cost and convenient
2. Measurement limitation (conditions that are difficult to measure at individual level)
(eg environmental contact, dietary exposure, fluoride content)
3. Other designs may be unable to measure
3. Scientists having interest on ecologic effect

Levels of measure

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Source of data could be

1. Aggregate measures:

Observations (mean, proportion) derived from individuals in a group. (eg proportion of smokers, mean family income)

2. Environmental measures:

Physical characteristics of a place in which each group lives or work (air-pollution, fluoride content of water)

3. Global measure:

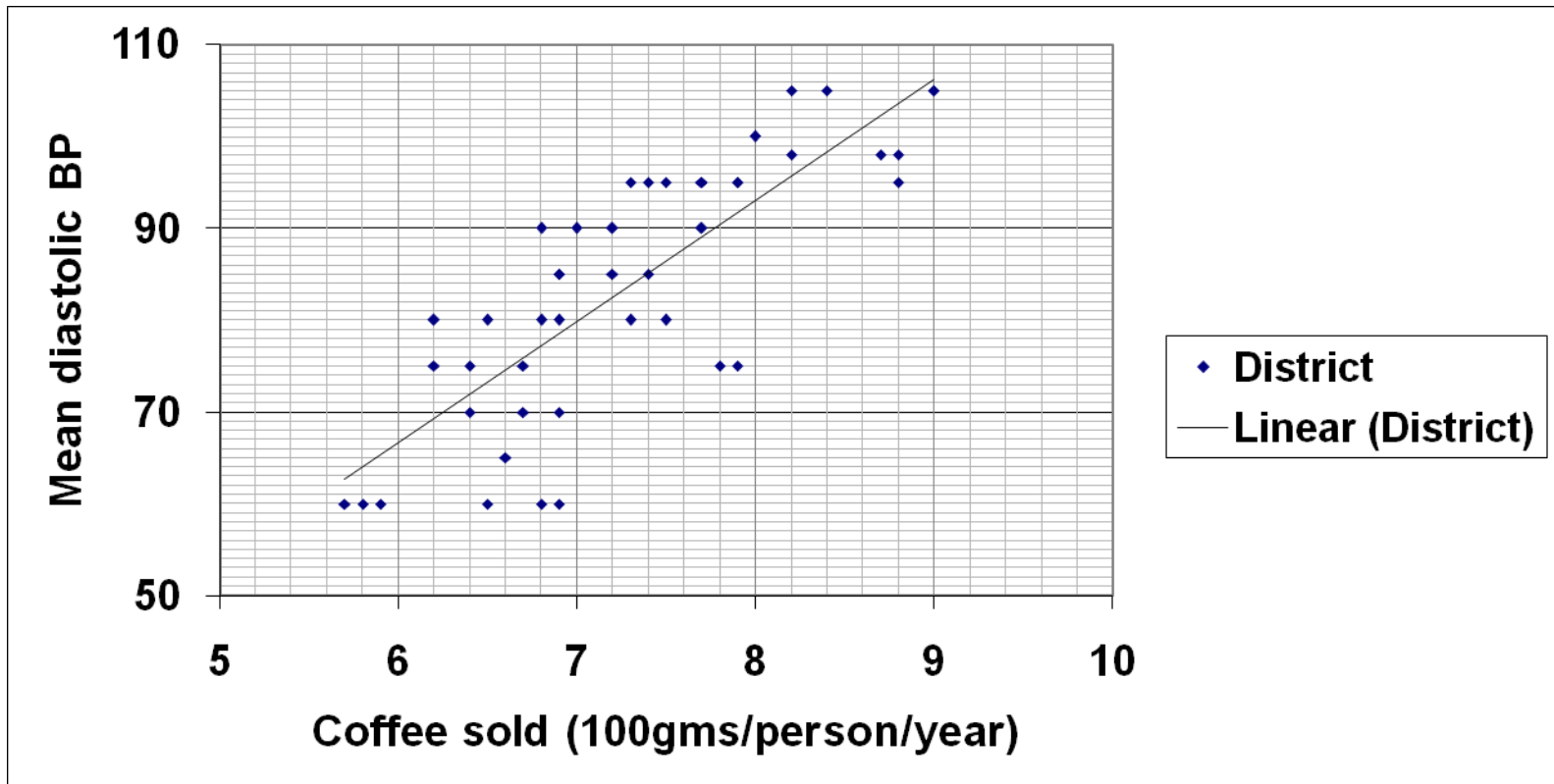
Attributes of groups, organizations or places, for which there is no measure at individual level,
(eg population density, presence of specific law)

Level of analysis

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- Completely ecologic analysis; all variables are ecologic measures and analysis is in a group.
- Partially ecologic analysis; addition of some individual variables and ecologic variables
- Measures of analysis in Correlational studies is using correlation coefficient (r)
- Correlation coefficient (r) is a descriptive measure between continuous variables that varies between -1 and +1)

Fig. Fictitious data to show correlation between coffee sold and mean diastolic BP. (positive $r \simeq 0.67$)



Levels of inference (*Generalization*)

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- It depends on the goal of inference wanted:
 - It can be to make **biologic (bio-behavioral) inference** about individual risks, or (such inferences are usually vulnerable to bias called “**Ecological/aggregate fallacy**”)
 - **AGGREGATIVE FALLACY** An erroneous application to individuals of a causal relationship observed at the group level. A type of ecological fallacy (sometimes just a synonym) and an antonym of the atomistic fallacy.
- It can also be **ecological inferences** about effects on group rates.
- It is also possible to infer to **contextual effect** that may be similar to biological effect.

Correlational/ecological studies (cont.)

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Strength

- ❑ Can be done quickly, inexpensively, and often using available data (routine records and reports-- for example, death rates, per capita income, national food consumption...)
- ❑ Low cost/cheap and convenience
- ❑ Measurement limitations at individual level, Eg environmental contact.
- ❑ Other designs may be unable to measure
- ❑ Interest on ecologic effect

Correlational/ecological studies (cont.)

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Limitation

- ❑ Inability to link exposure with disease in a single individual.
 - Data on exposure and outcome are not linked at the individual level; association found with aggregate data (average values) may not apply to individuals (**Prone to ecological fallacy**)
- Average exposure levels rather than actual individual values, ecological fallacy or bias.
- Lack of ability to control for effects of potential confounding factors.
- Conversely, lack of correlation does not necessarily imply absence of valid statistical association (**Berksons fallacy**)
- **Berkson's paradox**, also known as Berkson's bias, or collider bias
- It may mask a non-linear relationship between exposure and disease.
- **E.g. non-linear relationship in alcohol consumption and mortality from CHD ("J" shaped curve)**

Correlational/ecological studies (cont.)

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Example:

- Average per capita fat consumption and breast cancer rates compared between countries.
- Comparing incidence of dental cares in relation to fluoride content of the water among towns in the rift valley.
- Mortality from CHD in relation to per capita cigarette sales among the regions of Ethiopia.

2. Case reports or case series

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- Useful for the recognition of new diseases,
- Useful for constructing of the natural history of a disease,
- Use to formulate a hypothesis and to detect an epidemic (*because of clues about diseases determinants*).
- Studies are prone to **atomistic fallacy or AF** (the opposite of ecological fallacy).
- **Example of AF:** at an individual level a high income may be associated with lower rate of suicide but this does not mean that societies which are rich have a lower rate of suicide.

A. Case report/study

130

- It is the study of health profile of a single individual using a careful and detailed report by one or more clinicians.
- Detailed descriptions of a few patients or clinical cases (frequently, just one sick person) with an unusual disease or complication, uncommon combinations of diseases, an unusual or misleading semiology, cause or outcome (maybe a surprising recovery ...).
- **Are preliminary observations that are later refuted.**
- **Cannot estimate disease frequency or risk.**
- It is common form that is published in articles
- It is made using
 - Simple history,
 - Physical examination and
 - Lab. / radiological investigation.

CR...

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- Report is usually documented if there is unusual medical occurrence, thus it may be first **clue for identification of** a new disease.
- It is useful in constructing **a natural history** of individual disease.
- It **was a single case report that formulated the hypothesis** of oral contraceptive use increases venous thrombo embolism.

Case Reports...

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- A most basic type of descriptive study of individuals comprising of a careful and detailed report of a single patient by one or more physician
 - ▣ E.g. the progress of symptom and sign, response of treatment
- It is the study of health profile of a single individual using a careful and detailed report by one or more clinicians
 - ▣ It is common form that is published in articles
- It is made using
 - ▣ Simple history, physical examination and Lab./ radiologic investigation

Case Reports ...

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- Useful for the recognition of new disease
- Important for constructing the natural history of disease
- Help to formulate a research hypothesis and detection of epidemics
 - ▣ It was a single case report that formulated the hypothesis of oral contraceptive use increases venous thrombo-embolism

Case report :Limitations

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- No appropriate comparison group
- Cannot be used to test for presence of a valid statistical association
- Since based on the experience of one person:
 - ✓ Presence of any risk factor may be coincidental
 - ✓ Not a true epidemiologic design

B. Case series

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- It's a collection of patients with common characteristics used to describe some clinical, pathophysiological, or operational aspect of a disease, treatment, or diagnostic procedure.
- Some are similar to the larger case reports and share the virtues of this design.
- The number of subjects does not attenuate the limitations of the design.
- A case series does not include a comparison group and is often based on prevalent cases and on a sample of convenience.
- Common selection biases and confounding severely limit their power to make causal inferences.
- Individual case report can be expanded to a case series, which describes characteristics of a number of patients (usually 5-12) with a similar disease.
- Similar to case report, it is usually made on cases having **new and/ or unusual disease** (giving interest to clinicians)
- It is often used to detect the emergence of **new disease** or **an epidemics**.
Eg. The first five AIDS cases in USA.

Case series...

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- A sequence of case reports with common elements such as similar clinical features and suspected common exposures
 - ▣ Individual case report can be expanded to a case series, which describes characteristics of a number of patients (usually 5-12) with a similar disease
- Similar to case report, it is usually made on cases having new and/ or unusual disease (giving interest to clinicians)
- It is often used to detect the emergence of new disease or an epidemics

Example

137

Five young, previously healthy homosexual men were diagnosed as having *Pneumocystis carinii pneumonia* (PCP) at Los Angeles hospital during a six month period from 1980 to 1981 that's caused by the fungus *Pneumocystis jirovecii*.

This form of pneumonia had been seen almost exclusively among older men and women whose immune systems were suppressed.

This unusual circumstance suggested that these individuals were actually suffering with a previously unknown disease, subsequently it was called **AIDS**.

Uses of case series studies

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- Can be valuable early evidence for associations between exposures and diseases which can be studied in more detail
- Useful for the recognition of new diseases
- Useful for constructing of the natural history of a disease,
- Use to formulate a hypothesis and to detect an epidemic

Case series

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Strength:

- useful for studying signs and symptoms (new syndromes) and creating case definitions for epidemiological studies. Example: AIDS
- Case-series that include cases at various stages of an illness from mild cases to dead supplemented by investigation of the past medical history of these cases and observing them to death (doing autopsy as appropriate) can help build up a picture of the natural history of a disease
- Very useful in providing critical information, for hypothesis generation.

Case series...

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Limitation:

- Report is based on single or few patients, which could happen just by coincidence
- Lack of an appropriate comparison group
- Rates can not be calculated since the population corresponding to the source of cases can not be defined well
- Detailed and complete risk factor information is difficult to obtain for all cases from records
- Studies are prone to **atomistic fallacy** (the opposite of ecological fallacy).
 - ▣ **Example:** at an individual level a high income may be associated with lower rate of suicide but this does not mean that societies which are rich have a lower rate of suicide.

Limitations of series studies...

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- Are able to formulate a hypothesis but are not able to test for presence of valid association
 - ▣ Their fundamental limitation is presence of a risk factor that is simply coincidental (by chance)
- It is difficult to test for association because there is no relevant comparison group
- Case-series lack a control (comparison) group, and so can result in false leads, wasted energy and resources

CR and CS...

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- Both case report and case series are able to formulate a hypothesis but are not able to test for presence of valid association.
- Fundamental limitation of case report is, presence of a risk factor could be simply **coincidental (chance)**
- You are **not able to test for association** because there is **no relevant comparison group**

Cross-Sectional Studies /prevalence survey/

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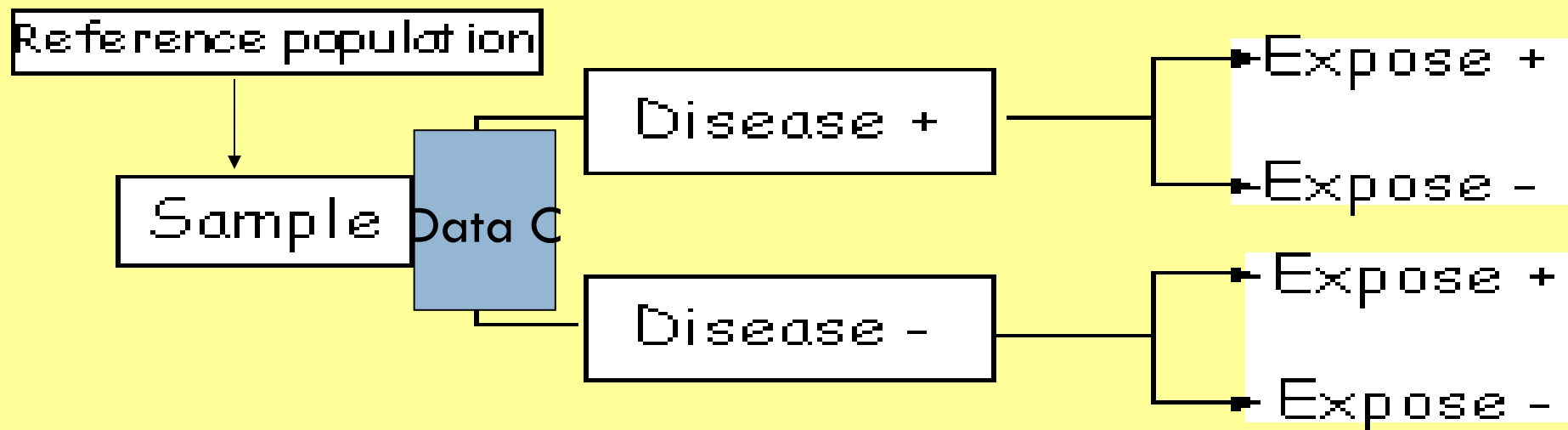
- Measure disease and exposure status simultaneously among individuals in a well-defined population at a point in time
 - Is generally called **prevalent study**
 - **Snapshot** of the health status of populations at a certain **point in time**

Steps in the conduct of cross-sectional studies:

1. Define a population of interest (reference population)
2. Recruiting a representative sample (adequate size, random selection)
3. Measure the variables of interest (disease &or exposure) at the same point in time
4. Sorting (if necessary) & Analyze the data

Cross-Sectional Studies (cont.)

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Cross-Sectional Studies (cont.)

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Characteristics of cross-sectional studies:

- Assess both exposure and outcome simultaneously
- Are based on point prevalence rates with only few exceptions
- Are frequently made on total population samples
- Subdivision of the total population takes place after data collection unlike in cohort & case- control studies
- Are cheaper, easier & useful **for studies of exposures that are unalterable over time** or relatively permanent features for individuals
E.g.-blood group, ethnicity, sex, etc.

Cross-Sectional Studies (cont.)

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- Data are obtained only once
 - ▣ Individuals with a defined disease, risk factor, or other condition of interest are identified at a point in time
- May be used first step in longitudinal or case control studies
- **Can have descriptive or analytic purposes**
 - ▣ The descriptive type is carried out to study prevalence of health related events at a **point in time/snapshot**
 - Diseases, risk factors, coverage of interventions, health service utilization, knowledge, attitude and practice
 - ▣ The analytic type is carried out to assess association between exposure and outcome
 - Exposure and disease status are assessed simultaneously among individuals at the same point in time
 - Measures of association is **made using odds ratio**

Cross-Sectional Studies (cont.)

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- Since exposure and disease status is assessed at a single point in time, **temporal relationship between exposure and disease can not be clearly determined**
 - ▣ Egg and hen phenomena
- Study prevalent cases rather than incident cases
 - ▣ The person might have changed the behaviour that resulted to the disease
 - ▣ Prevalent cases represent survivors who may be atypical with respect to exposure status
 - Incidence-prevalence/selective survival/Neyman's bias
- Therefore identify a mix of risk factors (incidence) and prognostic factors (duration)
- Best used for screening hypotheses if in etiological search
- Generally provide weak evidence regarding causation

Cross-Sectional Studies (cont.)

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Numerators and denominators in cross-sectional studies:

Numerators-two probabilities: -

- Referring to the exposure-prevalence rate of the outcome is compared between exposed & non-exposed subgroups
- Referring to the outcome-the prevalence rate of the exposure is compared in those with & without the outcome

Denominator-total population in the sample, based on the comparison being made (exposure prevalence or outcome)

Cross-Sectional Studies (cont.)

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Example:

		<u>Respiratory problems</u>		
		<u>Yes</u>	<u>no</u>	<u>total</u>
<u>Smoking</u>	<u>Yes</u>	70	50	120
	<u>no</u>	30	70	100
	Total	100	120	220

Prevalence of smokers (among respiratory problems)

$$= \frac{70}{100} \times 100 = 70\%$$

Prevalence of respiratory problems (among smokers)

$$= \frac{50}{120} \times 100 = 58.3\%$$

Factors Influencing Ds Prevalence

Increased by:

Longer duration of the disease

Prolongation of life of patients without cure

Increase in new cases (increase in incidence)

In-migration of cases

Out-migration of healthy people

In-migration of susceptible people

Improved diagnostic facilities

(better reporting)

Decreased by:

Shorter duration of disease

High case-fatality rate from disease

Decrease in new cases (decrease in incidence)

In-migration of healthy people

Out-migration of cases

Improved cure rate of cases

Cross-Sectional Studies (cont.)

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Advantages

- Less expensive: a one-step, one-time collection of data
- It starts with reference population, thus making possible generalization
- Provide much information useful for planning health services & medical programs
- Show relative distribution of conditions, Diseases, injury & disability in groups & populations
- Are based in a sample of a major population & do not relay on individuals that present themselves for medical treatment.
- May study several outcomes
- Controls over selection of subjects
- Relatively short duration
- Good first step for cohort study
- Yields prevalence

Cross-Sectional Studies (cont.)

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Disadvantages

- ❑ It may not show strong cause-effect relationship if sample size is small
- ❑ Potential bias in measuring exposure
- ❑ Not feasible for rare diseases
- ❑ Does not yield incidence or true relative risk
- ❑ Prevalent rather than incident (new) cases are used – the exposure could be associated with survival after dx occurrence, rather than development of the disease
- ❑ Temporal sequence between exposure and disease cannot be established *
i.e. which came first, chicken or the egg?
- ❑ Doesn't establish sequence of events, (temporal relationship)
- ❑ Potential survivor bias
- ❑ Not feasible for rare diseases

	Diseased	Non-diseased	Total	
Exposed	a	b	a+b	Exposed
Not exposed	c	d	c+d	Not exposed

$$\text{Prevalence Ratio (PR)} = \frac{a/(a+c)}{c/(c+d)}$$

Cholesterol	CHD	No CHD	Total
High	100	400	500
Low/normal	50	450	500

$$\text{Prevalence Ratio (PR)} = \frac{100/(100+400)}{50/(50+450)} = 2$$

Interpretation: In this study population, the prevalence of CHD is 2 times higher among those with high cholesterol, compared to the prevalence in those with normal or low cholesterol

2. Analytical Epidemiology

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Contents

- Purpose of analytical epidemiology
- Observational vs. experimental analytical studies
- Case control studies
- Cohort studies
- Intervention studies
- Types of intervention studies
- Analysis and interpretation

PURPOSE

- To test hypothesis about causal relationship
- To search for cause & effect
- To compare treatment regimens / prevention programs
- To assess diagnostic tests
- To quantify the association b/n exposure & outcome (measure of association)
- Focuses on determinants of diagnosis by testing hypothesis

Basic Question in Analytic Epidemiology

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- Are exposure and disease linked?

- Exposure  Disease

- Look to link exposure and disease

- What is the exposure?
 - Who are the exposed?
 - What are the potential health effects?
 - What approach will you take to study

- The relationship between exposure and effect?

Prospective Study

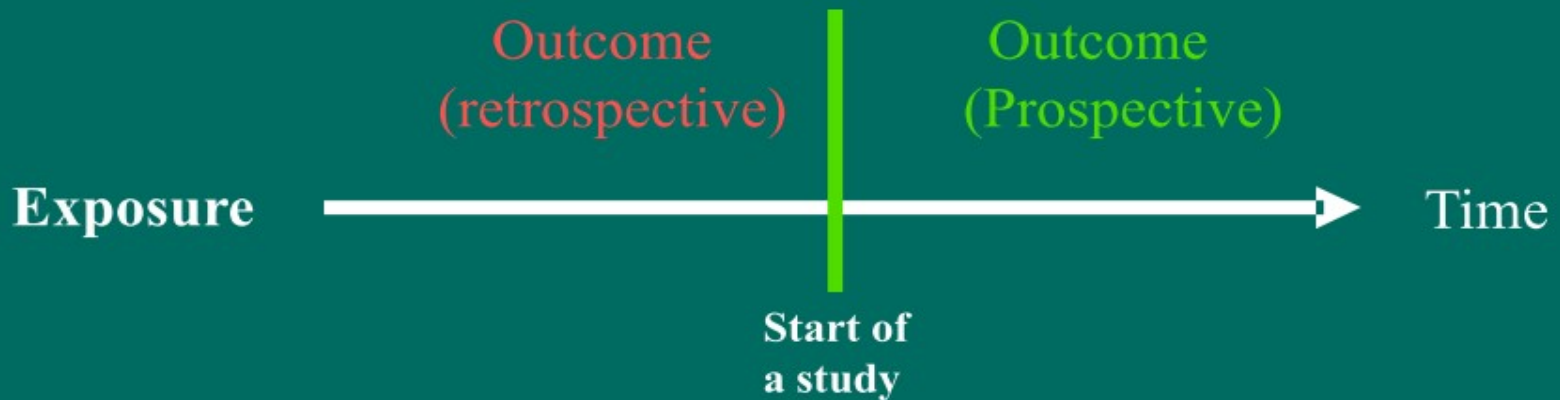
- ▣ looks forward,
- ▣ looks to the future, examines future events, follows a condition, concern or disease into the future

Retrospective Study

- ▣ to look back',
- ▣ looks back in time to study events that have already occurred

Prospective Vs Retrospective Study

- In prospective study outcome has not occurred at the beginning of the study.*



Types of analytic study design

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There are two types of analytic study designs

- ▣ Observational- to simply observe
- ▣ Interventional - manipulation or determination

OBSERVATIONAL STUDIES

- ▣ Information is obtained by simple observation of the event (two basic types): -
 - ▣ Case control study
 - ▣ Cohort study
 - ▣ Cross sectional (some times)

Case-Control Studies

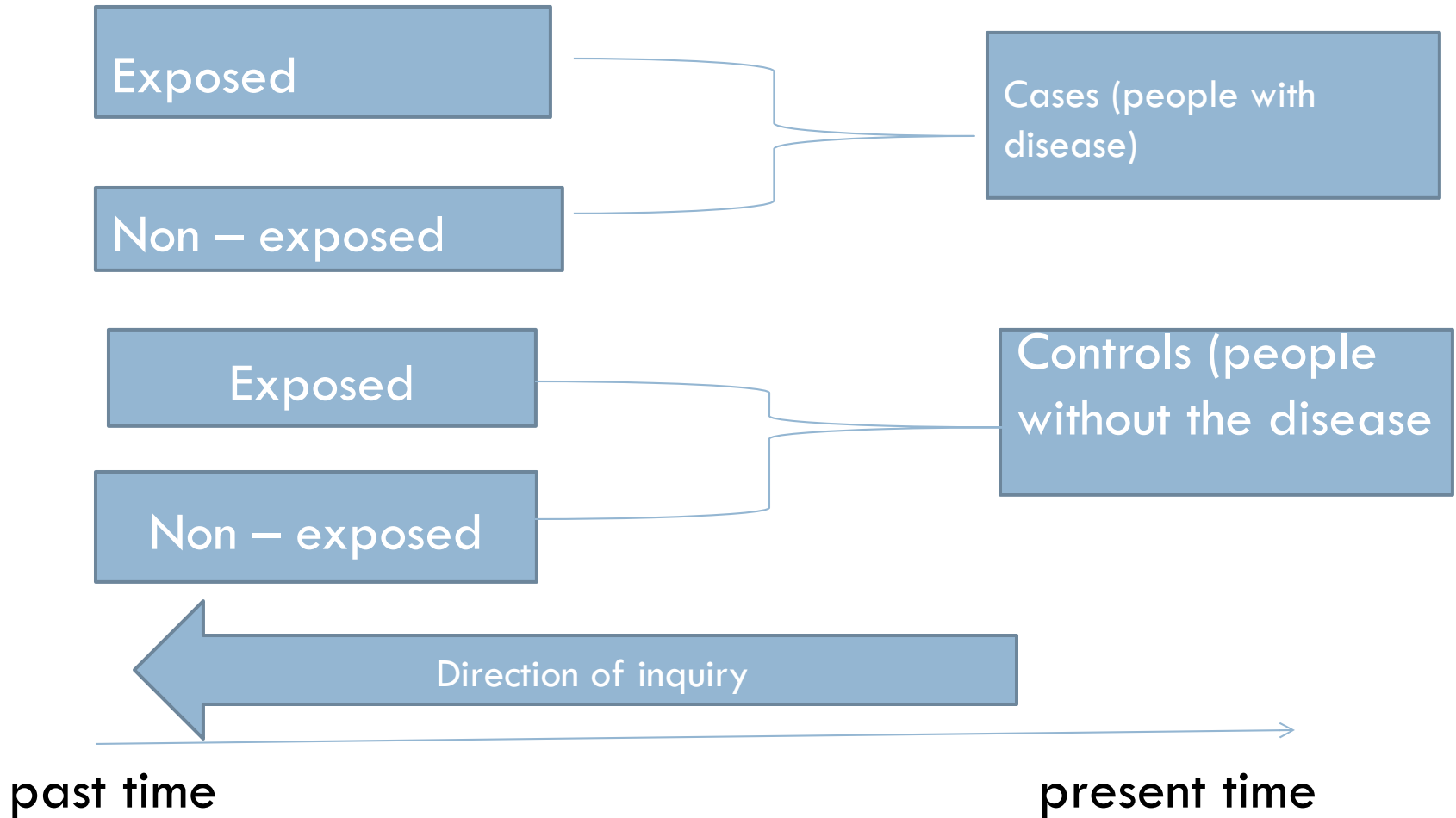
160

- **Syn: case-comparison study, case-referent study, retrospective study**
- **Defn: past history of exposure is compared in cases and their controls for their exposure status**
 - ▣ Look back at exposure (retrospective) - It assess retrospectively on exposure status.
- Case control studies aim to identify potential associations between 'risk' factors or exposure and a particular disease or outcome

Case-Control Studies...

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- Unit of observation and analysis: Individual (not group)
- Case-control studies are the most frequently undertaken analytical epidemiological studies
- They are the only practical approach for identifying risk factors for **rare diseases**
- They are best suited to the study of diseases for which medical care is sought, **such as cancers or hip fracture**
- It is relatively cheaper
- Measure of association is using **OR**



Case-control design

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- Select a group of subjects with disease (cases) and
 - ▣ Try to characterize their pre-disease exposure status
- Select a group of subjects without disease (controls)
 - ▣ try to characterize their exposure status
- Compare how common exposure was among cases with how common it was among controls, as a measure of exposure-disease association (measure of association)

Outcome in case control study

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- *Can we study outcome other than disease ?*
- The **outcome** has traditionally been the **presence or absence of disease**.
- However, other outcomes can also be studied e.g. **disability, smoking cessation etc.**

Start by defining a **CASE**

- Clinical definition insufficient
- list of criteria
- Reproducibility
- certainty of diagnosis
- Severity

Selection of Cases

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- Select cases after the diagnostic criteria and definition of the disease is clearly established
 - Study cases should be representative of all cases
 - The study need not include all cases in the population
- Cases may be located from hospitals, clinics, disease registries, screenings, etc
- Incident cases are preferable to prevalent cases for reducing
 - Recall bias
 - over-representation of cases of long duration

Hospital-based Vs population-based cases

□ Hospital-based

- Easy and inexpensive to conduct but it is prone for selection bias.

□ Population-based

- Avoids selection bias,
- Allows the description of a disease in the entire population
- The direct computation of rates of disease in exposed and non exposed persons.

Selection of Cases...

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Incident Vs Prevalent cases

- ▣ The ideal set of cases would be new (incident) and representative of all cases of the health problem under study

▣ **Prevalent cases:**

- ▣ Increase sample size available for rare disease
- ▣ Difficult to establish temporal sequence between exposure and outcome. E.g. Coffee consumption and peptic ulcer disease.

▣ **Incident cases:**

- ▣ Helpful to establish temporal relationship between exposure and outcome.
- ▣ Records are easily obtainable and recall is not a serious problem.

Selection of Controls

168

- Main challenge in case-control studies
- Controls should come from the same population at risk for the disease as the cases
- Should be similar to cases, except for disease status
 - ▣ E.g., hospital controls for hospital cases
- Should be representative of all persons at risk for the disease in the population
 - ▣ E.g., population-based controls
- Multiple controls can be used to help add statistical power when cases are unduly difficult to obtain
 - ▣ Using more than one control group lends credibility to the results
 - ▣ More than 3 controls for a case is usually not cost-efficient

Selection of Controls...

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Non-hospitalized Controls

- Population-based (random sample)
 - ▣ Costly and difficult
- School rosters, neighborhood, friends
 - ▣ Similar socioeconomic status (SES)

Hospitalized Controls

- Captive source (compliant and economical)
- Better recall of exposures
- But hard to define reference population
- Are a selected group
 - ▣ E.g., more smokers, drinkers and oral contraceptive users
- Select specific “other diagnosis?”

How many cases and controls?

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Based on:

- Availability of cases and controls
- List the main risk factors
- Review the prevalence of the risk factor in the controls
 - What would be the **expected prevalence** of the risk factor in controls

- ❑ A single control group is optimal in most studies
- ❑ Add more control groups only
 - ❑ when you are not confident with the control group or
 - ❑ when you see a clear deficiency in your control group or
 - ❑ when there is a clear advantage by adding another control
- ❑ Conditions for multiple controls:
 - ❑ when a single control group is considered to be not appropriate
 - ❑ when the selected group has a specific deficiency that could be avoided by inclusion of another control group
- ❑ **Control-case ratio**
 - ❑ Efficiency of a study can be maximized up to control-case ratio of 4:1

Selection of controls

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Considerations:

- ▣ Avoiding selection bias
- ▣ Avoiding information ('recall') bias

Advantages:

- Easily identified and readily available-**reduced cost**.
- More likely than healthy individuals to be aware of antecedent exposures or events – **minimize recall bias**
- Controls are also likely to have been subject to the same intangible selection factors -**minimize selection bias**
- More likely to be cooperative -**reduce non-response bias**

Disadvantages:

- Ill individuals are different from healthy
- Danger of altering the direction of association or **masking a true association** b/n exposure and outcome

Advantages:

- Generalizability is possible
- Good when cases are selected to represent affected individuals in a defined population.
 - ▣ For example, if cases to that particular hospital are coming from a geographically defined area selection of controls from the entire population could be possible.

Disadvantages:

- Costly and time-consuming
- Recall bias - controls may not recall exposures with the same level of accuracy
- People might be less motivated to participate, which increases non-response rate

Strengths and Weaknesses of C-C Studies

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Strengths

- Can use prevalent cases
- Retrospective design is inexpensive and quick to complete
- Can study rare diseases
- Can examine multiple exposures
- Usually requires a smaller study population than a cohort study
- Relatively cheap
- Quick and easy to complete, cost effective

Weaknesses

- Uncertainty of exposure-disease time relationship
- Inability to provide a direct estimate of risk
- Very prone to biases (recall & selection bias)
- Misclassification of exposures and outcomes
- Selection of controls
- Cannot yield a population level measure of disease incidence/prevalence.

Sampling

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- Sample size - for test of significant difference between two proportions, the following formula can be used:

$$n = \frac{(Z_{\alpha} + Z_{2\beta})^2 \{p_1(1-p_1) + p_2(1-p_2)\}}{(p_1 - p_2)^2}$$

Parameters:

n - size of sample in each group

P_1, P_2 —estimated population prevalence in the comparison groups

$\beta = 1 - \text{Power}$ (the probability that if the two proportions differ the test will produce a significant difference)

- Usually a power of 80% is used

- ▣ Relative Odds or Odds Ratio (OR)
- ▣ Attributable Risk Percent (AR%) among exposed
- ▣ Population Attributable Risk Percent (PAR %) :
- ▣ *Attributable risk (AR) or Risk difference (RD)
- ▣ *Population attributable risk (PAR)

Measure of association of case-control study is **odds ratio(OR)**

$$\text{OR} = \frac{\text{odds of disease in exposed}}{\text{odds of disease in unexposed}}$$

- ▣ Odds Ratio = 1, null hypothesis
- ▣ When $\text{OR} > 1$, risk factor
- ▣ When $\text{OR} < 1$, it is protective factor

COHORT STUDIES- Definition

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- Cohort
 - ▣ A group of people who share a common experience or condition
 - ▣ E.g. Birth cohorts, cohort of smokers, occupational exposures
- Cohort studies
 - ▣ The observation of a cohort over time to measure outcome(s)
 - ▣ Because the data on **exposure and disease** refer to different points in time, cohort studies are longitudinal, like case-control studies
 - ▣ Longitudinal, follow-up or incidence studies , **prospective study??**

Definition ...

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- They have 2 primary purposes:
 - ▣ **Descriptive:** to describe the incidence rates of an outcome
 - ▣ **Analytic:** to analyze associations between the outcomes and risk factors
 - Usual type
- Begin with a group of people free of disease
- Who are classified into subgroups according to exposure to a potential cause of outcome
- The whole cohort is followed up to see how the subsequent development of outcome differs between the groups with and without exposure (Figure 1)

Definition...

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BASIC ELEMENTS

- Disease free at entry
- Selected by exposure status rather than outcome
- Follow up is needed to determine the incidence of the outcome
- Compares incidence rates of exposed against unexposed group

a. EXPOSURE STATUS

- Study subjects should be disease free.
- Define study subjects using inclusion /exclusion criteria

b. POSSIBLE OUTCOME IN COHORT

1. No disease
2. Diseased
- 3 Lost to follow up

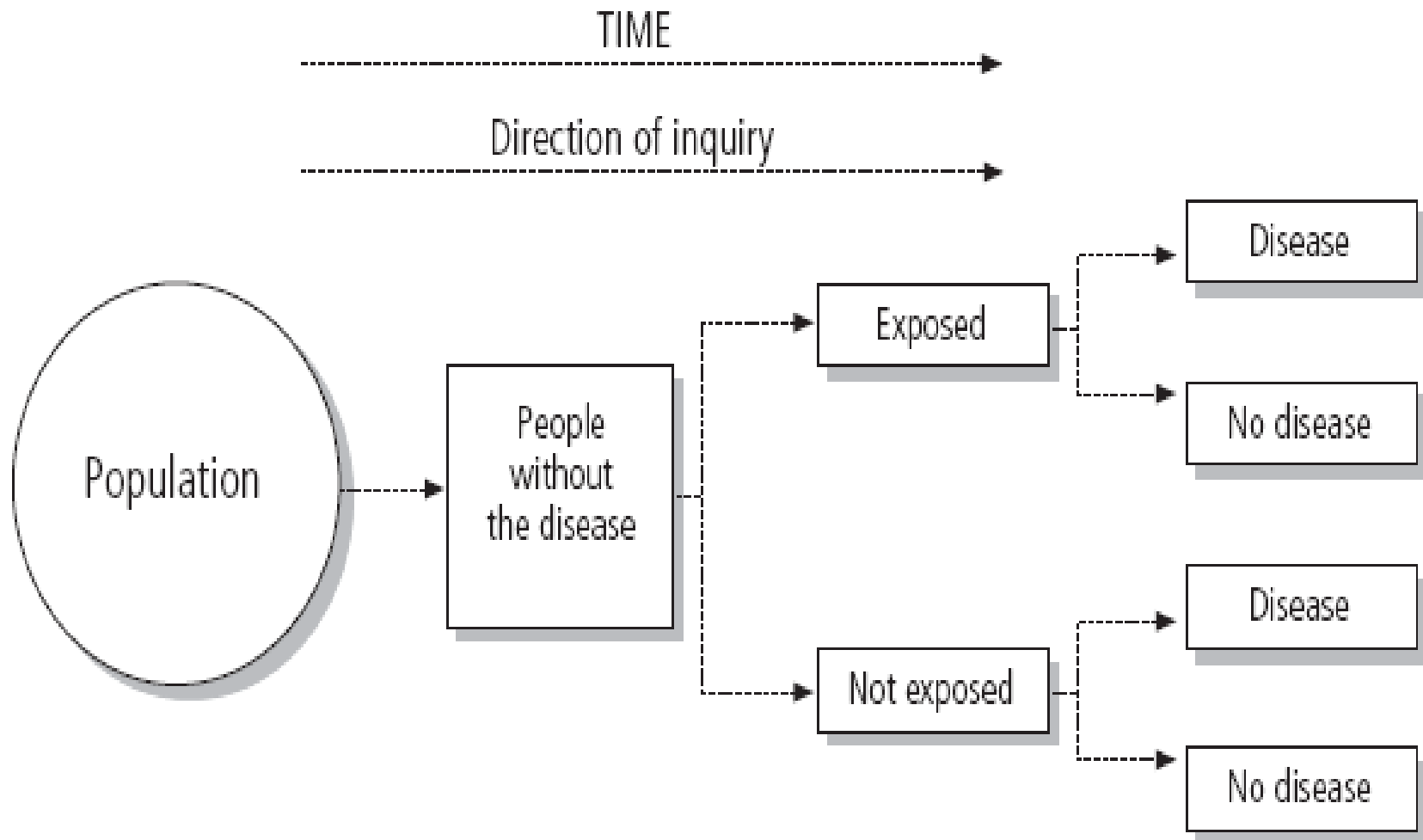


Figure 1- design of cohort study

Types

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- Prospective vs. Retrospective (Concurrent vs. Non-concurrent)
 - ▣ Depending on temporal relationship between **initiation of the study /timing of data collection/ and occurrence of the disease**
 - ▣ Cohort studies have been called **prospective studies**, but this terminology is confusing and should be avoided
 - the term “prospective” refers to the timing of data collection and not to the relationship between exposure and effect

Prospective cohort studies

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- Exposure and outcome data is collected after start of the study
 - ▣ cohorts Identified in the present →
 - ▣ exposure status or possible explanatory/prognostic factors determined in the present →
 - ▣ Cohorts followed-up to identify outcome →
 - ▣ Ascertainment of outcome done in future →

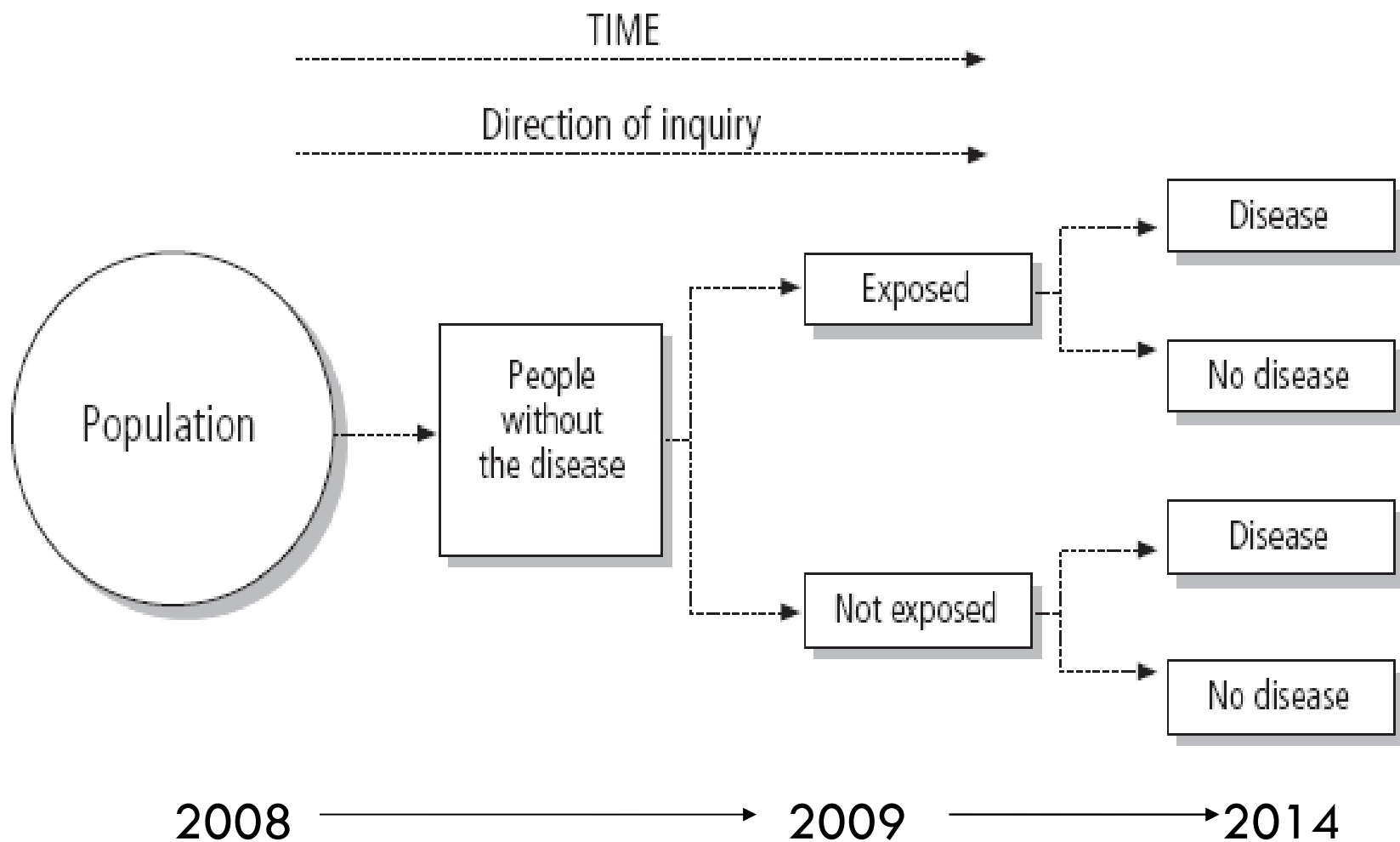


Fig. 2. Design of prospective cohort studies

Retrospective cohort

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- All the exposure and effect data have been collected before the actual study begins
- This type of investigation is called **a historical cohort study**
- Conduct
 - ▣ Identify cohort in the past using records/databases
 - ▣ Determine exposure or prognostic factors in the past using again records or databases then →
 - ▣ Identify outcome in past or present or future (in case of mixed cohort) →

Retrospective cohort...

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- Costs can occasionally be reduced by using a historical cohort (identified on the basis of records of previous exposure)
- This sort of design is relatively common for studies of cancer related to occupational exposures
- For example, records of military personnel exposure to radioactive fall-out at nuclear bomb testing sites have been used to examine the possible causal role of fall-out in the development of cancer over the past 30 years

Retrospective vs. Prospective

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- Can be conducted more quickly and cheaply
 - ▣ All relevant events have occurred
 - ▣ Efficient for disease with long latency periods
- Depend on availability of routine data
 - ▣ Incompleteness
 - ▣ Lack of data on confounding variables

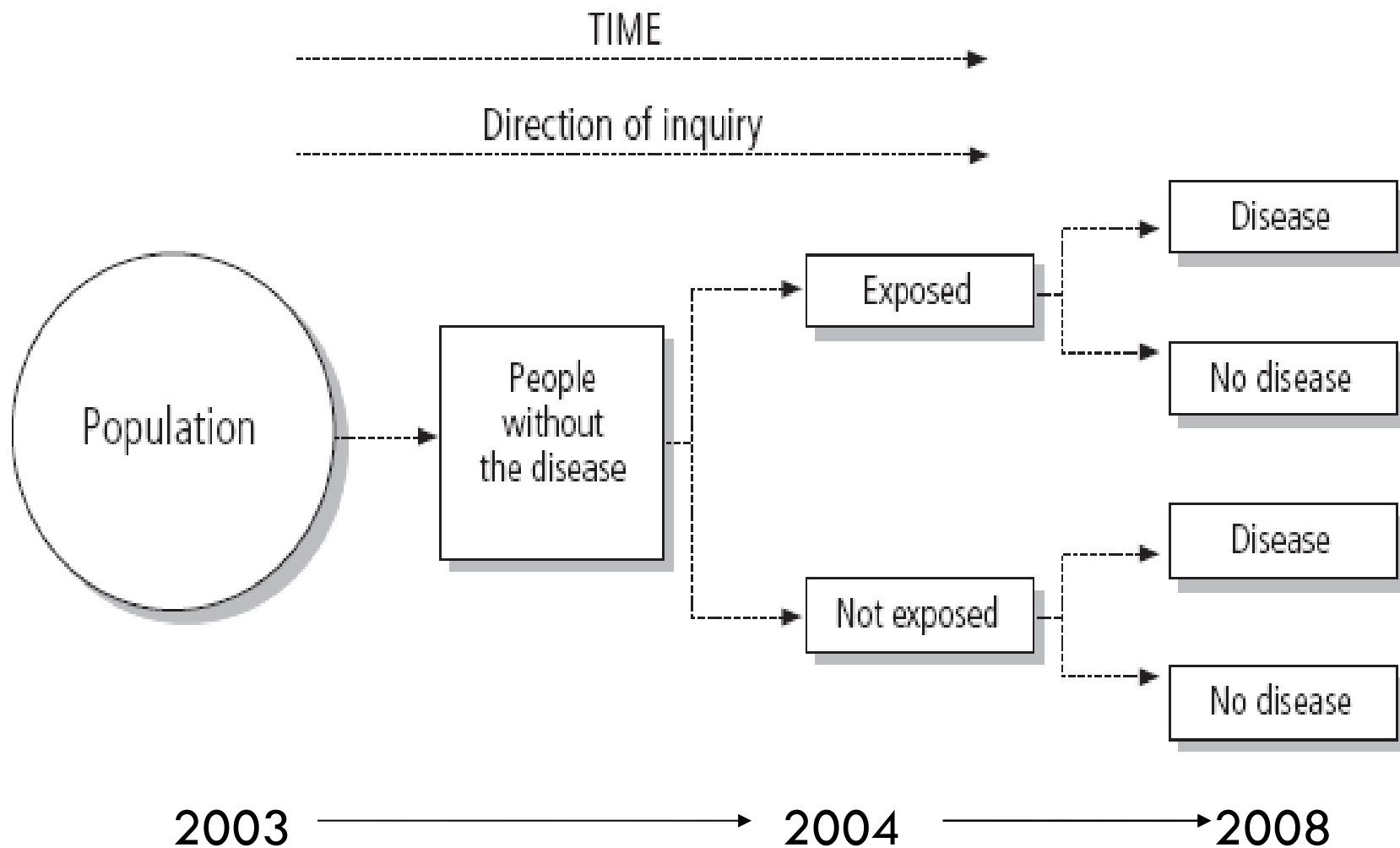


Fig. 3. Design of retrospective cohort studies

Design and data collection

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1. Define and identify cohorts

1.1. Identify population at risk

□ Selection of Exposed Population

- Depends on research question
- Outcome must not be rare in exposed
- Accessible and compliant subjects
 - E.g., Nurse's Health Study, Physicians

□ Selection of Non-exposed Group

- Similar to exposed
- Control for confounding factors

Design and data collection...

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- 1.2. Screen identified subjects for the disease and omit the prevalent cases
2. Define, assess, identify and classify exposure
3. Follow-up and ascertain outcome
 - ▣ Timing of outcome events-case definition

□ Self Report (interviews, surveys, etc.)

□ Advantages:

- Opportunity to question subjects on as many factors as necessary.
- Good for collecting information on exposures not routinely recorded.

□ Disadvantages:

- Subject to response bias (e.g. due to stigma, response expectations, etc.).
- Subject to interviewer bias

□ Pre-existing Records:

□ Advantages:

- **Inexpensive** and Relatively **easy** to work with
- Usually unbiased -data were collected for other purposes

Sources of Exposure Information...

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□ **Disadvantages:**

- Exposure information may not be precise enough to address the research question.
- Records frequently do not contain data on potential confounding factors.

□ **Direct Measurement**

- Can provide objective and unbiased exposure ascertainment (e.g. blood pressure, serum samples, environmental measurements, etc.).

□ **Death certificates** (National Death Index) –for some causes

□ **Clinical history**

□ **Medical examination** (periodic re- examination of the cohort)

□ **Hospital discharge logs**

Sampling

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Sample size - for test of significant difference between two proportions, the following formula can be used:

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Parameters:

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$\beta = 1 - \text{Power}$ (the probability that if the two proportions differ the test will produce a significant difference)

- ▣ Usually a power of 80% is used

- $RR = \frac{\text{INCIDENCE EXP}}{\text{INC NONEXP}}$
- Presence of association
 - $RR=1$ – no association; $RR<1$ – negative association; $RR>1$ – positive association
 - $AR \& AR\%$, PAR & $\%$,
 - $P\text{-value} < 0.05$ – statistically significant association
- Strength of association
 - ▣ Weak – RR close to zero; Strong – RR far from zero

Advantages

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- Elucidates temporal relationship between exposure and outcome (It shows temporal sequence)
- it is possible to examine a range of outcomes of particular value when exposure is rare
- Minimize selection bias – **prospective**
- Can examine multiple effect of a single exposure
- Allow direct measure of incidence of disease in the exposed & none exposed.

Disadvantages

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- **Time-consuming - prospective**
 - ▣ Require long periods of follow-up since disease may occur a long time after exposure
 - ▣ Many exposures investigated are long-term in nature and accurate information about them requires data collection over long periods
- **Expensive - prospective**
 - ▣ Often involve following large number of individuals for many years
- **Incompleteness of records – retrospective**
 - ▣ Require the availability of adequate records
- **Loss-to-follow-up – bias**
 - ▣ Validity of results - seriously affected by losses to follow up
- **Inefficient for rare disease**

INTERVENTION STUDIES

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- Investigator determines who is exposed, ideally using random methods
- Investigator allocates the exposure and follows for an outcome
- Intervention trials could be done for various purposes:
 - ▣ **Proof of concept trial:** designed solely to produce knowledge about cause and effect, does not test the efficacy of the intervention in actual practice.
 - ▣ **Prevention trial:** interventions are to prevent disease and study participants are persons without disease.
 - ▣ **Clinical trial:** interventions are treatment based on drugs and study participants are persons with disease.

Key Features of Experimental Design

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- **Investigator manipulates the condition under study**
- **Always prospective**

□ **Based on population**

- **Clinical trial** - usually performed in clinical setting and the subjects are patients.
- **Field trial** - used in testing medicine for preventive purpose and the subjects are healthy people. E.g. vaccine trial
- **Community trial**- unit of the study is group of people/community. E.g. fluoridation of water to prevent dental caries.

□ **Based on design**

- **Uncontrolled trial** - no control group. Control will be past experience (history).
- **Non-randomized controlled**- there is control group but allocation into either group is not randomized.
- **Randomized controlled** - there is control group and allocation into either group is randomized.

- Based on objective
 - **Phase I – clinical pharmacologic study**
 - Trial on small subjects to test a new drug with small dosage to determine the toxic effect, safe dose range.
 - Scale: **20-80** healthy individuals
 - **Phase II – efficacy studies**
 - Trial on small group to determine the therapeutic effect
 - Scale: **100-200** patients
 - **Phase III - study on large population to test effectiveness**
 - **Randomized Controlled Trial (RCT)**
 - Often it is multi-centered licensing
 - **Phase IV**
 - Post marketing surveillance
 - Long term prospective assessment of effects & side-effects

INTERVENTION STUDIES...

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- Although experimental studies come in many types, principles are the same and clinical trials dominate the field
- A clinical trial is a prospective study evaluating the effect and value of intervention(s) in human beings under pre-specified conditions.
 - ▣ It is the most definitive tool for evaluation of the applicability of clinical research
- It represents a key research activity with the potential to improve the quality of health care and control costs through careful comparison of alternative treatments.

RANDOMIZED CONTROLLED TRIAL

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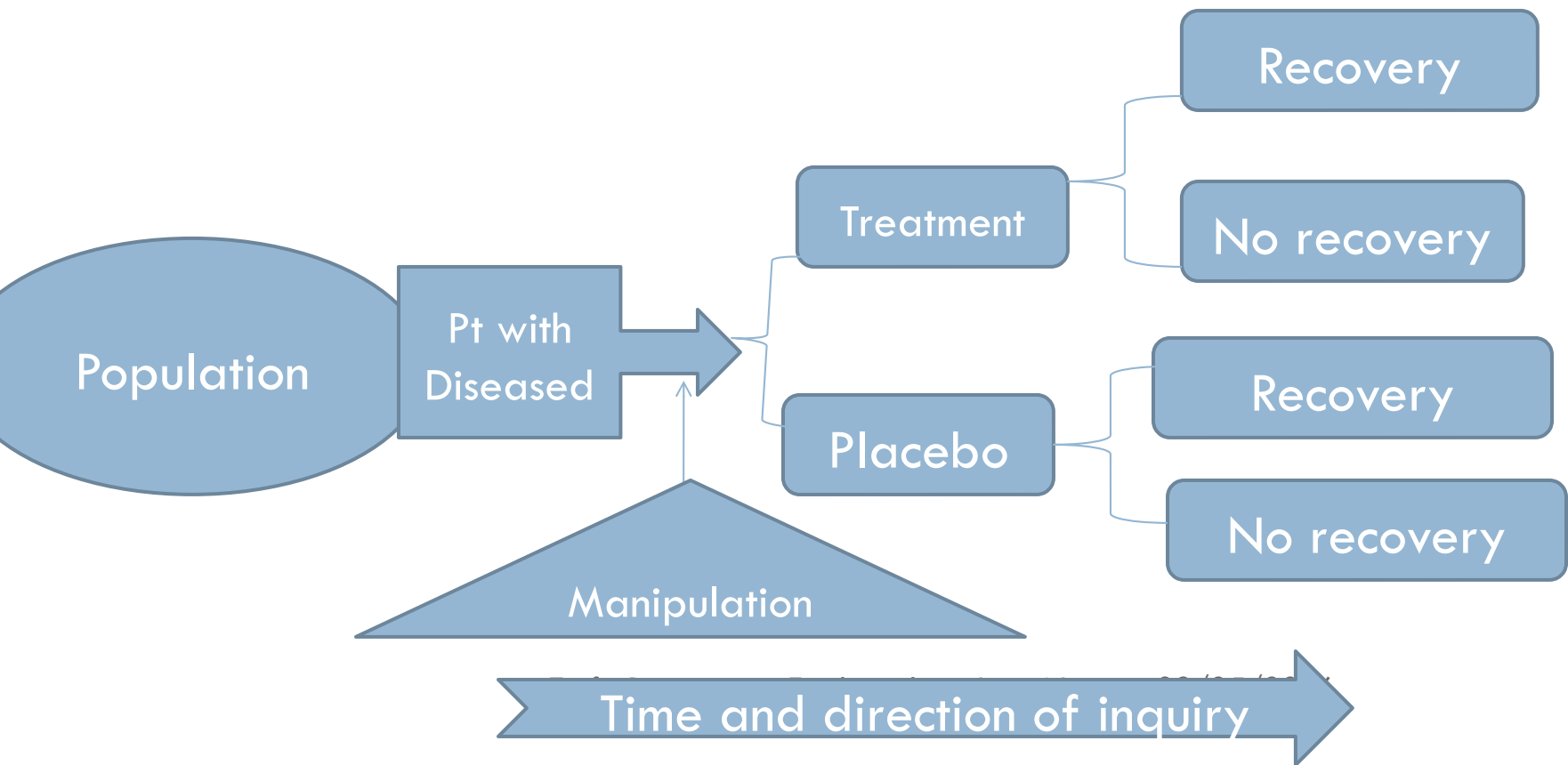
- An investigation involving **intentional** change in some aspect of the status of the subjects, e.g., introduction of a preventive/therapeutic regimen,
 - ▣ It is a design that could bring **high quality of a research** that resembles a controlled experiment **“gold standard”**
- It is similar to cohort study design that individuals are enrolled on the bases of their exposure.
- ***What is the difference b/n cohort and experimental studies?***
 - ▣ Investigator assigns subjects
 - To exposure & non exposure
 - Makes follow up to measure for the occurrence of a disease

- **Based on population it can be classified as**
 - ▣ **THERAPEUTIC** (*secondary prevention*)
 - Interventions made among patients
 - Usually made in health institutions =(clinical trial)
 - ▣ **PREVENTIVE** (*primary prevention*)
 - Interventions performed among healthy people.
 - If individuals are considered as study subjects
 - *field trial*
 - If the community are considered as study subjects
 - community trial

INTERVENTION STUDIES...

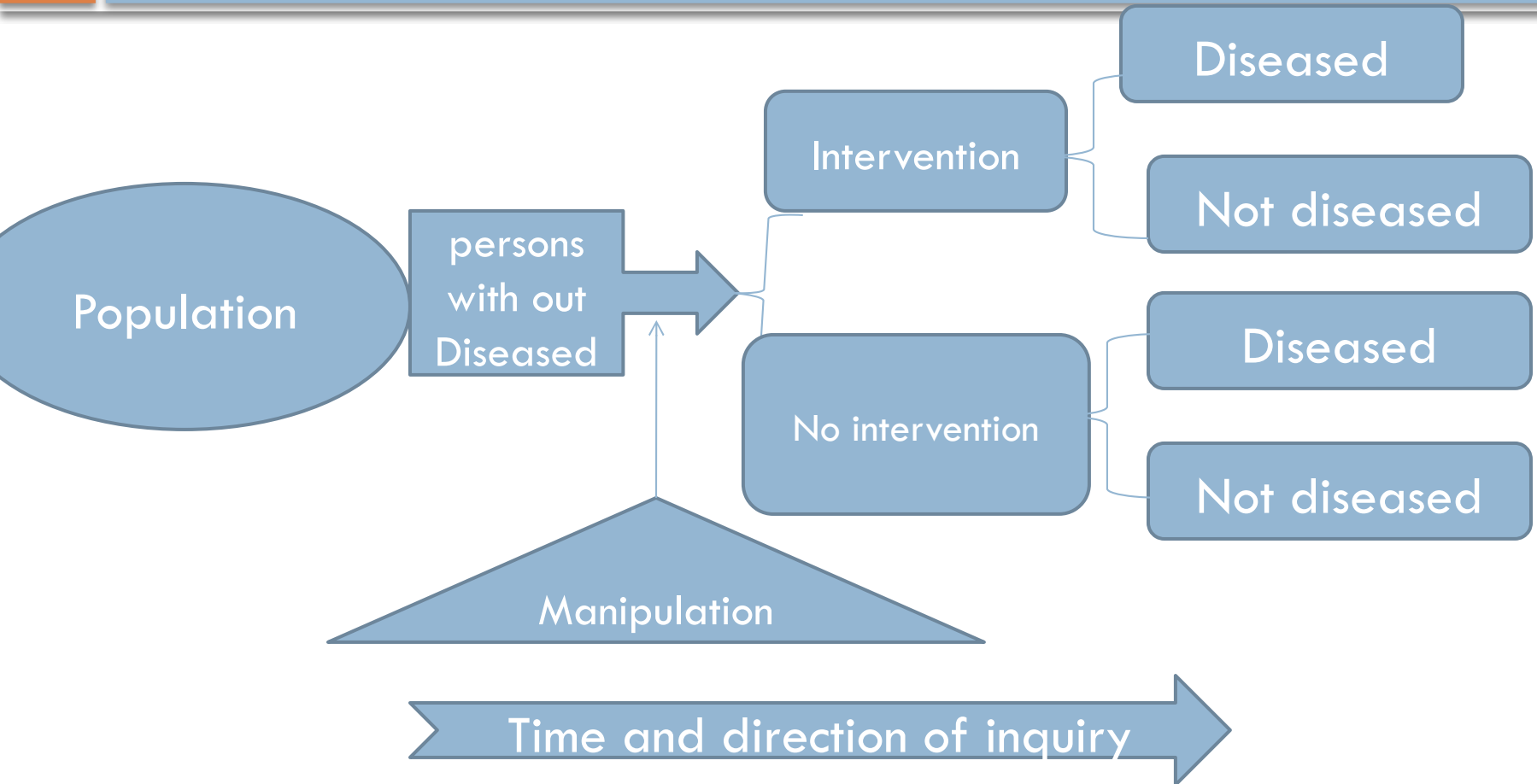
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1. **CLINICAL TRIAL:** (therapeutic trial) A research activity that involves the administration of a test regimen to humans to evaluate its efficacy & safety.



2. FIELD TRIAL

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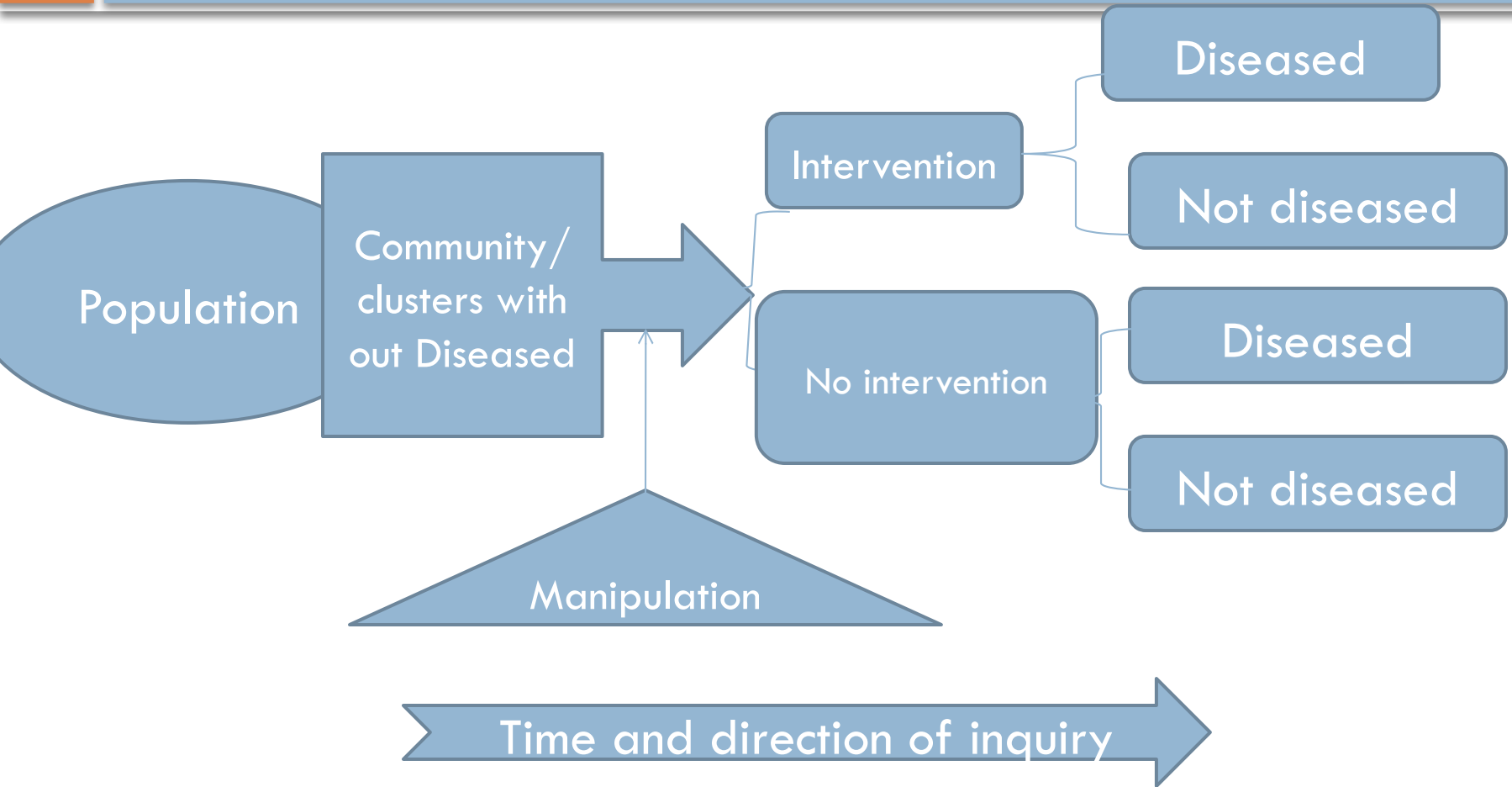
3. COMMUNITY TRIAL

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- Experiment in which the unit of allocation to receive a preventive or therapeutic regimen is **an entire community or political subdivision**.
- Examples include the trials of fluoridation of drinking water and of heart disease prevention in Ethiopia and Eritrea

COMMUNITY TRIAL

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Selection of Participants

Source population

threats to external validity



Study population



Randomize

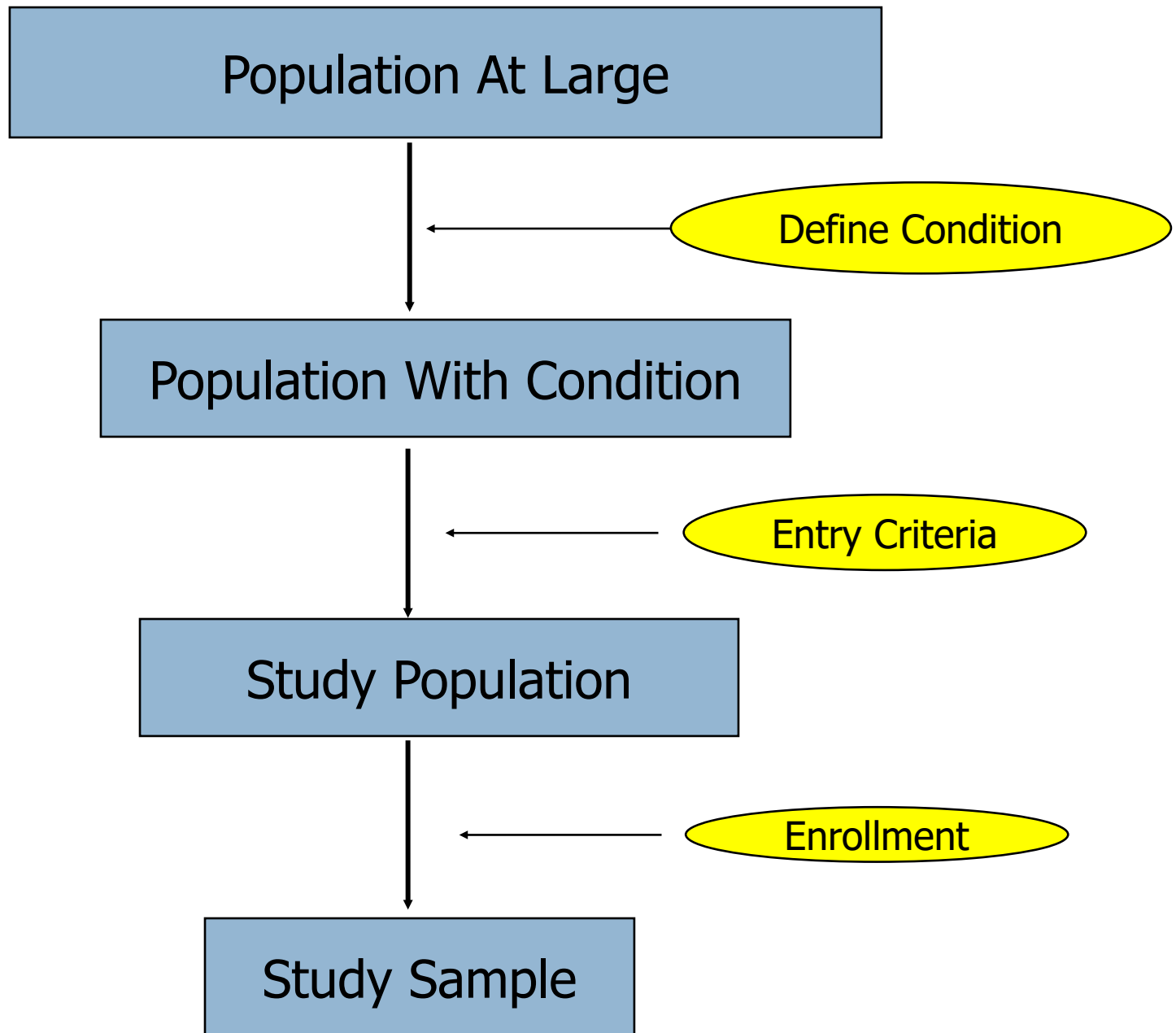


Treatment

No treatment



threats to
internal validity



Sample size

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- ❑ Calculating sample sizes for trials with dichotomous outcomes (eg, sick vs well) requires four components:
 - type I error (α),
 - power,
 - event rate in the treatment group (p_1),
 - RR or event rate in the control group (p_2), $RR = P_1 / p_2$

Panel 2: The simplest, approximate sample size formula for binary outcomes, assuming $\alpha=0.05$, power=0.90, and equal sample sizes in the two groups

n =the sample size in each of the groups

p_1 =event rate in the treatment group (not in formula but implied when R and p_2 are estimated)

p_2 =event rate in the control group

R =risk ratio (p_1/p_2)

$$n = \frac{10.51[(R+1)-p_2(R^2+1)]}{p_2(1-R)^2}$$

- **The quality of "gold standard" in intervention studies can be achieved through**
 - ▣ **Randomization**
 - ▣ **Use of placebo**
 - ▣ **Double Blinding**

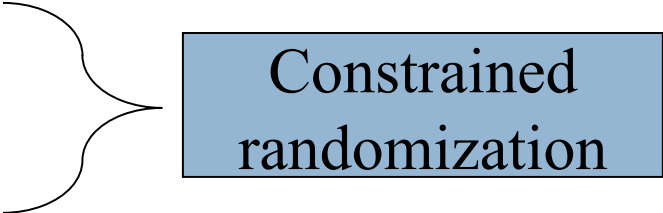
Randomization

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Methods of randomization

□ Fixed allocation/Randomization

- simple Randomization
- Stratified Randomization



Constrained
randomization

Advantages of Randomization

- Removes the potential of bias in the allocation of participants.
- Prevents confounding
- produce comparison groups

Applying Intervention

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- Intervention strategy compared with
 - ▣ **Placebo:** Need to control for therapeutic aspects of prescribing a medication or procedure not directly due to the medication or procedure itself
 - ▣ **Standard treatment**
- May have more than one comparison group
 - ▣ Advantages, disadvantages of “placebo”?
 - Ethical issues
 - ▣ Advantages, disadvantages of “standard treatment” control?
 - Interpretation of findings?

Blinding

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- **Blinding:** Means that the subject, investigator or both do not know to what group the subject is assigned to.
 - ▣ Controls for
 - Information bias (e.g., observer bias)
 - Reduce loss to follow up (reduce selection bias)
- Blinding is when the observers and/or subjects are kept ignorant as to the group to which the subjects are assigned
 - ▣ Subjects who can be masked/blinded
 - Study participants , Caregivers/treaters , Data collectors/assessors of outcome , Data analysts
 - Investigators

Single blind

The patients do not know which treatment they receive

Double blind

The patient and the observer or the physician do not know

Triple blind

The patient, the observer and the analyst do not know

- The basic analyses are:
 - ▣ comparison of the two groups with baseline characteristic to ensure similarity.
 - ▣ Calculation and comparison of rates of the incidence of the outcome for exposed and non-exposed.
 - *Relative risk of disease*
 - *Efficacy of a drug or vaccine*
- **Results of Randomized Trials**
 - ▣ Relative risk $RR = \text{incidence in exposed} / \text{incidence in non-exposed}$
 - ▣ Efficacy of treatment or vaccine

Experimental.....

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□ **ADVANTAGES**

- Randomization minimize bias & confounding
- Blinding minimizes observer bias
- Placebo could blind patients
 - Ability to demonstrate causal associations when appropriately designed
 - ❖ Use of controlled conditions
 - ❖ Powerful techniques – randomization & blinding
 - Increase subject comparability
 - Decrease potential bias and confounding
 - Investigators control level of exposure
 - ❖ Permits them to establish precise doses or procedures most appropriate to experiment

Disadvantage/weakness

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- Generalizability limited by selection of participants
- Rare/late Side effects of intervention may not be recognize
 - Limited applicability
 - Ethical concerns limit use
 - This is why RCTs focus on interventions that have the potential to reduce rather than increase the risks of morbidity or mortality
 - Bias that can result from differential rates of compliance, withdrawal, or losses to follow-up between experimental and control groups
 - Can alter the magnitude or direction of the relationship between the intervention and study outcome

Disadvantage/weakness

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- ❑ Long-term commitment
- ❑ Considerable expenses necessary to implement and complete a trial
- ❑ Some address narrowly focused questions in artificial settings
- ☞ Applicability to real-world situations may not always be clear
- ❑ Reliance on volunteers and use of strict exclusion criteria
- ☞ Make it difficult to recruit sufficient subjects for a study
- ☞ Can limit external validity of findings
- commercial, political and/or academic pressure

Exercise

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1. Imagine you are involved in two experiments

1) COVID_19 vaccine trial

2) Developing new drug for malaria

- ☐ How will you conduct these two experiments?
- ☐ What are the challenges?

Exercise...

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2. Group 1:

- Design an intervention trial to evaluate the effectiveness of Early Targeted Postnatal Follow-up on neonatal mortality and relevant morbidity.

3. Group 2:

- a. Design a multi-hospital intervention trial to evaluate the efficacy of adjunct treatment with zinc in young infants with serious bacterial infections.
- b. Design a hospital-randomized intervention trial to evaluate the effectiveness of adjunct treatment with zinc in young infants with serious bacterial infections.

4. Group 3:

- Design an intervention trial to evaluate the efficacy of a pneumococcal conjugate vaccine given together with EPI vaccines.

5. Group 4:

- Design an intervention trial to compare the efficacy of skin incision with scalpel or cutting diathermy on wound complication rate, blood loss, incision times and pain scores in women with cesarean sections.

Conclusion

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The randomized clinical/field trial is considered the “gold standard” for evaluating the efficacy of therapeutic, preventive, and other measures in both clinical medicine and public health

Conclusion...

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Most appropriate study design(s)	
Objective	Study Design
Aetiology or risk factors of disease	Cohort, Case-Control
Extent of disease in population	Survey
Natural history and prognosis of disease	Cohort
Evaluate preventive and therapeutic measures	RCT
Develop public health policy	All

6. Measures of association

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Content

- Strength measure
 - Odds ratio (OR)
 - Relative risk (RR)
- Impact measure
 - Attributable risk
 - Attributable risk percent
 - Population Attributable risk
 - Population Attributable risk percent

Brain Storm

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Comparison of populations:

	Lung cancer	No lung cancer	Total
Smokers	100	900	1000
Non-smokers	20	980	1000
total	120	1880	2000

How strong is the association?

Introduction

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- Association is a Statistical relationship between two or more variables
- The strength of association is assessed by Calculating RR,OR or other measures of association.
- Involve comparison of incidence in two or more groups, e.g., exposed and unexposed, to some hazard
- Target group can be either the exposed group or the whole population
- Reference group is the unexposed
- The two incidences can be divided or subtracted

Why do we need them?

- To move from descriptive to analytic epidemiology
- Comparisons within and between populations
- Risk factor identification

Measures of strength of association

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- ❑ Summarize the frequency measures of two variables into a single summary parameter that estimates the degree of association.
- ❑ Calculations used to measure disease frequency relative to other factors
- ❑ Quantifies or expresses the strength of the relationship between an "exposure" and "outcome" of interest
- ❖ Requires comparing two groups:
 - ❑ Exposed Vs Unexposed
 - ❑ With outcome Vs Without Outcome

Exposures and Outcomes

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Exposures

- Exposure in usual sense
e.g., ingestion of contaminated food,
droplets from someone with active PTB
- Behaviours
e.g., sharing needles, drinking, alcohol etc
- Treatment
e.g., intervention - education program
- Trait
e.g., genotype

Outcomes

- Disease
e.g., malaria, TB, diabetes
- Event
e.g., injury from land mine,
car accident
- Condition
e.g., blindness
- Death
- Other

Measure of strength...

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- Variables can be related +/- or strong/weak/significant/insignificant.
- Indications of how more or less likely one is to develop disease as compared to another
- **Types of Measures**
 - *Absolute difference*
 - Risk difference
 - Exposed - unexposed
 - *Relative difference*
 - Risk ratios
 - Odds ratios
 - Exposed / unexposed

Measure of strength...

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- Measures of association between risk factors and disease are often calculated from data presented in a two by two tables:
- **Two-by-two Table Showing Association i.e. r-by-c table**
- Every epidemiologic study can be summarized in a 2-by-2 table.

Exposure	Disease	
	Ill (D+)	Well (D-)
Yes (+)	a	b
No (-)	c	d

Exposure	Disease status		Person years unit
	Yes(+)	No (_)	
Yes	A	-	PY1
No	C	-	PY2

Measure of Strength...

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Key

- A # of individuals who are exposed and have the disease
- B # exposed and do not have disease
- C # not exposed and has the disease
- D # non exposed & do not have the disease
- $A + B$ -Total number of individuals exposed
- $C + D$ -Total number none exposed
- $A + C$ -Total number with disease
- $B + D$ -Total number with out disease
- $A+B+C+D$ -Total number of participant/volunteerr/subject

A. Relative Risk/Risk/Rate Ratio

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The relative risk or risk ratio compares the risk of some health-related event (often disease or death) in two groups, typically in persons exposed to the disease to those not exposed (**used in cohort study**):

The ratio of the risk of an event among the exposed to the risk among the unexposed.

The ratio of the incidence rate in the exposed to the incidence rate in the unexposed

Exposure	Disease		Total
	Yes (+)	No (-)	
Yes (+)	A	B	A+B
No (-)	C	D	C+D
Total	A+C	B+D	A+B+C+D

RR...

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It is a direct measurement of a risk

It is usually used in **Cohort** and **Experimental** study design.

RR = incidence of a disease among exposed
incidence of a disease among non-exposed

□ In short

or $RR = I_e / I_o$

$$RR = \frac{\left(\frac{a}{a + b} \right)}{\left(\frac{c}{c + d} \right)}$$

Strength of association in RR

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- 1.0-----No association(null value)
- 1.1-1.3-----Weak
- 1.4-1.7-----Mild
- 1.8-2-----Moderate
- 3-8 -----Strong
- ❖ For inverse associations (incidence ratio is <1.0).
- ❖ For example: reciprocal of 0.5 is 2.0, which corresponds to a “moderate” association.

Examples-1

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- Among 2390 women aged 16 to 49 years who were free from bacteriuria, 482 were OC users at the initial survey in 1973, while 1908 were not. At a second survey in 1976, 27 of the OC users had developed bacteriuria, as had 77 of the non users.
- Construct a 2 by 2 table, and
- Calculate the measure of association and
- interpret the finding.

Examples-2

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- A study on postmenopausal hormone use and coronary heart disease among postmenopausal female nurses showed that after a total of 54,308.7 person-years of follow-up, 30 women who reported that they had used hormones developed CHD. For the “never users”, 60 developed CHD among 51,477.5 person-years of follow-up.
- Draw a two-by-two table and
- calculate the measure of association and

B. Odd or Cross Product Ratio

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- It is an indirect measure of a risk in a disease of rare occurrence (non-proportional ratio).
- It is usually used in a case-control and cross-sectional analytic study designs.
- Calculated as dividing odds of having the disease if exposed to odds of having the disease if non-exposed.
- $$OR = \frac{\text{Odds of exposed among cases}}{\text{Odds of exposed among controls}}$$
- $$OR = a/b/c/d = ad/bc$$
- $$ODDS\ RATIO = \frac{\text{Ratio of diseased to non-diseased in exposed}}{\text{Ratio of diseased to non-diseased in unexposed}}$$

or

$$= \frac{\text{Ratio of exposed to non-exposed in those with the disease}}{\text{Ratio of exposed to non-exposed in those without the disease}}$$

OR...

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Derivation of OR in Cohort study

- $P D+ | E+ = (\text{exposed developed the disease}) = a/(a+b)$
 - $P D- | E+ = (\text{exposed did not develop the disease}) = b/(a+b)$
 - Odds of developing disease among exposed =
 - $\frac{D+ | E+}{D- | E+} = \frac{a/(a+b)}{b/(a+b)} = \underline{a/b}$
 - $P D+ | E- = (\text{non-exposed developed the disease}) = c/(c + d)$
 - $P D- | E- = (\text{non-exposed did not develop the disease}) = d/(c + d)$
 - Odds of developing disease among non-exposed =
 - $\frac{D+ | E-}{D- | E-} = \frac{c/(c+d)}{d/(c + d)} = \underline{c/d}$
 - **Odds ratio** = $a/b : c/d = \underline{ad/bc}$
- RR can be estimated by OR if the following conditions are fulfilled:
- The controls are representative of the general population
 - The selected cases are representative of all cases
 - The disease is rare

Example-3

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- A case control study was conducted on a total 3276 pre-menopausal female nurses to determine using OC as a risk factor for MI. Of 156 women with Myocardial Infarction (MI), 23 were current OC users at the time of their hospital admission. Of the 3120 control women without MI, 304 were current OC users.
- Draw a two by two table
- Calculate the measure of association and
- Interpret it.

Absolute/impact measures

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- It's used to place the association between an exposure and an outcome into a meaningful public health context.
- Whereas a measure of association quantifies the relationship between exposure and disease, and thus begins to provide insight into causal relationships.
- Measures of public health impact reflect the burden that an exposure contributes to the frequency of disease in the population.
- **They all assume (require) a cause-effect relationship exists between the exposure and the outcome.**

Attributable/Excess Risk (AR/ER/RD)

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- Synonyms: causal rate difference, causal risk difference, Cumulative incidence difference, Incidence density difference
- The proportion of the rate (risk) of a disease or other outcome in exposed individuals that can be attributed to the exposure.
- This measure is estimated by subtracting the rate (risk) of the outcome (usually, incidence or mortality) among the unexposed from the rate (risk) among the exposed individuals.
- $AR = \text{Incidence among exposed (Ie)} - \text{Incidence among non-exposed (Io)}$
- AR quantifies the risk of disease in the exposed group that is attributable to the exposure by removing the risk of disease that occurs due to other causes.

AR/RD...

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- Attributable Risk: Measure of the potential for prevention of disease if the exposure could be eliminated (assuming a causal relationship).
- It tells us how many cases of disease in exposed people could have been prevented by eliminating the exposure
- Among the EXPOSED: How much of the disease that occurs can be attributed to a certain exposure?
- Attributable Risk: important for
 - Policy decisions
 - Funding decisions (e.g. prevention programs)

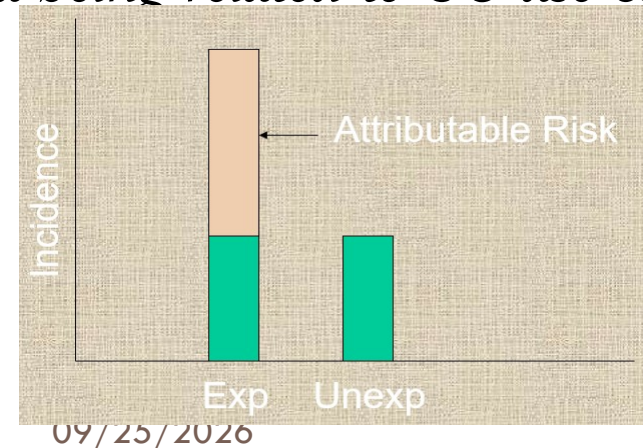
Example-4

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- From the previous example 1 Calculate the AR and interpret it.

Answers

- **Interpretation:** The excess occurrence of bacteruria among OC users attributable to their OC use is **1566** per 100,000 OC users
- In a population of 100,000 OC users, **5602** would be expected to develop bacteruria, 1566 of those who developed bacteruria being related to OC use & the remainder, 4036, to other factors



Attributable Risk Percent (ARP)

245

- Attributable fraction expressed as a % of the total rate or risk rather than as a proportion.
- Estimates the proportion of the disease among the exposed that is attributable to the exposure, or
- The proportion of the disease in the exposed group that could be prevented by eliminating the exposure

$$AR \% = \frac{(I_e - I_o)}{I_e} \times 100$$

Refer to example 1 and calculate AR%

$$AR \% = \frac{1566/105}{27/482} \times 100 = 27.96 \%$$

- **Interpretation:** If OC use causes bacteruria, about 28 % of bacteruria among women who use OC can be attributed to their OC use and can be eliminated if they did not use oral contraceptives

AR%...

246

- Estimates the proportion of the disease among the exposed that is attributable to the exposure, or
- proportion of the disease in the exposed group that could be prevented by eliminating the exposure

$$AR \% = \frac{(I_e - I_o)}{I_e} \times 100$$

- PAR shows the effect of eliminating the exposure on the population as a whole,
- PAR takes into account not only the actual incidence rate of the outcome but also the prevalence rate of the exposure
- $PAR = AR \times \text{prevalence rate of the exposure}$
- PAR shows the effect of eliminating the exposure on the population as a whole,
- PAR takes into account not only the actual incidence rate of the outcome but also the prevalence rate of the exposure
- $PAR = AR \times \text{prevalence rate of the exposure}$
- **Example**
 - **Cigarette smoking and death from lung cancer.**
 - $AR = 89 \text{ per } 100,000 \text{ per year}$
 - $\text{Prevalence rate of cigarette smoking} = 20 \%$
 - **Calculate PAR.**
 - $PAR = 89 \text{ per } 100,000 \text{ per year} \times 20 \%$
 $= 17.8 \text{ per } 100,000 \text{ per year}$

PAR...Interpretation

248

- In a population of 100,000 smokers, 89 deaths from lung cancer per year could have been avoided by preventing them from smoking (this refers to AR).
- In a general population of 100,000 with a prevalence rate of cigarette smoking of 20 %, about 18 deaths from lung cancer per year would be prevented by eliminating cigarette smoking (this refers to PAR).
- Both AR and PAR are used to estimate the effect on disease incidence of eliminating a given risk factor,
- AR estimates reduction in disease incidence only in those exposed,
- PAR estimates reduction in disease incidence in the population as a whole.
- **PAR = Incidence rate in total population minus
incidence rate in non-exposed population**

Example-4

249

- **Cigarette smoking and death from lung cancer for x population is.**
 - ▣ AR = 89 per 100,000 per year
 - ▣ Prevalence rate of cigarette smoking = 20 %
 - ▣ Calculate PAR.
- **PAR = 89 per 100,000 per year X 20 %**
= 17.8 per 100,000 per year

Interpretation

250

- In a population of 100,000 smokers, 89 deaths from lung cancer per year could have been avoided by preventing them from smoking (this refers to AR)
- In a general population of 100,000 with a prevalence rate of cigarette smoking of 20%, about 18 deaths from lung cancer per year would be prevented by eliminating cigarette smoking (this refers to PAR).
- Both AR and PAR are used to estimate the effect on disease incidence of eliminating a given risk factor,
- AR estimates reduction in disease incidence only in those exposed,
- PAR estimates reduction in disease incidence in the population as a whole.
- $PAR = \text{Incidence rate in total population} - \text{incidence rate in non-exposed population}$

Population Attributable Risk Percent (PARP)

251

- PARP: Is the attributable fraction in the population expressed as a percentage.

$$\text{PAR \%} = \frac{\text{PAR}}{\text{Incidence rate in total population}} \times 100$$

Example-5: If PAR = 17.8 per 100,000 per year

Mortality rate in non-smokers = 7 per 105

Mortality rate in the total population = 24.8 per 105 per year

Calculate PAR %

$$\text{PAR \%} = \frac{17.8 \text{ per } 105 \text{ per year}}{24.8 \text{ per } 105 \text{ per year}} \times 100 = 71.8\%$$

Interpretation: 72% of deaths from lung cancer occurring in the general population could be prevented by eliminating cigarette smoking.

Possible Outcomes in Studying the Relationship

252

- No association between exposure and disease
 $AR=0, \quad RR=1$
- Positive association between exposure and disease (more exposure, more disease)
 $AR>0, \quad RR>1$
- Inverse/negative association between exposure and disease
(more exposure, less disease)
 $AR<0 \text{ (negative)}, \quad RR < 1 \text{ (fraction)}$

7. Cause-effect Relationships and Evaluation of Evidence

253



OBJECTIVES

- ▣ Describe basic concepts of causal inference
- ▣ Explain criteria for making causal inference



CONTENTS

- ▣ Causal models
- ▣ Philosophy of scientific inference
- ▣ Causal inference in epidemiology
- ▣ **Analysis of cause effect relationship**
 - Validity of studies
 - Role of chance
 - Role of bias
 - Role of confounding factors
 - Effect modifiers
 - Evaluation of overall evidence for a cause-effect relationship



*The relationship between exposure and outcome may be {association **or** causal association}*

CAUSATION

254

- Definition of cause
 - ▣ Something that is necessary for a disease to develop at the moment that it occurs
- One of the major purposes of epidemiological studies
- ***Discovering the causes of disease***
- ***Judge whether an association of an exposure and a disease is causal.***
- Concepts

Principles of causation:

- **The single germ theory:** the mere existence of microorganism in an individual is sufficient for the development of the Ds.
- **The ecological approach:** The three discrete sectors of the environment (biological, social and physical) are closely related to each other and to host factors.

Cause of...

255

- Cause of disease: is an event, condition, characteristics or a combination of these factors plays an important role in producing the disease.
- The cause of disease may be primary(necessary/etiological agent) or Risk factors -could be related to HAE!
- Etiology=Primary factor +EA

Three ecological models

256

The Epidemiologic Triangle

- Was widely used for many years
- Still referred frequently in epidemiological literatures
- Consists of three components:
 - ▣ Host, Environment and Agent
- Each component must be analyzed and understood for prediction of patterns of a Ds
- This model highlights the agent of Ds as a separate component

The Web of Causation

257

- Effects never depend on single isolated causes, but rather develop as the result of chains of causation in which each link is the result of “a complex genealogy of antecedents”. E. g.: the web of causation underlying coronary heart Ds.
- Multiple factors identified without emphasis on the agent of the Ds

The Wheel Model

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- Another approach to depicting human-environment relations
- Consists of a hub (the host or human), which has genetic make-up as its core, surrounded by the environment, schematically divided into the three sectors mentioned above - biological, social, and physical.
- The relative size of the different components of the wheel depend upon the specific Ds problem under consideration

Sufficient Cause Model

259

- ▣ Multifactorial etiology
 - Sufficient cause
 - Constellation of minimal causes that produce the disease
 - Chance exists in every sufficient cause (presented as U in Rothman's pies and is regarded as everything that we do not yet know)
 - Component cause
 - Specific component of sufficient cause
 - Necessary cause
 - Component cause which is part of every sufficient cause

CAUSATION

260

- Interaction among causes

- Interaction of component causes is based on their membership in the same sufficient causes
- Extent of interaction depends upon prevalence of other members of those causes

- Induction period

- Induction period (period from time that a component cause acts until the disease occurs) depends upon when the last sister component cause acts
- latent period is specific to disease, and may be modifiable

Philosophy of Scientific Inference

261

- How does knowledge develop?
 - ▣ Through the process scientific inference/reasoning
- Causal inference - type of scientific inference
- Scientific inference philosophies-----**Individual Reading**
 - ▣ Inductivism
 - ▣ Refutationism
 - ▣ Consensus
 - ▣ Bayesianism

Inductivism

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- Making generalizations (inductions) from observations to general law of nature (F.Bacon, 1620 and Mill, 1862)
- E.g. 1– Jenner
 - ▣ Observation - lack of smallpox among milkmaids
 - ▣ Generalization/theory - cowpox confers immunity to smallpox

Cont'd

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- Criticism (Hume)
 - Inductivism has no logical force
 - What happened in the past doesn't imply the future.
 - Coincidences may explain observed phenomenon.
 - Observers can only see events, not causal connections.
 - **Rooster** crowing does not make the **sun rise**.

Refutationism

- Scientific hypothesis can be found to be consistent with observation but can never be proven (Popper)
 - Consistency is not proof
 - Where do hypothesis come from?
 - A hypothesis is formed.
 - It has testable propositions, otherwise not a valid scientific hypothesis
 - If the hypothesis is refuted, then another hypothesis needs to be formed.

Refutationism..

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- If a valid observation is inconsistent with a hypothesis, it implies the hypothesis is false
 - ▣ E.g. boiling water at high altitude
 - ▣ The observation refutes the hypothesis
- Science advances by “conjecture and refutation”
 - ▣ Refutationism/falsification – uses deductive logic

Consensus

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- The scientific community has role in determining validity of scientific theories (Kuhn)
 - ▣ The basis for refuting observations could be invalid observations
 - ▣ Observations which falsify theories are sometimes treated as anomalies
 - in the hope that they might eventually be explained
 - E.g. – shallow-inhaling smoking and lung cancer
 - But some may eventually lead to overthrow of the theory

Bayesianism

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- Thomas Bayes (1702–1761) probability claims that science can be true, but the knowledge upon which the truth is based needs to be evaluated.
- *BP known as “prior probabilities”*: formulation of probabilities of hypotheses before the data under analysis are taken into account.
 - Inferences can never be known for certain since the observations can never be known for certain since they are subject to error.
 - Bayes Theorem – One's posterior probability is based upon one's prior probabilities.

Cont'd

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- ▣ Focus upon “degrees of belief” rather than truthfulness of hypothesis.
- Even though truth can never be truly known, Epidemiologists can still use the scientific method in practical applications.
- Theories stand on how well they predict future observations.

Bayes'...

269

- In epidemiology, it is often used to obtain the probability of disease in a group of people with some characteristic on the basis of the overall rate of that disease (the prior probability of disease) and of the likelihoods of that characteristic in healthy and diseased individuals.

$$P(D|S) = \frac{P(S|D)P(D)}{P(S|D)P(D) + P(S|\bar{D})P(\bar{D})}$$

- **Where** D = disease, S = symptom, and $\bar{D}$ = no disease. The formula emphasizes what clinical intuition often overlooks—namely, that the probability of disease given this symptom depends not only on how characteristic of the disease that symptom is but also on how frequent the disease is among the population being served.

Causal inference in Epidemiology

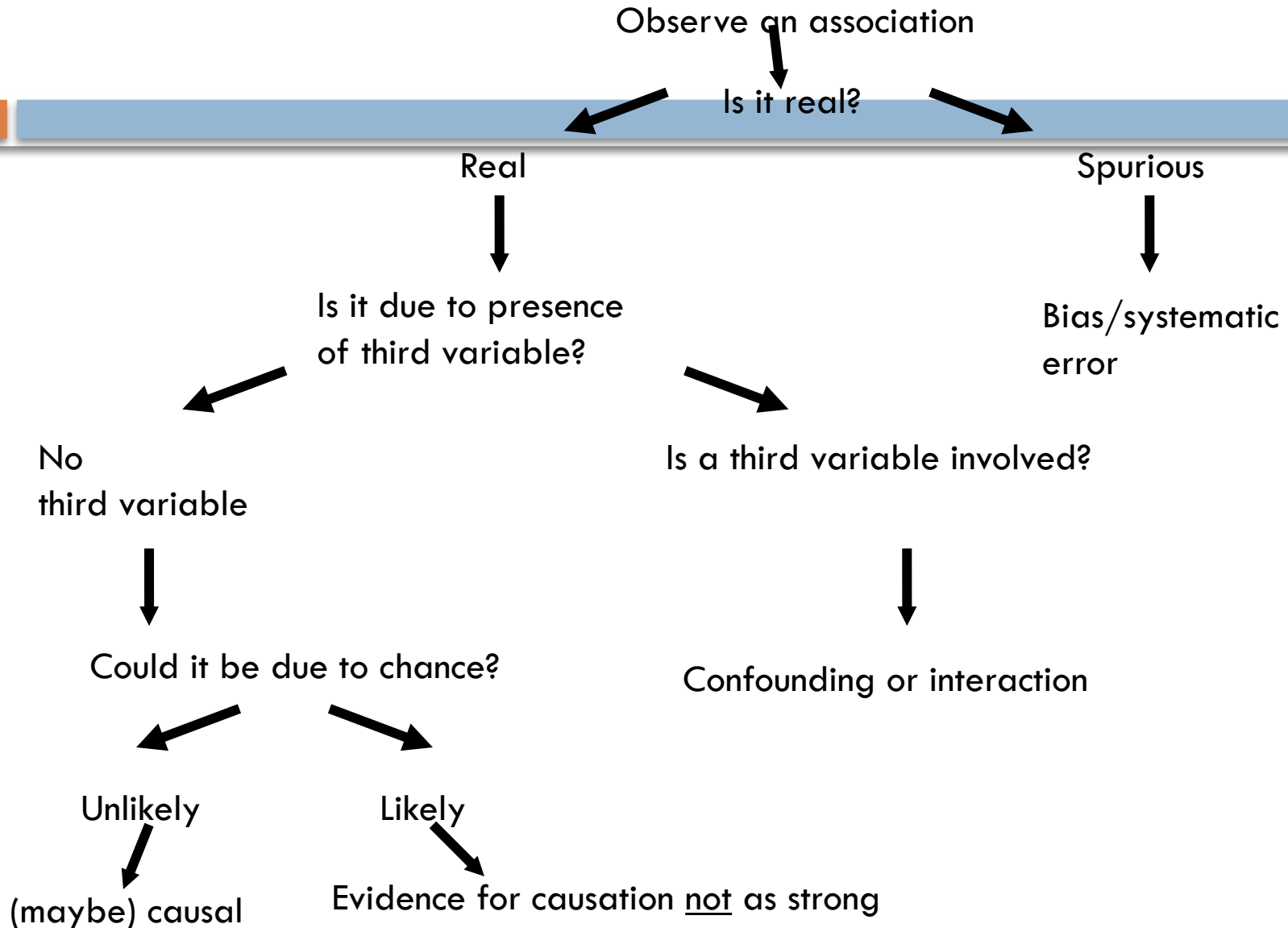
270

- Developing a hypothesis
- Testing the hypothesis
- Criteria for causation

Testing a Hypothesis

271

- An observed association could be
 - ▣ Non-causal
 - Artifactual
 - Sampling error
 - Confounding
 - Bias
 - Reverse causality-bias
 - Coincidence
 - Other causes
 - ▣ Causal



Criteria for Causation

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- Once we found that chance, bias and confounding are all determined to be unlikely, then we can conclude that a valid statistical association exists.
- We should then apply for judgment of causality
- Offered by Bradford Hill, formulated in 1965 by Austin Bradford Hill (1897-1991)
 - ▣ Guides, not **rules**
 - ▣ Strength, consistency, specificity, temporality, biologic gradient, plausibility, coherence and experimental evidence.
- **CAUSAL INFERENCE:** The thought processes and methods that assess or test whether a relation of cause to effect does or does not exist.
- **CAUSALITY:** The relating of causes to the effects they produce/ the property of being causal/the presence of cause.

1. Strength of association

274

- Refers to high magnitude of the measure of association
- Strong association more likely to be causal
 - ▣ If due to another factor, the effect of that factor should be more stronger
 - If so it would have been evident
 - ▣ Weak associations does not necessarily indicate absence of association
 - E.g. smoking and cardiovascular disease
 - ▣ Even strong associations might not be causal
 - E.g. – down syndrome and birth order confounded by maternal age

The stronger the association the more likely that it is a causal.
strong --- The more it is far from unity.

$$\begin{array}{ll} \text{---RR} \geq 2 & \text{Weak} < 2 \\ \text{---RR} < 0.5 & \text{weak} > 0.5 \end{array}$$

2. Consistency

275

- Refers to repeated observation of association in different populations under different circumstances
- Serves to rule out the role of a factor that varies across studies
- Lack of consistency does not rule out causal association
 - ▣ Unknown unusual component cause might be needed
- Special methods of combining of a number of well designed studies exist, **Meta Analysis**

3. Specificity

276

- ❑ Refers to that a cause leading to a specific effect
- ❑ Invalid criterion
 - ▣ Single events may be component causes of many effects
- ❑ Single exposure single disease
- ❑ This works more to most living organisms as causes.

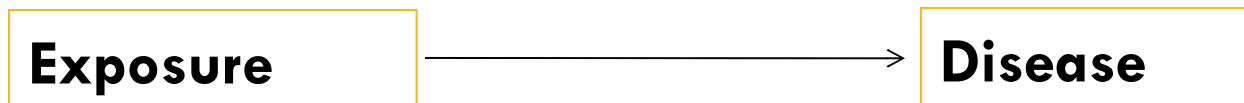
Plasmodium Sp. → Malaria

HIV → AIDS

4. Temporality

277

- Refers to the necessity that the cause precede the effect in time
- Such time sequence does not in itself establish causation
 - ▣ E.g. - night follows day but is not caused by it
- Reverse time order does not rule out causal relationship
-



5. Biologic gradient(Dose-Response)

278

- Refers to presence of dose-response relationship between hypothesized cause and effect
 - ▣ E.g. – smoking and lung ca
- However some causes have all-or-none effect
- Dose-response relationship could also be due to confounding
 - ▣ E.g. – relationship between birth order and down syndrome and maternal age

Dose-Response example...

279

Death due to ischaemic heart disease among a group of male doctors below 45 years followed in a cohort study.

Current smoking pattern (cigarettes per day)	Death rate Per 100,00 men-year	Rate ratio
Non-smokers	7	1.00
1-14	46	6.57
15-24	61	9.71
25+	104	14.86

6. Plausibility

280

- Refers to biologic plausibility of the hypothesis
- Bases on prior belief-----**believable**: believable and appearing likely to be true, usually in the absence of proof.
- Knowledge about **physiology, biology and pathology** should support the cause-effect relationship.
- **BIOLOGICAL PLAUSIBILITY**: The causal criterion or consideration that an observed, presumably causal association is plausible on the basis of existing biomedical knowledge(Pathophysiology).

7. Coherence

281

- Implies that a cause- and-effect interpretation of an association does not conflict with current knowledge on the natural history and biology of disease----**logically or aesthetically consistent:**
 - ▣ E.g. – smoking and lung ca; substances in the cigarette smoke have been shown to cause histopathological changes in respiratory epithelium of animals.
 - ▣ Biological coherence requires more than just biological plausibility or compatibility with biological knowledge derived from studies of nonhuman species or experimental systems

8. Experimental evidence

282

- Intervention to prevent disease →
 - ▣ By eliminating the harmful agent
 - ▣ Modifying host response to the exposure E.g. – through immunization
- Decrease in incidence of the disease
- Consider **reverse** and **reciprocal** causality?

Judging the evidence

283

- ❑ There are no completely reliable criteria for determining whether an association is causal or not.
- ❑ In judging the different aspects of causation,
- ❑ **The correct temporal relationship is essential,**
- ❑ Once this has been found, weight should be given to
 - ❑ Plausibility,
 - ❑ Consistency,
 - ❑ dose-response relationship and
 - ❑ Strength of the association

Purpose of Evaluation of Evidence

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To determine if what is observed is a reflection of the truth?

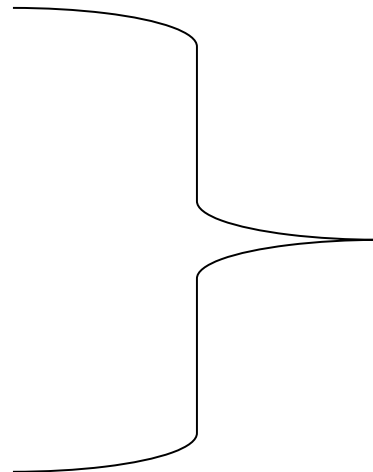
Observed:

Prevalence

Incidence

Relative Risk

Odds Ratio...



Are they true?

Are there alternative explanations?

Concepts

Concepts of Interaction/ EM

How do you assess EM and Confounding?

How can you present your finding if there is EM and Confounding?

Judging Observed Association

Could it be due to selection or measurement bias?



No

Could it be due to confounding?



No

Could it be A result of chance?



Probably NOT

Could it be Causal?



Apply the criteria and make judgment of causality

Common Problems in observed findings

- ❑ Inadequacy of the observed sample
- ❑ Inappropriate selection of study subjects
- ❑ Inappropriate/unfair data collection methods

Alternative explanations for the observed association other than cause and effect relationships

288

- A. The association may be the result of **chance**
- B. The association may be the result of **bias**
- C. Association can be the result of a **confounding effect**.
- D. An apparent cause can be an effect, rather than a cause (**reverse causation**)
- E. Cause can be both a cause effect (**reciprocal causation**)

e.g. Vitamin A deficiency can cause diarrhea or diarrhea can cause Vitamin A deficiency

Validity and Reliability

289

- ▣ Validity of Epidemiological studies
 - Validity assessment
- ▣ The role of chance
- ▣ Bias
- ▣ Confounding
- ▣ Effect modification and Mediation

Validity

290

- **Validity:** is a measure of the degree to which something truly represents **what it is being measured**/
- It's defined as the extent to which an instrument (for example, a questionnaire or a laboratory test) measures what it is intended to measure.
- Validity can be determined only if there is a reference procedure or 'gold standard': a definitive procedure to determine the characteristic being measured.
 - Ex: information on birth weight obtained from an interview can be validated against hospital records,
 - food-frequency questionnaires against food diaries and biological measurements.
- If the measures are binary – they can be measured by **Specificity, Sensitivity, Positive & negative Predictive values**

Validity...

291

Two **types of validity**

- A. Internal validity** - is the degree to which the results of the study are correct for the particular group of people studied
- b. External validity (Generalizability)** - is the extent to which the results of study apply to people not in it.
 - True value – standard or reference value or a theoretical value
 - **Accuracy** – means how close is the measurement with respect to “true” value.
 - **Precision** – is related to the ability of an instrument to measure reproducible values.
 - When talking about the quality of measurement validity types are:
 1. Construct validity
 2. Criterion-related validity
 3. Face and Content validity
 4. Predictive validity
 5. Concurrent validity
 6. Convergent validity
 7. Discriminant validity

1. Construct Validity

292

- Evidence is accumulated from a number of procedures when there is no standard method.
- It refers to the fact that the measure captures the major dimensions of the concept under study
- Eg. When we want to measure quality of a service, the instrument should contain all important dimensions of the concept like process, structure, outcome and satisfaction.

2. Face validity

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- When little or nothing is known about the variable being measured, the level of validity obtained is called face validity
- Face validity is the lowest level of validation and is used only when you are beginning to study a particular variable and have no prior research literature to refer to.
- If there is literature on the variable, either theory or research, then face validity is not sufficient.

3. Content Validity

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- Refers to the representativeness of the response categories used to represent each of the dimensions of the concept
- Content validity involves comparing the content of the measurement technique to the known literature on the topic and validating the fact that the tool does represent the literature accurately.
- Content validity is frequently estimated from the review of the literature on the topic or
- Through consultation with experts in the field who have become experts by having done research in the area.
- After you have critically reviewed the literature, you construct your questions or instruments to cover the known content represented in the literature.

4. Criterion-related validity

295

- Definition:- Validation procedure involving comparing results with the Gold Standard (you check the performance of your operationalization against some criterion.)

5. Concurrent validity

296

- This may be tested by comparing results of one measurement with those of a similar measurement administered to the same population and approximately the same time.
- If both measurements yield similar results, then concurrent validity can be established.

6. Predictive validity

297

- It can be examined by comparing the results obtained from the measurement with some actual later-occurring evidence that the measurement aims at predicting
- When there is a high degree of correspondence between the prediction and the actual event then predictive validity is established

Assessing validity of a categorical variable

298

- The validity of a categorical variable is carried out by:-
- Sensitivity = proportion of individuals classified as positives by gold standard and who are correctly identified by the study test
$$= a/(a+c)$$
- Specificity = proportion of individuals classified as negatives by gold standard and who are correctly identified by the study test
$$= d/(b+d)$$

Reliability of measurement

299

- **Reliability** is the consistency of measurements, or the degree to which an instrument measures the same way each time it is used under the same condition with the same subjects.
- A measure is considered reliable if a person's score on the same test given twice is similar.
- **Reliability** = repeatability or reproducibility,
 - ▣ Is a measure of the **consistency** of the performance of a test when used under similar circumstances.
 - To be valid, a measurement must be reliable.
 - ▣ However, reliability is not in itself sufficient for validity: in other words, a test may yield the same result consistently, but the result may not be the true (valid) one.
- Poor reliability of a measurement may be due to variation
 - ▣ When a subject is tested on different occasions (biological variation), or
 - ▣ To errors in the measurement technique (observer and instrument variation).

Reliability...

300

- There are three methods of testing or assessing the reliability of research instruments:
 1. **Test-retest reliability**:-tests for the stability of the instruments (how stable it is over time)
 2. **Inter rater reliability** (tests for equivalence) :- consistency of the results by different investigators or similar tests at the same time
 3. **Split-half reliability** (Internal consistency) :- the measurement of the concept is consistent in all parts of the test.

1. Test-retest reliability /Tests of Stability

301

- A stable research instrument is one that can be repeated on the same individual more than once and achieve the same results.
- Testing for stability, however, can be done only when you assume that the aspect being measured has remained constant.
- Repeated observations and test/retest procedures are used to test the stability of an instrument.

2. Inter rater reliability /Tests of Equivalence

302

- Tests of equivalence attempt to determine if the same results can be obtained using different observers at the same time or if similar tests given at the same time yield the same results.
- In observational methods, when the characteristic being observed is expected to change over time, a test of stability cannot be used.

3. Split-half reliability /Test of Internal Consistency

303

- ❑ Internal consistency refers to the extent to which all parts of the measurement technique are measuring the same concept
- ❑ To test the internal consistency of a questionnaire, the split-half method is often used.
- ❑ In this method, a questionnaire is divided in half by some random method, and the two halves are correlated. If they consistently measure the same concept, a high correlation will be obtained.

Cont'd...

304

- Simply test/retest is that you should get the same score on test 1 as you do on test 2 conducted at different times
- The primary difference between test/retest and internal consistency estimates of reliability is that test/retest involves two administrations of the measurement instrument, whereas the internal consistency method involves only one administration of that instrument.

Cont'd...

□ The three main components of test-retest method are as follows:

- 1) implement your measurement instrument at two separate times for each subject;
- 2) compute the correlation between the two separate measurements; and
- 3) assume there is no change in the underlying condition (or trait you are trying to measure) between test 1 and test 2.

Reliability in other context

306

- Checks of the repeatability of measurements of the main exposures and outcomes should usually be included in an epidemiological study.
 - ▣ Across time (test-retest)
 - ▣ Across items within a questionnaire (split half; alpha) or between questionnaires (parallel forms)
 - ▣ Across observers (intercorder reliability)
- These checks can take various forms
 - ▣ *Intra-observer or intra-measurement reliability*
 - having the same observer perform the same measurements on the same subjects on two or more separate occasions.
 - Ex: data from medical records may be extracted by the same abstractor on two occasions;
 - the same interviewer may re-interview subjects after a time interval;
 - duplicate biological samples may be reprocessed by the same laboratory technician

Reliability...

307

- *Intra-observer or intra-measurement reliability*
 - These separate measurements are then compared.
 - The **appropriate time interval** b/t measurements varies according to the type of outcome or exposure measurement.
 - If it is **too short**, subjects and/or observers may recall the previous result;
 - if it is **too long**, the subject's exposure or outcome status may have changed (*this is not a problem when data are extracted from medical records*).

Reliability...

308

□ *Inter-observer reliability*

- can be assessed by having the same subjects measured by two or more independent observers.
 - EX, the performance of two or more data abstractors may be compared using information extracted independently from the same medical records, or
 - the performance of two or more interviewers may be compared using independent interviews of the same subjects on two different occasions.
- Again, the **interval used between** measurements needs careful consideration

Reliability...

309

□ *Inter-observer reliability*

- ▣ The agreement b/t pairs of measurements carried out by two independent observers on the same subjects can be presented as a 2×2 table

- *observed agreement (O) or mean pair agreement index = (No. of agreements/Total no. of pairs)*

$$= (a + d) / N$$

	Observer B		
	+	-	
Observer A	+	a	a+b
	-	c	c+d
	a+c	b+d	N

Reliability...

310

- Some agreement would be expected even if both observers simply guessed the result.
- The **kappa statistic** (K) is an alternative measure that takes account of the agreement expected solely on the basis of chance.
 - To calculate the kappa statistic, the number of pairs of observations that would be expected on the basis of chance in cells **(++)** and **(--)** must first be calculated.
 - The expected value in any cell is given by:
 - $[(\text{Total of relevant row}) \times (\text{Total of relevant column})] / \text{Grand total}$

Reliability...

311

- Thus
 - for cell (+ +), the expected value will equal:
$$[(a+b) \times (a+c)]/N$$
 - for cell (– –), the expected value will equal:
$$[(c+d) \times (b+d)]/N$$
- The expected agreement on the basis of chance (E) is
 - [Expected value for cell (+ +) + Expected value for cell (– –)] / N
- The **actual agreement** beyond chance is therefore:
 - Observed agreement (O) – Expected agreement (E)

How do we assess reliability?

312

- **One way is to look simply at percent agreement.**
- **Percent agreement is the proportion of all diagnoses classified the same way by two observers**

Example

313

- Two physicians are each given a set of 100 X-rays to look at independently and asked to judge whether pneumonia is present or absent.
- When both sets of diagnoses are matched, it is found that 95% of the diagnoses are the same

		MD#1	
		Yes	No
MD#2	Yes	43	3
	No	2	52

Is percent agreement good enough?

314

- **Do these two physicians exhibit high diagnostic reliability?**
- **Can there be 95% agreement between two observers without really having good reliability?**

Use of the **kappa** statistic to assess reliability

315

- Kappa is a widely used test of inter or intra-observer agreement (or reliability) which **corrects for chance** agreement.
- Cohen's kappa coefficient is a statistic that is used to measure inter-rater reliability for qualitative items.
- It is generally thought to be a more robust measure than simple percent agreement calculation, as K takes into account the possibility of the agreement occurring by chance.

kappa

316

It varies from $+1$ to -1

- $+1$ means that the two observers are perfectly reliable. They classify everyone exactly the same way.**
- 0 means there is no relationship at all between the two observer's classifications, above the agreement that would be expected by chance.**
- -1 means the two observers classify exactly the opposite of each other. If one observer says yes, the other always says no.**

Calculating Kappa

317

1. Calculate **observed agreement** (cells in which the observers agree/total cells).
2. Calculate **expected agreement** (**chance agreement**) based on the marginal totals
 - **Chance agreement:** Defined as the probability of two raters coding the same behavior at the same time by chance

$$K = \frac{\text{Observed agreement} - \text{Expected Agreement}}{1 - \text{Expected Agreement}}$$

Expected = $\frac{\text{Row total} \times \text{Column total}}{\text{Total sample size (n)}}$, thus $E_{ij} = \frac{R_i \times C_j}{n}$

OBSERVED		MD#2		TOTAL
		YES	NO	
MD#1	YES	43	3	46
	NO	2	52	54
TOTAL		45	55	100

95% AGREEMENT

EXPECTED		MD#2		TOTAL
		YES	NO	
MD#1	YES	20.7	25.3	46
	NO	24.3	29.7	54
TOTAL		45	55	100

50.4 % AGREEMENT

Formula for calculating Kappa from expected agreement

319

Observed agreement - Expected Agreement

1 - Expected Agreement

$$\frac{95\% - 50.4\%}{1 - 50.4\%} = \frac{44.6\%}{49.6\%} = .9$$

- You can use this [Cohen's Kappa Calculator](#) to automatically calculate Cohen's Kappa for any two raters.

Guide to use of Kappas in epidemiology and medicine

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- Kappa > 0.80 is considered **Excellent**
- Kappa $0.60 - 0.80$ is considered **Good**
- Kappa $0.40 - 0.50$ is considered **Fair**
- Kappa < 0.40 is considered **Poor or**

Cohen's Kappa	Interpretation
0	No agreement
0.10 - 0.20	Slight agreement
0.21 - 0.40	Fair agreement
0.41 - 0.60	Moderate agreement
0.61 - 0.80	Substantial agreement
0.81 - 0.99	Near perfect agreement
1	Perfect agreement

Measures to improve reliability & validity

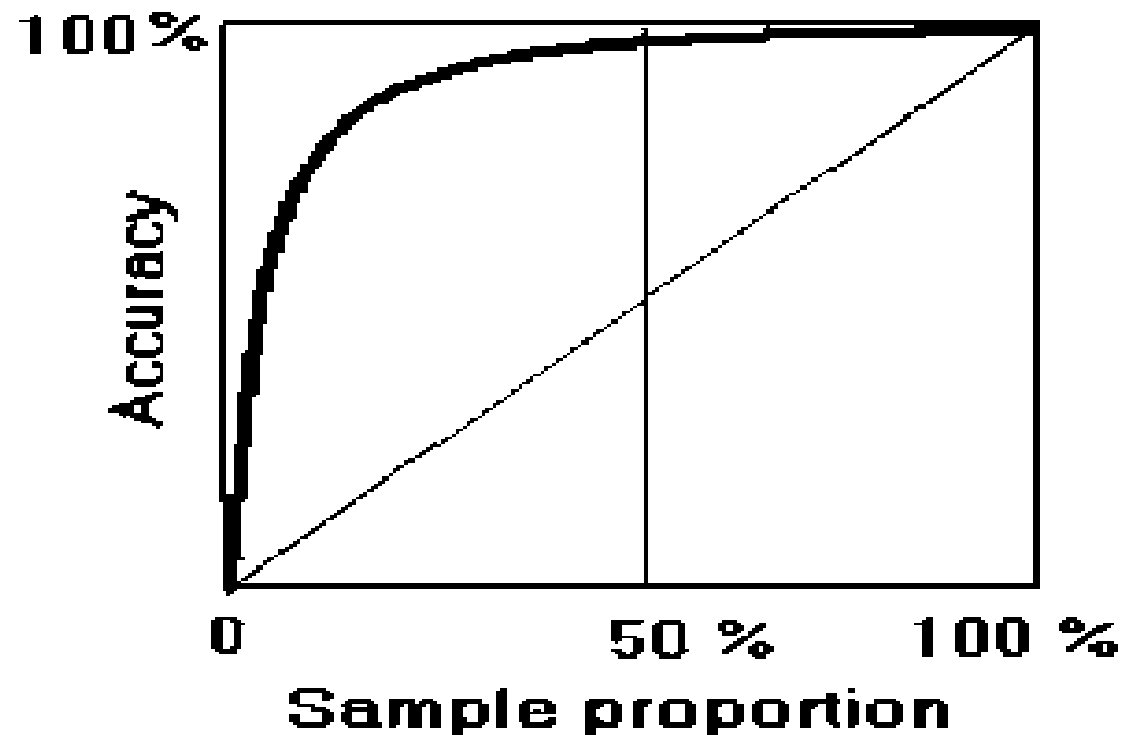
321

- One is to pilot test your instruments, getting feedback from your respondents.
- Train the interviewers thoroughly.
- All data entry for computer analysis should be "double-punched" and verified.
- Finally, use multiple measures of the same construct if the different measures don't share the same systematic errors,
- Close supervision of the data collectors

Accuracy as a function of sample size

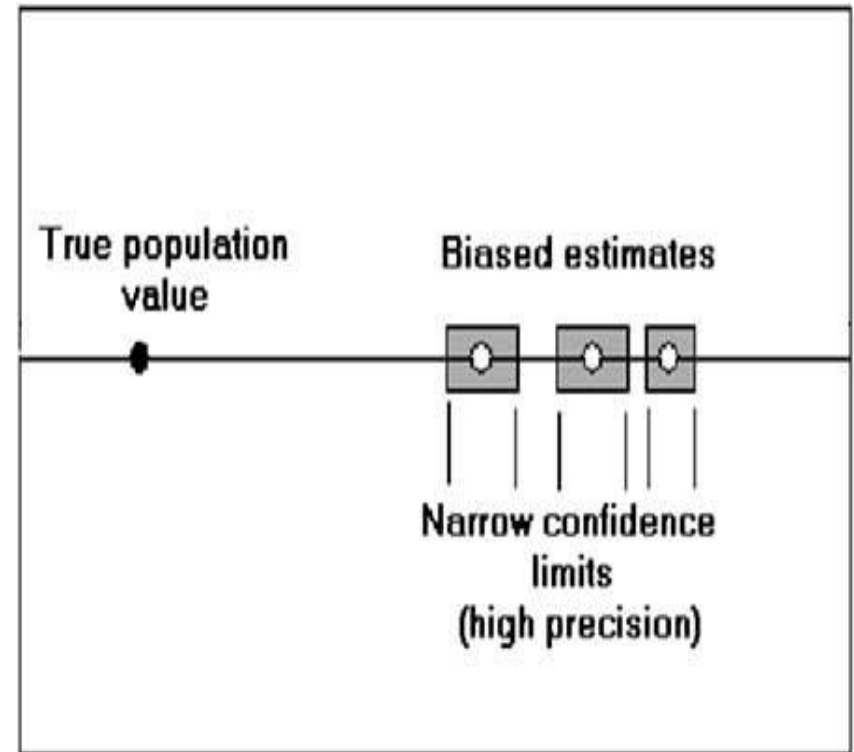
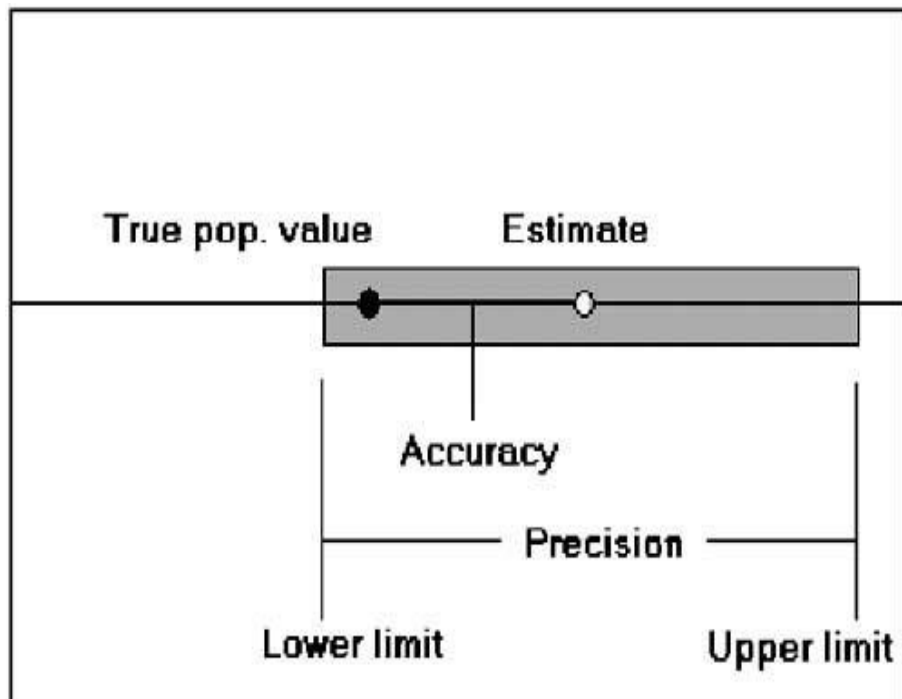
322

- The following diagram illustrates the pattern of accuracy growth when sample size increases



In example below accuracy is bad but precision is misleadingly good and this is indicated by the narrow confidence limits; this is b/c of biased estimate

323



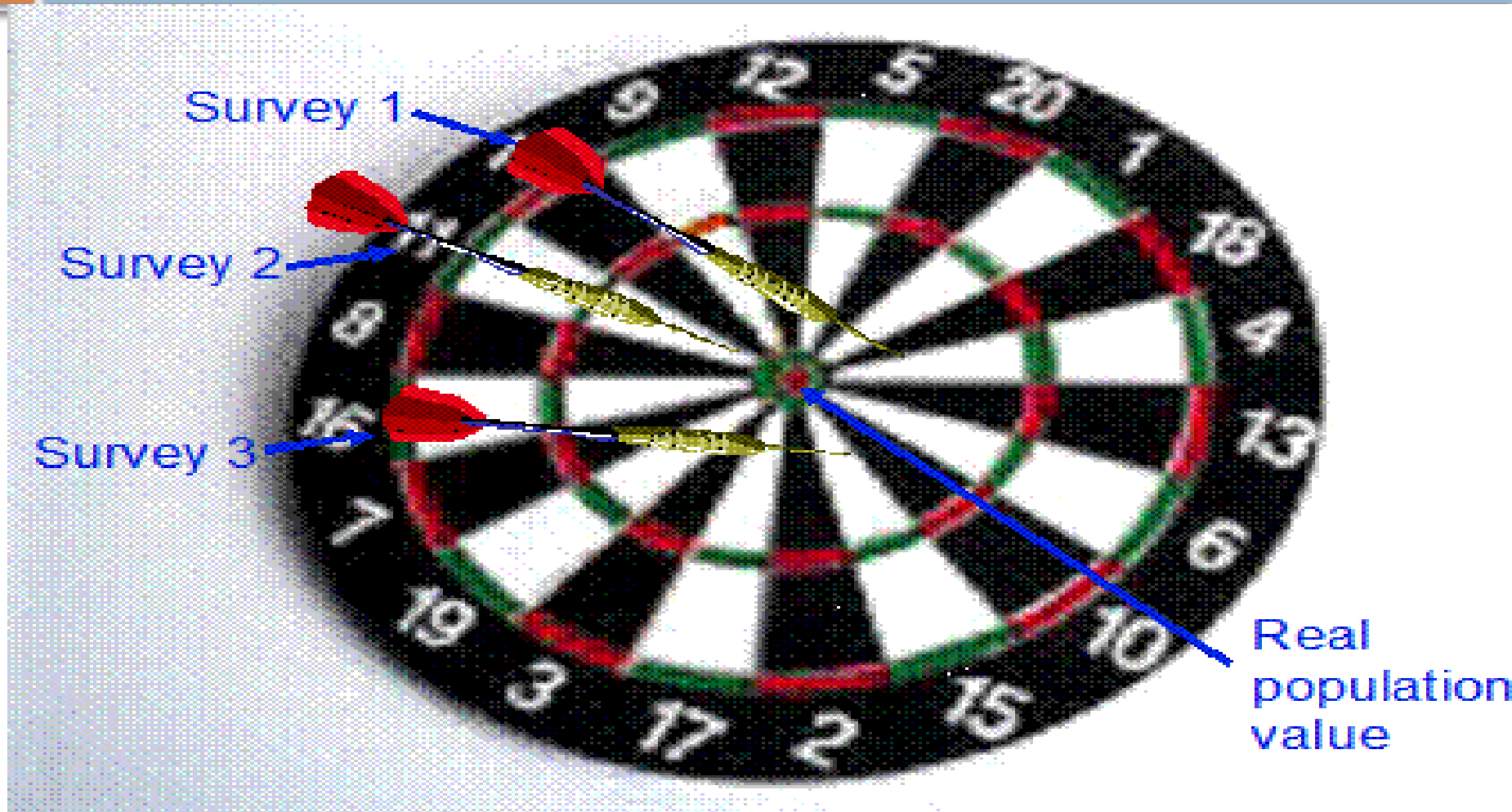
Precision

324

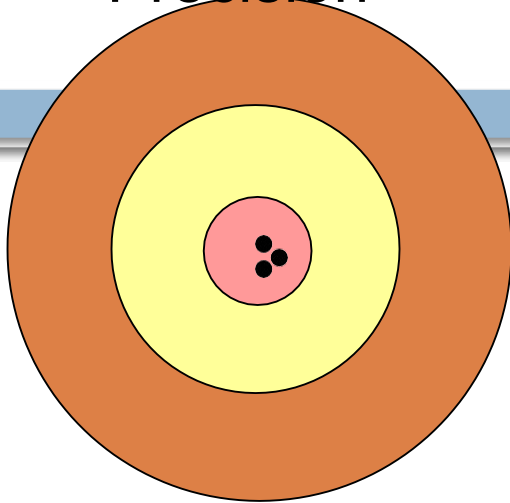
- Biased estimates are lower or higher than the true population value, generally because they are derived from samples that are not representative of the data population.
- Bias is not easily detectable or not detectable at all.
 - ▣ Consequently users may be unaware of the problem since they also do not know the true population value.
- The best approach to the reduction of bias is through the application of appropriate stratification.
 - ▣ Need to reduce the overall variability of the estimates
 - ▣ Need for representative samples

Precision and Accuracy...

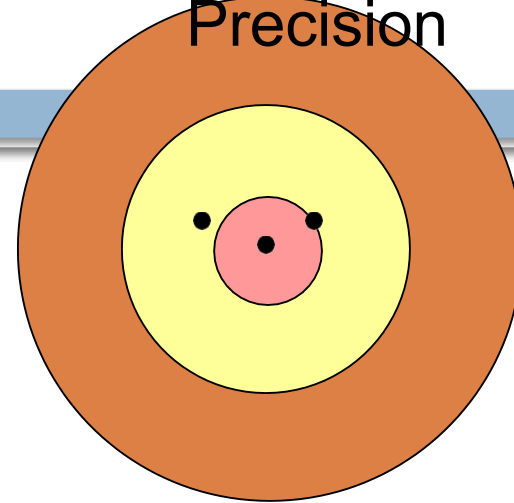
325



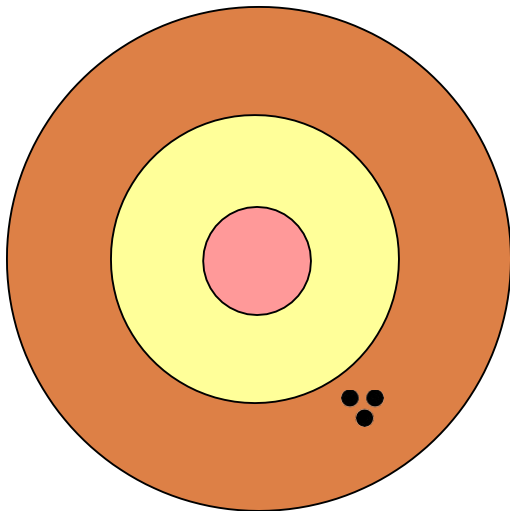
Accuracy &
Precision



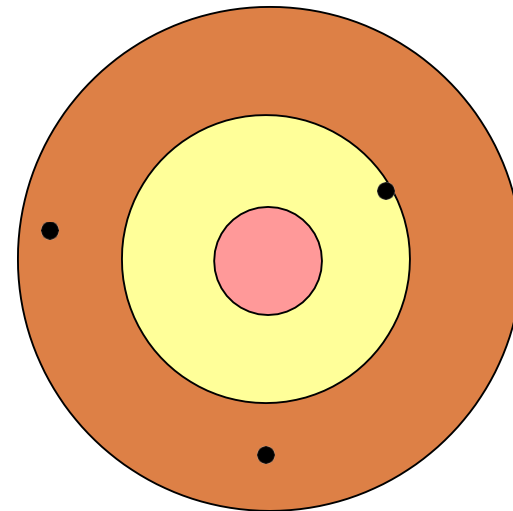
Accuracy without
Precision



Precision without Accuracy



No Precision & Accuracy



Can you hit the bull's-eye?

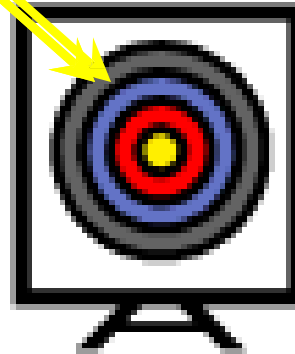
327

Three targets with three arrows each to shoot.

How do they compare?



Both
accurate
and
precise



Precise
but not
accurate



Neither
accurate
nor
precise

Can you define accuracy vs. precision?

Accuracy and Precision

328

- *Accuracy* = is the degree of closeness to the truth?
- **Precision** = is the finding due to sampling variation?
 - ✓ Precision in measurement and estimation corresponds to the reduction of random error.
 - ✓ mostly related to sampling variation/error. Solution:
 - ✓ Increase sample size
 - ✓ Improving the efficiency of

Accuracy and Precision

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- **Accuracy:** The degree to which a measurement represents the true value of something. ***Simply put: How close a measurement is to the true value***
- **Precision:** The degree of resemblance among study results, were the study to be repeated under similar circumstances.
Simply put: How close the measurements are to each other
- Lack of precision is referred to as '**random error**'.
- **Validity:** The extent to which the study measures what it is intended to measure. ***Simply put: Are the values describing what was supposed to be measured?***

Accuracy and Precision...

330

- **Internal validity:** *Are the results valid for the study subjects?*
- **External validity (Generalizability):** *Are the results valid for the population from which the sample was drawn?*
- Lack of validity is referred to as '**Bias**' or '**systematic error**'.
- **Reliability:** A measure of how dependably an observation is exactly the same when repeated. It refers to the measuring procedure rather than to the attribute being measured. ***Simply put: Will one get the same values if the measurements are repeated?***
- **Note:** Many authors use the terms in the following sense:
- **Accuracy (validity):** Validity and accuracy are used synonymously
- **Precision (reliability):** Precision and reliability are used synonymously

Precision and accuracy...

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- The usual situation is you do one survey in a population in which you do not know the true value, so you end up with a single dart (or the result from a single survey)

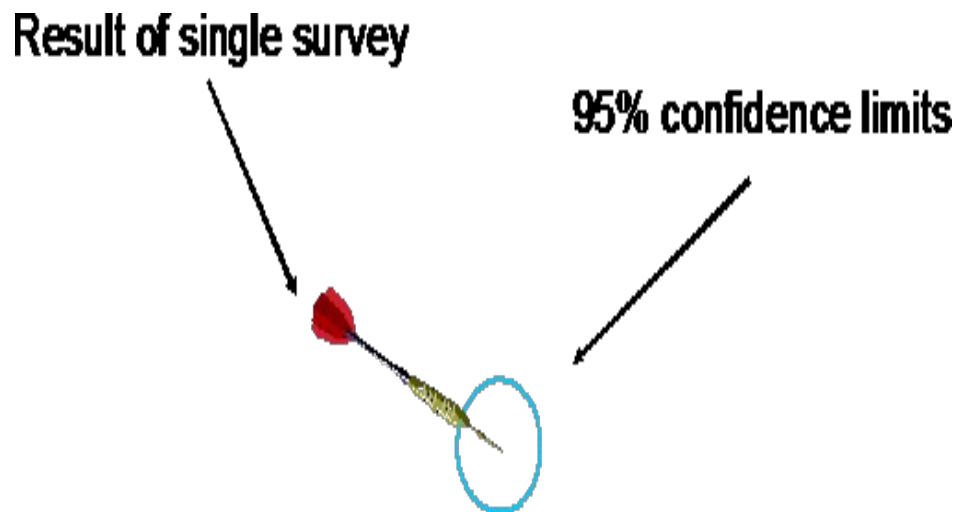
Result of single survey



- But this one dart doesn't tell us much.
- Where is the true population value? Is this a useful estimate?
- Can we make programmatic decisions based on this survey result?
- If we calculate the 95% confidence interval, we can at least get some idea of the precision of our estimate

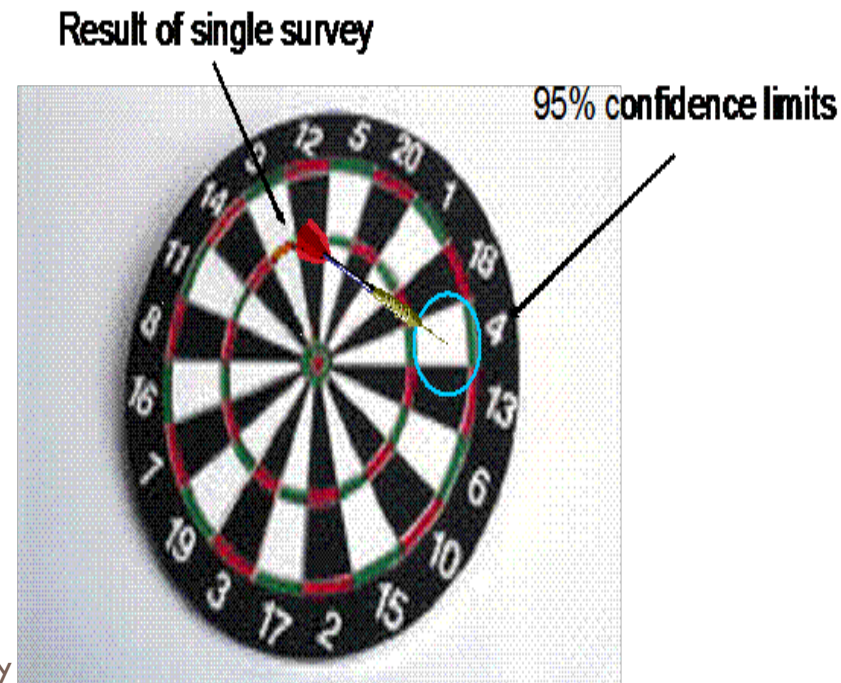
This point is very, very important.

- Just having narrow CI producing good precision **DOES NOT NECESSARILY** mean that the survey result is close to the true population value.
- If there is bias which produces inaccuracy,
- In this case, you would draw very poor conclusions if you assumed that the dart indicated the true population value.



Tesfa B.

Epidemiology



Precision and accuracy...

333

- ***Bias and sample size determines accuracy***
- How do we prevent bias?
 - ▣ Be sure measurement instruments are functioning well
 - ▣ Be sure to use correct technique in all measurements
 - ▣ Carry out the sampling correctly and randomly
- How do you do these things?
 - ▣ Seek expert advice and have others review your plans
 - ▣ Provide good, thorough training to all survey workers
 - ▣ Provide good field supervision
 - ▣ Do quality checks on the data as they come in

Precision and accuracy...

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- It is noted that:
 - ▣ Accuracy is 100% in the case of a census.
 - ▣ The pattern of accuracy growth is not linear.
 - ▣ The accuracy of a sample equal to half the data population size is not 50% but very near to 100%.
 - Good accuracy levels can be achieved at relatively small sample sizes, if the samples are representative.
- The result of this relationship is that beyond a certain sample size the gains in accuracy are negligible, while sampling costs increase significantly.

Summary

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The real difference between reliability and validity is that:-

- Reliability estimates the consistency of your measurement/ more simply the degree to which an instrument measures the same way each time it is used in under the same conditions with the same subjects.
- In short, a measurement is precise if it obtains similar results with repeated measurement.
- A measurement is accurate if it is close to the truth with repeated measurement.
- A **faulty** measurement may be expressed **precisely** but may **not** be **accurate**.

Summary...

- **Validity**, on the other hand, involves the degree to which you are measuring what you are supposed to, more simply, the accuracy of your measurement.
- **Validity** is more important than reliability because if an instrument does not accurately measure what it is supposed to, there is no reason to use it even if it measures consistently (reliably).

The role of chance

- we can draw **inferences** about the experience of an entire population based on **evaluation of only a sample**.
- One of the major problems in drawing such inference is that the play of chance may always affect the results observed simply because of **random variation** from sample to sample.
- One of the major determinants of the degree to which chance affects the findings in any particular study is **sample size**
- The smaller the sample , the more variability there will be in the estimates and the less likely the findings will reflect the experience of the total population.

Role of chance..

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- ❑ The larger the sample on which the estimate is based, the less variability and the more reliable the inference.
- ❑ It is important to quantify the degree to which chance variability may account for the results observed in any individual study.
 - ❑ This is done by performing an appropriate test of statistical significance.
 - ❑ A measure that is often reported from all tests of statistical significance is the P value.
 - ❑ $P \leq 0.05$ - statistically significant.
 - ❑ $P > 0.05$ – no statistically significant association (chance can not be excluded as a likely explanation)

Hypothesis Testing (Test of statistical Significance)

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1. Make explicit statement of hypothesis

☐ Null Hypothesis

$H_0: P_1 = P_2$ (p= proportion developed outcome of interest)

$H_0: RR = 1$

☐ Alternate Hypothesis

$H_1: P_1 \neq P_2$

$H_1: RR \neq 1$

Hypothesis Testing

Perform Test of Significance

- Use appropriate test
- **P-value ≤ 0.05 → Chance is unlikely explanation**
 - Reject the null hypothesis
 - There is Statistically significant difference
- It is always advisable to report the actual P value rather than merely that the results did or did not achieve statistical significance.
- confidence interval (CI) is far more informative measure than P value to evaluate the role of chance

Confidence Interval

341

1. Provide information that p-value gives.
 - ☐ If null value is included in a 95% CI, by definition the corresponding P-value is >0.05 .
2. Indicate the amount of variability (effect of sample size) by the width of the CI
 - ☐ This information can not be obtained from p-value.

Interpretation of CI

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Wide CI



indicate greater variability



suggest inadequacy of the sample size

Particularly important in interpreting **non-significant** results

Narrow CI → suggest that truly there is no association

Wide CI → suggest inadequacy of sample size to have adequate statistical power

The role of bias

343

- **Bias** - is any systematic error in the design, conduct, or analysis of a study that results in a distorted estimate measurement
- The key word in the understanding of the concept of bias is “**different**”.

Two main types of bias:

A. Selection bias

B. Information (Observation) bias

Selection Bias (examples)

344

- ☐ **Invitational (who gets invited into the study?)**
- ☐ **Acceptance (who accepted the invitation?)**
 - ☐ Loss to follow-up
 - ☐ Volunteer/Compliance bias
 - ☐ Non-response bias

☐ **Selection bias** is a particular problem in case control and retrospective cohort studies

1. Diagnostic bias

- Diagnostic bias occurs when a disease is more likely to be diagnosed in some one with **exposure to a suspected risk factor**.
 - Happen when knowledge of an exposure is used as a diagnostic criterion
 - For example women who take oral contraceptives (OCs) may be screened more often for breast cancer than women who do not take OCs because of the suspected link between OCs and breast cancer.
 - This would result in breast cancer being diagnosed more readily in those who are exposed to Ocs.

2. Self selection/ Volunteer bias/ Compliance bias

- People who accept to participate in a study, or people who refuse to participate are often quite different from the general population.

3. Non-response bias

- ▣ It reduces the effective sample size, resulting in loss of precision of the survey estimates.

4. Loss to follow up

- ▣ It is major source of bias in cohort & intervention studies
- ▣ it is also a problem in intervention studies before outcome of interest occurs

5. Healthy worker bias

- ▣ Bias in occupational health studies which tend to underestimate the risk associated with an occupation due to the fact that employed people tend to be healthier than the general population

Ways of minimizing selection bias

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- Population-based studies are preferable
- Avoid the inclusions as study subjects of people who have volunteered on their own
- In case-control , select several different control groups.
- In hospital-based case control study, controls are usually selected among patients with disease other than the disease studied.
- keep losses to follow-up to an absolute minimum.
 - For those who are lost, an assessment of as much outcome data as possible.

Ways of minimizing cont...

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- *Loss to follow up*
 - ▣ Calculate estimates of the exposure-disease association assuming the most extreme situations.
 - ▣ Assumptions:
 - all those who were lost to follow up developed the outcome of interest
 - none developed the outcome of interest.
- The results of these calculations provide a range within which the true association will lie.

B. Information bias / Observation bias

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- ❑ Refers to bias which arises during the data collection process
- ❑ It occurs b/c of mistakes in categorizing study subjects with respect to their exposure or disease status

1. Investigator bias/ Interviewer bias/ Observer bias

- ❑ Occurs when investigators collect information differently in different comparison groups

2. Response bias/ Recall bias

- ❑ Occurs as a result of difficulty to recall prior exposures

Information bias cont...

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3. Social desirability bias

- ▣ Occurs because subjects are systematically more likely to provide a socially acceptable response.

4. Placebo effect

- ▣ In experimental studies which are not placebo controlled, observed changes may be ascribed to the positive effect of the subjects belief that the intervention will be beneficial.

5. Hawthorn effect

- ▣ Refers to the change in the dependent variable which may be due to the process of measurement or observation itself

1. **Blinding:** Three types of blinding

- ☐ Single blind
- ☐ Double blind
- ☐ Triple blind

Ways of minimizing

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2. Use of Placebo

- ❑ Placebos are **inert treatments intended** to have no effect other than the psychological benefit of offering treatment.
 - ❑ Can only be used if there is no accepted treatment for the condition under study.
- ❑ Use of placebo minimizes the bias in the ascertainment of both subjective disease outcomes and side effects.
 - ❑ It facilitates that both groups in the study gain equal attention.
- ❑ **Placebo effect:** tendency for individuals to report favourable response to any therapy regardless of the physiologic efficacy of what they received.

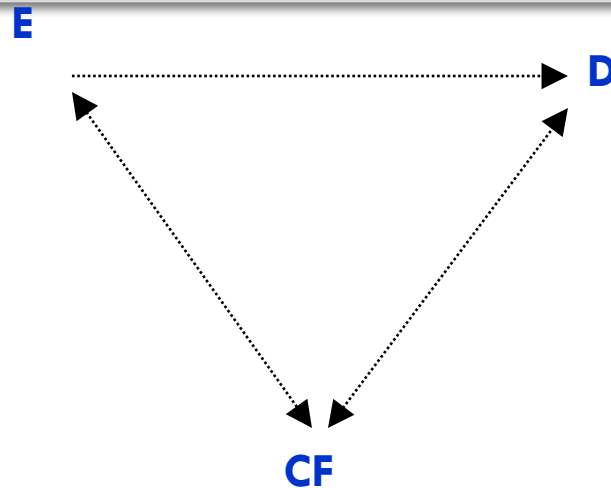
2. **Same standard procedures, instruments, questionnaires, interviewing techniques** should be used for data collection in both comparison groups
3. Classification of study subjects according to their outcome & exposure status should be based on the **most objective & accurate methods available**
4. When exposure status is determined by interview, it should be assessed in several different ways for both groups

- **Confounding** is distortion of the estimated effect of an exposure on an outcome, caused by the presence of an extraneous factor associated both with the exposure and the outcome,
- **Confounding variable, confounder** is a variable that can cause or prevent the outcome of interest, is not an intermediate variable, and is associated with the factor under investigation.
- Confounding may be considered a confusion of effects

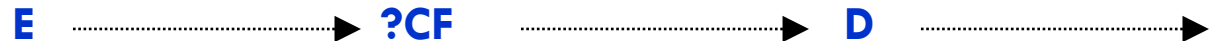
CONFOUNDING

355

Confounding IS
present



Confounding NOT
present



The role of confounding cont...

356

Example of confounding effect.

- An observed association between consumption of **coffee** and increased risk of **MI** could be due, at least in part, to the effect of **cigarette smoking**, since coffee drinking is associated with smoking and, independent of coffee consumption, smoking is a risk factor for MI.

Effect of Confounding

357

1. Totally or partially accounts for the apparent effect
 2. Mask an underlying true association
 3. Reverse the actual direction of the association
- Confounding must be dealt as quantitative problem; the amount of confounding rather than mere presence or absence that is important to evaluate.

Control for Confounding Variables

358

☐ In the design:

- ☐ Randomization
- ☐ Restriction
- ☐ Matching (at design and analysis stage)

☐ During analysis:

- ☐ Standardization
- ☐ Stratification
- ☐ Multivariate analysis

Confounding...

359

Hypothesis: High alcohol consumption is associated with stomach cancer (case-control study)

	D ⁺	D ⁻	
E ⁺	62	35	97
E ⁻	68	95	163
	130	130	260

$$OR = (62 / 68) / (35 / 95)$$

$$OR = 2.47$$

- The odds of being exposed to high alcohol consumption appear to 2.47 times higher for stomach cancer cases as compared to controls
- The risk of stomach cancer is 2.47 times higher in persons with high alcohol consumption as compared to persons without high alcohol consumption

Stratified analysis cont...

360

But what about smoking?

- Perhaps the cases were more likely to be smokers than the control subjects since heavy consumers of alcohol may also be likely to be smokers.
- In other words, may be high alcohol consumption has little to do with the risk of stomach cancer independent of smoking.

Stratified analysis cont...

361

NON-SMOKERS

	D ⁺	D ⁻	
E ⁺	18	20	38
E ⁻	42	80	122
	60	100	160

$$OR = (18 / 42) / (20 / 80)$$

$$OR = 1.71$$

SMOKERS

	D ⁺	D ⁻	
E ⁺	44	15	59
E ⁻	26	15	41
	70	30	100

$$OR = (44 / 26) / (15 / 15)$$

$$OR = 1.69$$

Is there evidence that smoking confounds the relationship between alcohol consumption and stomach cancer?

Epidemiological criteria (guidelines) for causality

362

- An association rarely reflects a causal relationship but it may.
- Criteria for causality provide a way of reaching **judgements** on the likelihood of an association being causal.

- Is there an association between a risk factor “X” and a disease “Y”???

The third variable

364

- The relationship b/n exposure and outcome variable may be complicated by ‘a third variable’

- The third variable could be
 - Effect modifier
 - Confounding

- What is different b/n Effect modifier and confounder???

CONFOUNDING

366

DEFINITION: A third variable (not the exposure or outcome variable of interest) that distorts the observed relationship between the exposure and outcome.???

- The three chx of a confounding variable.???

Effect modification

367

- **Interaction**- Two or more risk factors modify the effect of each other with regard to the occurrence or level of a given outcome
- The effect could be synergism or antagonism
- Synergistic (positive interaction) – potentiates the effect of the exposure of interest
- Antagonistic (negative interaction) – diminishes or eliminates the effect of the exposure of interest

Effect Modification...

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- The magnitude varies according to level of a third factor.
- occurs when the effect of a factor is different for different groups.
- Also called:
 - “Effect Measure Modification”
 - “Interaction”
- Note: Unlike confounding, effect measure modification should be **described and reported**, rather than controlled.

Effect modifier...

369

- 1-an effect modifier mustn't be a cause of a disease
 - 2-an effect modifier is not associated with the exposure
 - 3-an effect modifier is a variable that changes/alters the association between two other variables (exposure and outcome)
-
- For a binary outcome, “effect” can be measured as
 - risk difference
 - risk ratio
 - odds ratio

Why study effect modification? Why do we care?

- 370 ☐ to define high-risk subgroups for preventive actions,
- ☐ to increase precision of effect estimation by taking into account groups that may be affected differently,
- ☐ to increase the ability to compare across studies that have different proportions of effect-modifying groups, and
- ☐ to aid in developing a causal hypotheses for the disease

Assessment of effect modification

Assess measure of association within strata

Are stratum specific measures same?

Yes

No

Crude measure = stratum specific?

Statistical interaction; report stratum specific measures

Yes

No

No confounding or interaction
Report crude measure

Confounding
report measure
adjusted for
confounding

Assessment of EM...

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Stratification

Let us see using examples:

- ❖ Hypothesis: Smoking is associated with esophageal cancer (hypothetical cohort)

Effect Measure Modification

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	D ⁺	D ⁻	
E ⁺	82	2689	2771
E ⁻	28	10,223	10,251
	110	12,912	13,022

$$RR = (82 / 2771) / (28 / 10,251)$$

$$RR = 10.8$$

$$RD = 2.7\%$$

- ***Smokers appear to be at 10.8 times higher risk of developing E. cancer than non-smoker.***

However, is this elevated risk similar among drinkers and non-drinkers?

Effect Measure Modification

Alcohol consumed

Alcohol not consumed

	D ⁺	D ⁻	
E ⁺	64	863	927
E ⁻	22	7282	7304
	86	8145	8231

	D ⁺	D ⁻	
E ⁺	18	1826	1844
E ⁻	6	2941	2947
	24	4767	4791

$$RR = (64 / 927) / (22 / 7304)$$

$$RR = (18 / 1844) / (6 / 2947)$$

$$RR = 23$$

$$RD = 6.5\%$$

$$RR = 4.8$$

$$RD = 0.8\%$$

Does alcohol modify the relationship between smoking and esophageal cancer?

Effect Measure Modification

375

CRUDE

STRATA 1

STRATA 2

$$RR_{CA} = 10.8$$

$$RR_{al} = 23$$

$$RR_{nal} = 4.8$$

Unlike the assessment of confounding, the crude estimate is NOT USED to evaluate the presence of effect measure modification.

Instead, the stratum-specific estimates are compared directly to see if they are different (heterogeneous). Test of interaction in the above Strata: $P < 0.05$ (breslow-days test)

This example suggests “risk-ratio heterogeneity.”

Assessing confounding

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stratification

- First look at the stratum specific RR
- If the ratios are about the same, interaction is unlikely
- If the two RRs are about the same but different from cRR, the confounding may be present
- What is the relationship b/n smoking and CHD? Is sex confounder or effect modifier? See table below

Assessing confounding

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	CHD ⁺	CHD ⁻	
E ⁺	305	345	650
E ⁻	58	292	350
	363	637	100

$$RR = (305 / 650) / (58 / 350)$$

$$RR = 2.8$$

• *Smokers appear to be at 2.8 times higher risk of developing CHD than non-smoker.*

However, is this elevated risk similar among males and females?

Confounding...

Females

	CHD ⁺	CHD ⁻	
E ⁺	5	45	50
E ⁻	8	142	150
	13	187	200

$$RR = (5 / 50) / (8 / 150)$$

$$RR = 1.9$$

males

	CHD ⁺	CHD ⁻	
E ⁺	300	300	600
E ⁻	50	150	200
	350	450	800

$$RR = (300 / 600) / (50 / 200)$$

$$RR = 2.0$$

Specific RRs are 1.9 and 2, approximately the same?

The cRR is 2.8, may indicate the presence of confounding

Assessing confounding..

379

- Once it is decided that confounding is presence, **the average RR (the pooled RR)** should be presented
- Pooled RR can be calculated using **Mentel-Haenszel methods**

Assessing confounding...

380

From 2x2 table, we can calculate MH_{OR}

Mantel-Haenszel summary estimate

$$\begin{aligned} \text{MH summary OR} &= \frac{\sum ad / N}{\sum bc / N} \\ &= \frac{(ad/N)_{\text{male}} + (ad/N)_{\text{fem}}}{(bc/N)_{\text{male}} + (bc/N)_{\text{fem}}} \end{aligned}$$

Confounding...

381

- To determine the presence or absence of confounding, the aRR is compared to cRR to see if there is important difference
- No significant test for confounding
- To say a variable is confounder, some suggest, 10% or more differences b/n aRR and cRR

Formulae for detecting cRR and aRR

382

$$\frac{|cRR - aRR|}{aRR} \times 100$$

Confounding...

383

b/ assessing confounding using Logistic regression

- Once, EM is excluded, enter the confounder variable into the logistic regression model with the main exposure variable
- Then, Compare two OR: cOR and aOR

Strategies to evaluate interaction

384

- The absolute difference or attributable risk model (**additive**): the AR in those exposed to Factor A varies as a function of a third variable
- The relative difference or Ratio model (**multiplicative**): when the relative difference (ratio) in the risk of an outcome Y between subject exposed and those not exposed to a risk factor A differs as a function of a third variable

Multiplicative model...

385

Risk of E.cancer by
Smoking and alcoh

Risk ratio of E.c by
smoking and alc

	sm ⁺	Sm ⁻
alc ⁺	64/927=6.9%	22/7304= 0.3%
alc ⁻	18/1844= 0.98%	6/2947= 0.2%

	Sm ⁺	Sm ⁻
alc ⁺	6.9/0.2= 34.5	1.5
alc ⁻	4.9	1

Those who smoked and drank had 34 times
greater risk compared to those who neither
drink nor smoke

Among non smokers, the risk of cancer was 1.5 times

Greater among those consumed alcohol as

Compared to those who didn't consume alcohol

If smoking and alcohol operate independently in causing cancer, we could expect, under Multiplicative model, the expected risk be:

$$\text{Exp } R(AB) = R(AB_u) * R(A_uB) / R(A_uB_u)$$

Multiplicative...

386

- $R(AB)$ = risk of cancer in those exposed to smoking and alcohol
- $R(A_{\cup}B)$ = risk of cancer in smoker with no alcohol
- If there is perfect multiplicative relationship b/n smoking and alcohol, for those who had both exposure we would expect them to have risk of

$$\text{Exp } R(AB) = 0.98 * 0.3 / 0.2 = 1.47\%$$

Multiplicative...

387

- If the risk in those who had both exposure was near to 1.47%, there is no interaction under multiplicative model
- If the observed risk in those who had both exposure was significantly greater or less than 1.47, we say effect modification present. The observed result was 6.9% which is far greater than 1.47, indicating presence of EM under Multiplicative model (**Synergism**)

Multiplicative model...

388

- Similar approach using RR:

$$\text{Exp } R(AB) = RR(AB_u) * RR(A_u B)$$

$$RR(AB) = 4.9 * 1.5 = 7.4$$

The expected value was 7.4, but the observed value is 34.5 indicating
Presence of EM under MM

2/ Additive model using risk difference

389

- In the first example, cRD=2.7%, and Stratum specific RDs were 6.6 and 0.8%
- Since these stratum specific RDs differ very much, there was interaction
- Interaction using additive model:

$$\text{Exp } R(AB) = R(AB_u) + R(A_uB) - R(A_uB_u)$$

=0.98+0.3-0.2=1.08%, which far less than the observed value, 6.9%
,indicating interaction, **synergism**

Mortality by Age and CD4

	Age	Mortality Rate	Relative Risk
CD4 $\geq$ 200	< 35	1.9	1.00
CD4 $\geq$ 200	$\geq$ 35	5.0	2.63
CD4 < 200	< 35	26.5	1.00
CD4 < 200	$\geq$ 35	29.5	1.11

Is there any evidence of multiplicative interaction?

Additive Interaction – the association of older age and lower CD4 counts on Mortality

Observed Mortality	Age < 35	Age $\geq$ 35
CD4 $\geq$ 200	1.9	5.0
CD4 < 200	26.5	29.6

Joint expected AR
(24.6 + 3.1) = 27.7

Joint observed AR
29.6

Observed Attrib Mort	Age < 35	Age $\geq$ 35
CD4 $\geq$ 200	0	3.1
CD4 < 200	24.6	27.7

Is there any evidence
Of additive interaction?

Mortality by race and CD4 count

392

	Race	Mortality Rate	Relative Risk
CD4 $\geq$ 200	White	2.3	1.00
CD4 $\geq$ 200	Non-white	4.2	1.82
CD4 < 200	White	23.4	1.00
CD4 < 200	Non-White	30.4	1.30

Is there any evidence of multiplicative interaction?

Multiplicative Interaction – the association of Non-white race and lower CD4 counts on Mortality

Observed Mortality	White	Non-white
CD4 $\geq$ 200	2.3	4.2
CD4 < 200	23.4	30.4

Joint expected RR
(1.9 + 21.1) = 23.0

Joint observed RR
28.1

Observed Attrib Mort	White	Non-white
CD4 $\geq$ 200	0	1.9
CD4 < 200	21.1	28.1

Is there any evidence
Of additive interaction?

Models...

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- There may be interaction using both models
- There may be interaction using either of the model (not both)
- Which model is best to assess interaction?
- Many soft wares (including SPSS) use additive model
- For public health and disease control (and maybe everything else), should use the *additive* model (Rothman) (*ME2* doesn't mention Π)

EM and confounding using stratification in SPSS

395

- what is the effect of birth weight on infant survival?
- Is sex confounder or EM?

Survival outcome * Birth weight * Sex Crosstabulation

Sex				Birth weight		Total
				2500+	< 2500	
MALE	Survival outcome	Alive	Count	3433	214	3647
			% within Birth weight	91.8%	79.0%	90.9%
		Dead	Count	308	57	365
			% within Birth weight	8.2%	21.0%	9.1%
	Total		Count	3741	271	4012
			% within Birth weight	100.0%	100.0%	100.0%
FEMALE	Survival outcome	Alive	Count	3237	326	3563
			% within Birth weight	93.4%	82.5%	92.3%
		Dead	Count	229	69	298
			% within Birth weight	6.6%	17.5%	7.7%
	Total		Count	3466	395	3861
			% within Birth weight	100.0%	100.0%	100.0%

Risk Estimate

Sex		Value	95% Confidence Interval	
			Lower	Upper
MALE	Odds Ratio for Survival outcome (Alive / Dead)	2.969	2.168	4.066
	For cohort Birth weight = 2500+	1.116	1.067	1.167
	For cohort Birth weight = < 2500	.376	.286	.493
	N of Valid Cases	4012		
FEMALE	Odds Ratio for Survival outcome (Alive / Dead)	2.992	2.234	4.007
	For cohort Birth weight = 2500+	1.182	1.110	1.259
	For cohort Birth weight = < 2500	.395	.314	.498
	N of Valid Cases	3861		

Tests of Homogeneity of the Odds Ratio

	Chi-Squared	df	Asymp. Sig. (2-sided)
Breslow-Day	.001	1	.972
Tarone's	.001	1	.972

Test of interaction
Ho: the stratum ORs are the same

$P > 0.05$, Ho accepted
I.e, the ORs are the
Same, No interaction

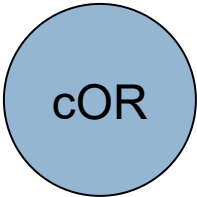
Since there is no interaction, assess the presence of confounding?

Survival outcome * Birth weight Crosstabulation

			Birth weight		Total
			2500+	< 2500	
Survival outcome	Alive	Count	6670	540	7210
		% within Birth weight	92.5%	81.1%	91.6%
	Dead	Count	537	126	663
		% within Birth weight	7.5%	18.9%	8.4%
Total		Count	7207	666	7873
		% within Birth weight	100.0%	100.0%	100.0%

Risk Estimate

	Value	95% Confidence Interval	
		Lower	Upper
Odds Ratio for Survival outcome (Alive / Dead)	2.898	2.342	3.586
For cohort Birth weight = 2500+	1.142	1.100	1.186
For cohort Birth weight = < 2500	.394	.330	.470
N of Valid Cases	7873		



Is sex associated with survival?

Criteria 1

Survival outcome * Sex Crosstabulation

			Sex		Total
			MALE	FEMALE	
Survival outcome	Alive	Count	3693	3608	7301
		% within Sex	89.9%	91.5%	90.7%
	Dead	Count	413	336	749
		% within Sex	10.1%	8.5%	9.3%
Total		Count	4106	3944	8050
		% within Sex	100.0%	100.0%	100.0%

Risk Estimate

	Value	95% Confidence Interval	
		Lower	Upper
Odds Ratio for Survival outcome (Alive / Dead)	.833	.716	.969
For cohort Sex = MALE	.917	.857	.982
For cohort Sex = FEMALE	1.102	1.014	1.197
N of Valid Cases	8050		

Is sex associated with birth weight?
Criteria 2

Birth weight * Sex Crosstabulation

			Sex		Total
			MALE	FEMALE	
Birth weight	2500+	Count	3741	3466	7207
		% within Sex	93.2%	89.8%	91.5%
	< 2500	Count	271	395	666
		% within Sex	6.8%	10.2%	8.5%
Total		Count	4012	3861	7873
		% within Sex	100.0%	100.0%	100.0%

Risk Estimate

	Value	95% Confidence Interval	
		Lower	Upper
Odds Ratio for Birth weight (2500+ / < 2500)	1.573	1.339	1.849
For cohort Sex = MALE	1.276	1.161	1.402
For cohort Sex = FEMALE	.811	.758	.867
N of Valid Cases	7873		

Sex has fulfilled the criteria for to be confounder

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- There fore, present the pooled OR(MH)

Mantel-Haenszel Common Odds Ratio Estimate

Estimate			2.981
ln(Estimate)			1.092
Std. Error of ln(Estimate)			.109
Asymp. Sig. (2-sided)			.000
Asymp. 95% Confidence Interval	Common Odds	Lower Bound	2.407
	Ratio	Upper Bound	3.693
	ln(Common	Lower Bound	.878
	Odds Ratio)	Upper Bound	1.306

The Mantel-Haenszel common odds ratio estimate is asymptotically normally distributed under the common odds ratio of 1.000 assumption. So is the natural log of the estimate.

In summary, the process is as follows

403

- Estimate a crude (unadjusted) estimate between exposure and disease.
- Stratify the analysis by any potential major confounders to produce stratum-specific estimates.
- Compare the crude estimator with stratum-specific estimates and examine the kind of relationships exhibited.

With a Confounder:

404

- the crude estimator (e.g. RR, OR) is outside the range of the two stratum-specific estimators
- If the adjusted estimator is importantly different (often 10%) from the crude estimator, the “adjusted variable” is a confounder.
- Statistical methods (Mantel-Haenszel method, multiple logistic regression, multivariable log. Reg., proportional hazards) are available to calculate the “adjusted” estimator, accounting for confounders.

With Effect modifiers:

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- the crude estimator (e.g. RR, OR) is closer to a weighted average of the stratum-specific estimators;
- the two stratum-specific estimators differ from each other
- Report separate stratified models or report an interaction term.

9. Screening in disease control

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Contents

- ❑ Definitions and concept of screening
- ❑ Objectives of Screening
- ❑ Principles of Screening in Public health programs
- ❑ Criteria for selection of screening program
- ❑ Types of Screening
- ❑ Criteria and Measures for evaluating Screening programs
- ❑ Public health importance of Se & Sp tests, and prevalence
- ❑ Quality control of screening test
- ❑ Common Biases in screenings
- ❑ Screening program evaluation and monitoring

Introduction

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- Screening has become an established part of medical practice.
- Routine recording of individual's characteristics, smoking habits, life style, blood pressure, urine testing, cholesterol measurement, preoperative chest x-ray, ..etc. may all be regarded as screening activities
- This lecture has been developed to help health practitioners in understanding and applying screening programs in their practice.
- The **PRESUMPTIVE** identification of **UNRECOGNIZED** disease or defect by the application of tests, exams or other procedures which can be applied **RAPIDLY** to sort out apparently well persons who **PROBABLY** have a disease from those who **PROBABLY** do not.

What is Screening?

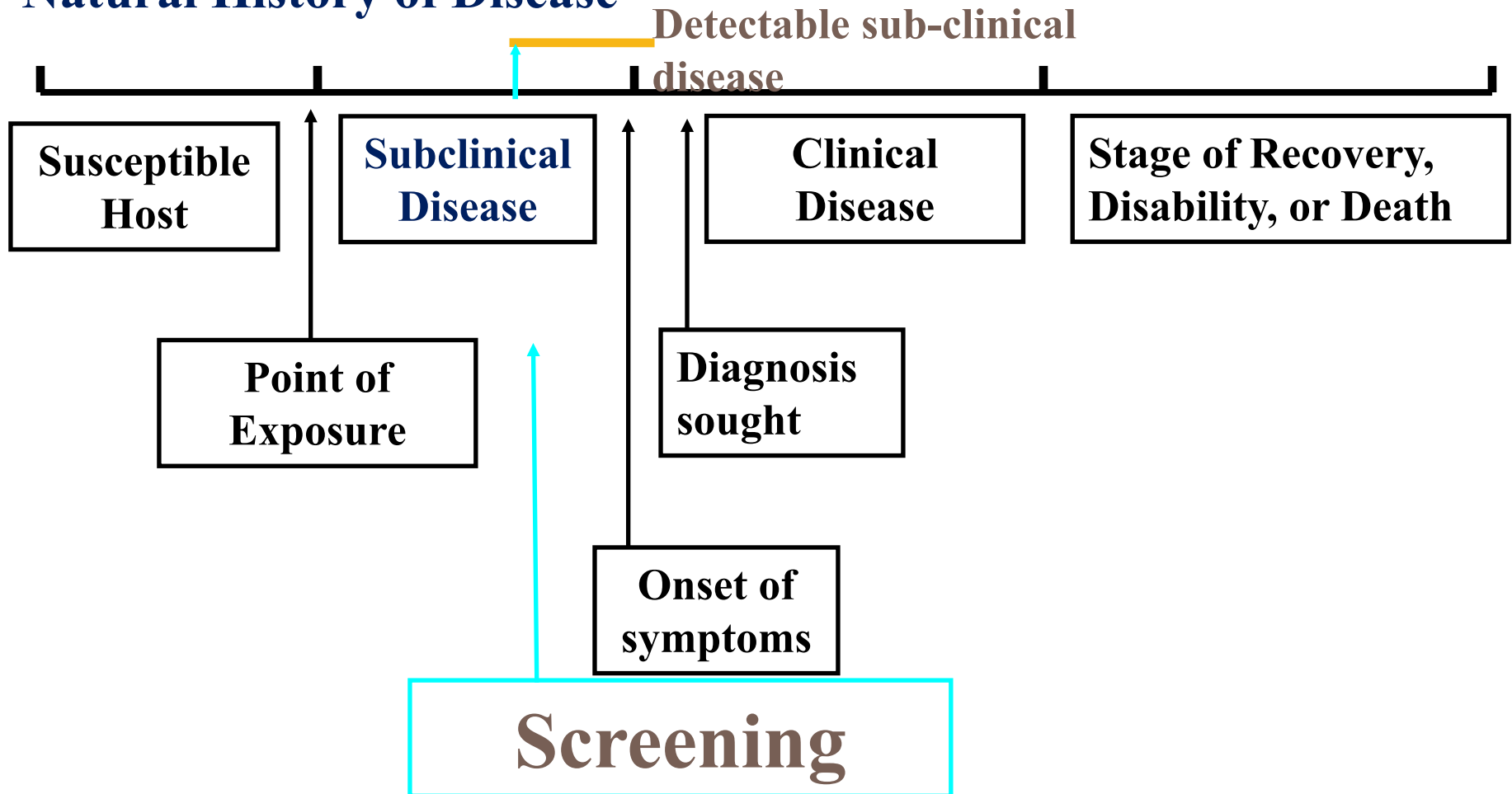
408

- Screening is the process to detect, among healthy people, disorders or risk factors of which they are unaware.
- **Screening** is the testing of apparently healthy populations to identify previously undiagnosed diseases or people at high risk of developing a disease.
- **Screening aims** to detect early disease before it becomes symptomatic.
- **Screening** is an important aspect of prevention, but not all diseases are suitable for screening.

Screening...

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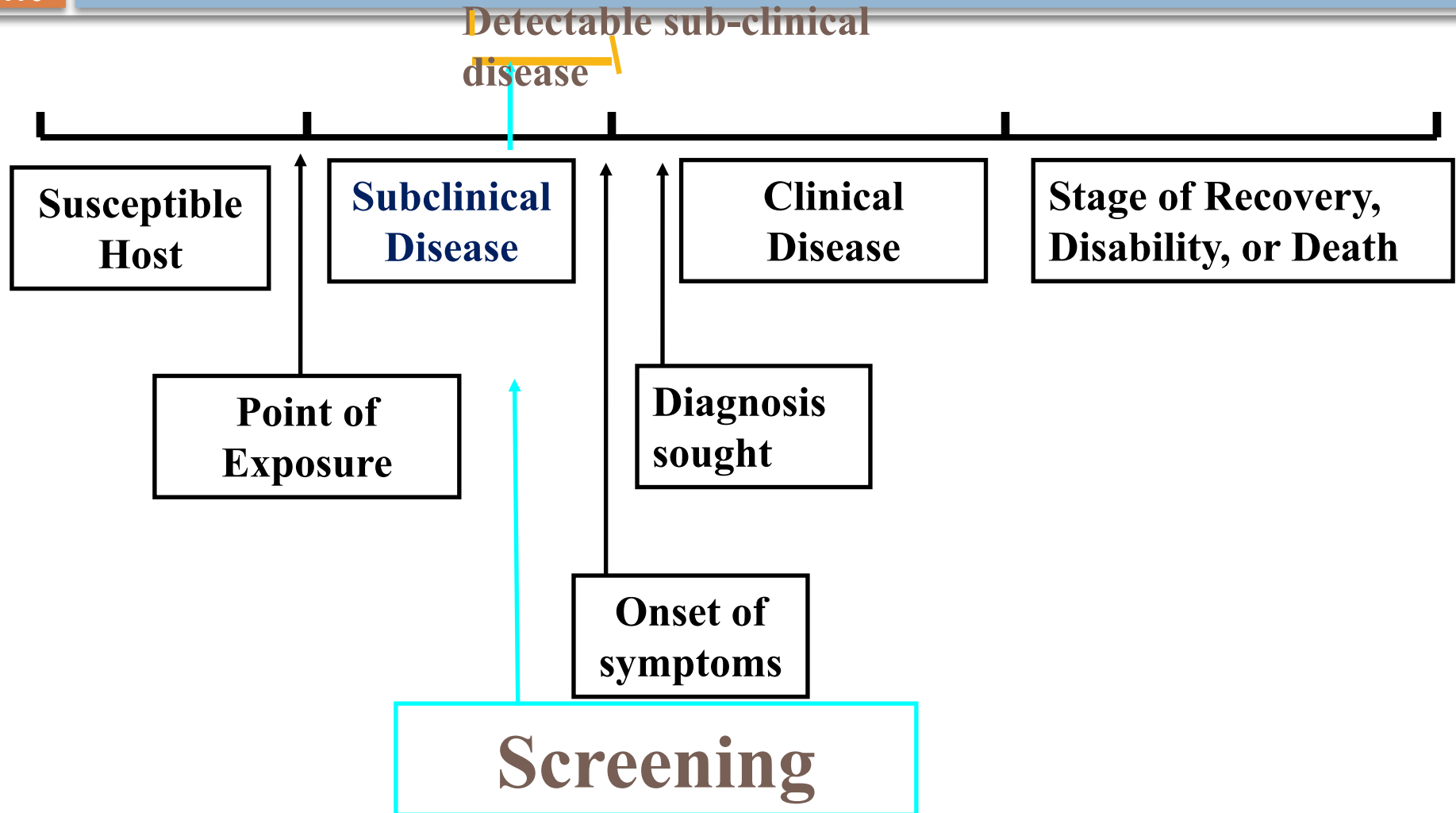
Natural History of Disease



Intro...

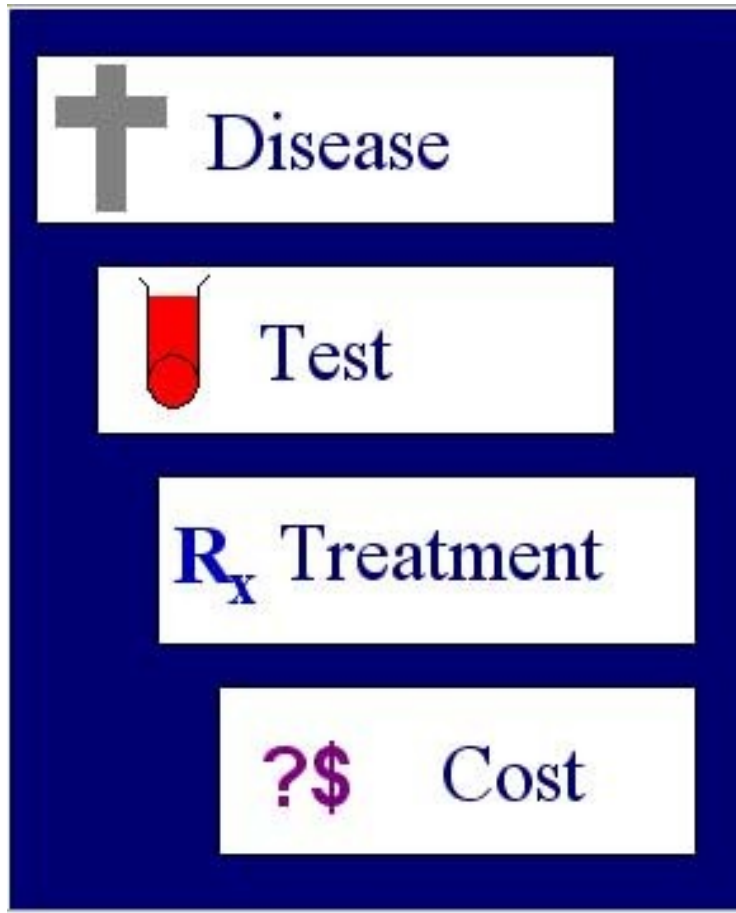
Natural History of Disease

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The Principles of Screening

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- The choice of disease for which to screen;
- The nature of the screening test or tests to be used;
- The availability of a treatment for those found to have the disease;
- The relative costs of the screening.

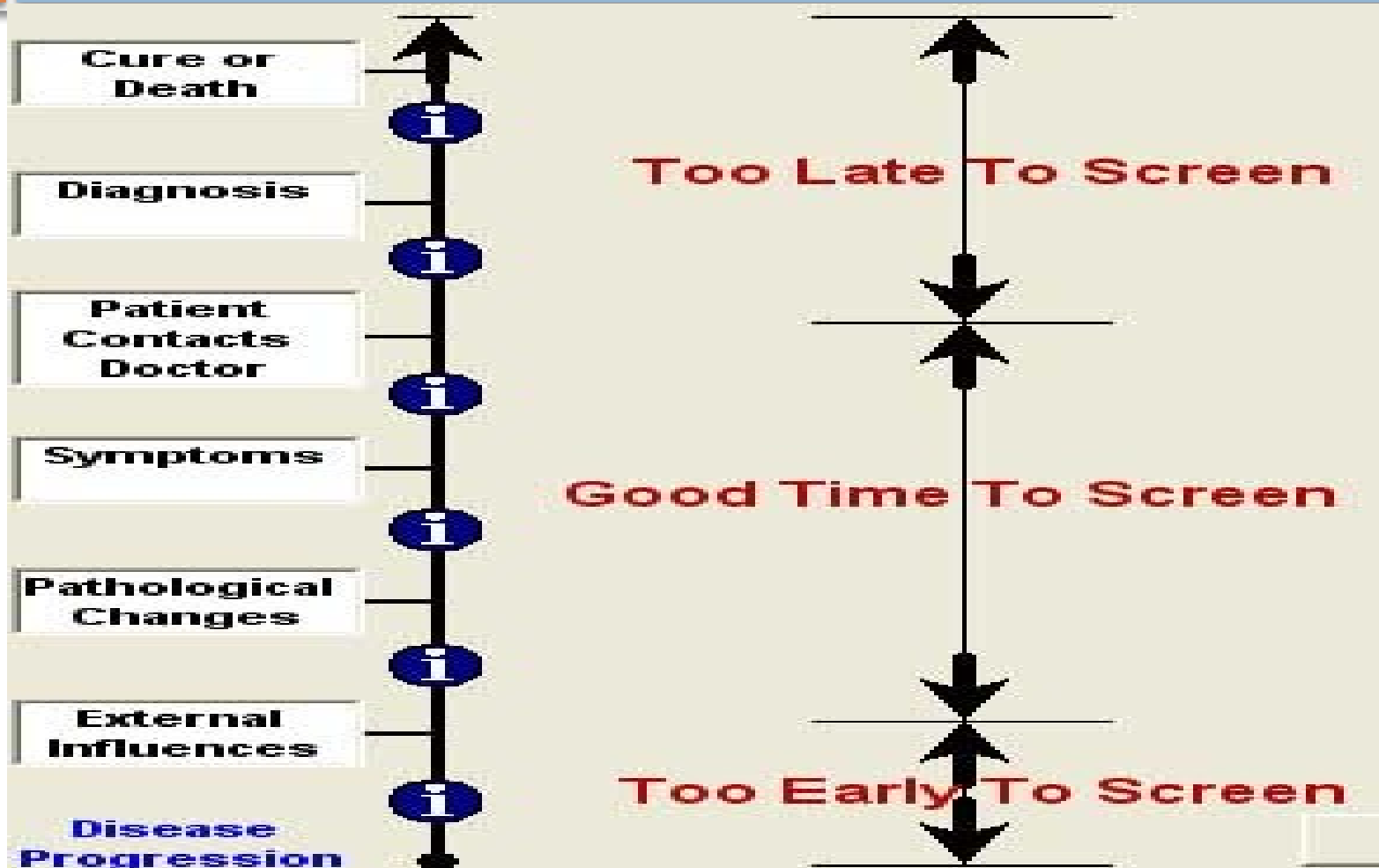
+Disease

412

- The disease must be an important health problem.
- There should be a recognizable latent or early symptomatic stage.
- The natural history of the disease, including latent to declared disease, should be adequately understood.

When to screen?

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Test

414

- There should be a **suitable** test or examination.
- The test should be **acceptable** to the population

Treatment

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- There should be an acceptable treatment for the patients with recognized disease.
- There should be facilities for diagnosis and treatment should be available.
- There should be an agreed policy on whom to treat as patients.

Cost

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- The cost of case finding (including diagnosis and treatment of patients diagnosed) should be economically balanced in relation to possible expenditure on medical care as a whole.
- Case finding should be a continuing process and not a "once for all" project.

1. The disease condition

- ▣ Important health condition (prevalent, serious)
 - *Has early stage*
- ▣ High prevalence at asymptomatic stage
- ▣ Adequately understood natural history

2. The test

- ▣ Simple, precise, validated
- ▣ Acceptable to population
- ▣ Agreed diagnostic test(s)
- ▣ Available, adequate, effective and acceptable
- ▣ Sensitive, specific, reliable, simple, cheap, safe

3. The treatment

- ▣ Effective treatment **Available, adequate, and acceptable**
- ▣ Evidence based policies on whom to treat

4. The program

- ▣ Acceptable to public and professionals
- ▣ Benefits outweigh risks
- ▣ Available facilities and staff
- ▣ Overall costs assessed

Characteristics to be evaluated

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□ These characteristics are:

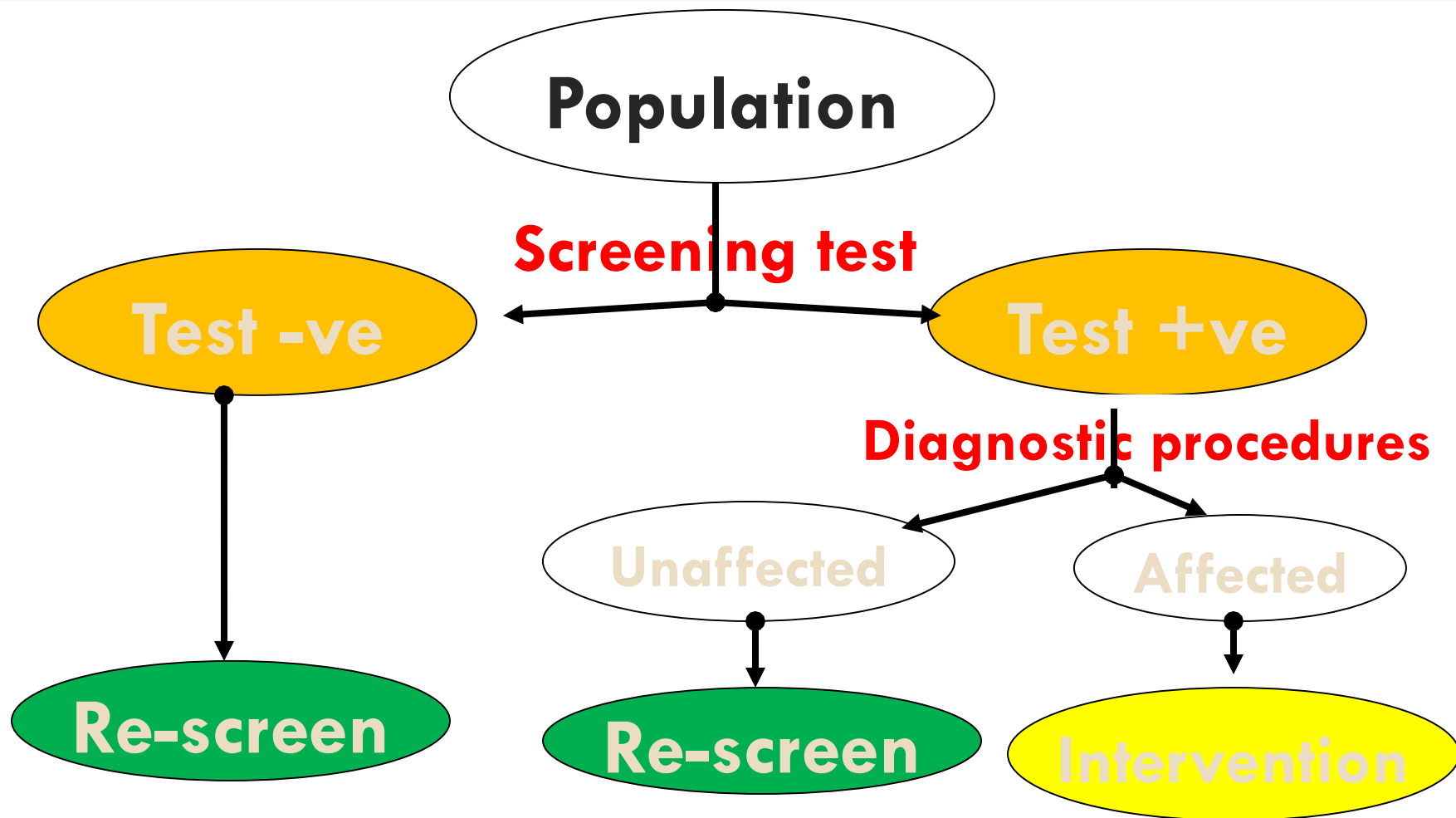
1. Validity
2. Predictive value
3. Reliability
4. Yield

1. **Validity is analogous to accuracy**

- The validity of a screening test is **how well the given screening test reflects** another test of **known greater accuracy**
 - Validity assumes that there is a **gold standard** to which a test can be compared
- **Validity**- does it truly measure what it sets out to measure?
- **Variability/reliability** -does it gives the same measurement each time

Flow diagram for a screening program

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Outcomes of a Screening Test

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Screening Test	True Disease Status		
	Positive (+)	Negative (-)	Total
Positive (+)	A True Positive [TP]	B False Positive [FP]	A+B TP+FP
Negative (-)	C False Negative [FN]	D True Negative [TN]	C+D FN+TN
Total	A+C TP+FN	B+D FP+TN	A+B+C+D TP+FP+FN+TN

Types of screening

422

A. Mass

- Large scale screening of **whole population/sub**groups where no selection of population groups is made/irrespective of a particular risk.
- E.g. checking all infants for hearing problems

A. Selective/high-risk/targeted

- For selected **high-risk groups in the population** which may still be large-scale and can be considered as one form of population screening. Example: ANC, PNC

A. Multiple or multiphasic

- Application of ≥ 2 screening test in combination to large groups of people
- Has evolved by combining single screening tests where much time and effort has been spent by population in attending a single test.
- Can be done **sequentially or simultaneously**

D. Case-finding or opportunistic

- Screening of individual coming for other service. E.g. PIHCT

Factors appropriate for screening

423

- ◆ Important health problem
- ◆ High prevalence
- ◆ Natural history understood
- ◆ Long latent period
- ◆ Early detection improves prognosis

Evaluation of a screening program

424

- ◆ **Reliability**
- ◆ **Feasibility**
- ◆ **Validity**
- ◆ **Performance**
- ◆ **Effectiveness**

Criteria for Evaluating a Screening Test

425

- In 1968 WHO proposed the following criteria for screening of diseases

J. M. G. Wilson and *Gunnar Jungner* **classic screening criteria.**

- Includes Validity, Reliability, and Yield

- **Validity**

- ❖ Provide a good indication of who does and does not have disease

- ✓ Sensitivity of the test

- ✓ Specificity of the test

- **Reliability**

- ❖ (Precision): gives consistent results when given to same person under the same conditions

- **Yield**

- ❖ Amount of disease detected in the population, relative to the effort

- ✓ Prevalence of disease/predictive value

Criteria for screening program (WHO, 2008)

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1. The screening program should respond to a recognized need - has potential to decrease in rate of disability /death or both
2. The objectives of screening should be defined at the outset.
3. There should be a defined target population.
4. There should be scientific evidence of screening program effectiveness.
5. The program should integrate education, testing, clinical services and program management.
6. There should be quality assurance, with mechanisms to minimize potential risks of screening.
7. The program should ensure informed choice, confidentiality and respect for autonomy.
8. The program should promote equity and access to screening for the entire target population.
9. Program evaluation should be planned from the outset.
10. The overall benefits of screening should outweigh the harm.

Validity of Screening Test (Accuracy)

427

- Sensitivity: Probability to test positive among truly affected.
 - ▣ Is the test detecting true cases of disease? (Ideal is 100%: 100% of cases are detected)
- Specificity: Probability to test negative among truly unaffected.
 - ▣ Is the test excluding those without disease? (Ideal is 100%: 100% of non-cases are negative)

Sensitivity and Specificity

428

- $$\text{sensitivity} = \frac{\text{True positives}}{\text{True positives} + \text{false negatives}} =$$
$$\frac{\text{True positives}}{\text{All persons with the disease}} * 100 = \frac{TP}{TP + FN}$$
- $$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negatives} + \text{False Postives}} =$$
$$\frac{\text{True Negatives}}{\text{All persons without the disease}} * 100 = \frac{TN}{TN + FP}$$
- $$\text{Sensitivity} = [a / (a + c)] \times 100$$
- $$\text{Specificity} = [d / (b + d)] \times 100$$

Percent False Negatives and Positives

429

- ☐ **Percent False negatives** = % of people with the disease who were not detected by the test.
 - ☐ $PFN = \frac{FN}{FN + TP} * 100$
- ☐ **Percent False Positives** = % of people without the disease who were incorrectly labeled by the test as having the disease.
 - ☐ $PFN = \frac{FP}{FP + TN} * 100$

Predictive Value

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□ In the clinical setting, a more important question is:

If the test results are positive (or negative) in a given patient, what is the probability that this patient has (or does not have) the disease?

In other words:

What proportion of patients who test positive (or negative) actually have (or do not have) the disease in question?

Predictive Value...

431

- Predictive value is the ability of a test to predict the presence or absence of disease from test results.
- There are two types
 - ▣ Predictive Value of a Positive test(PPV)
 - ▣ Predictive Value of a Negative Test(NPV)
- As prevalence increases, positive predictive value increases.
- As specificity increases, positive predictive value increases.
- As sensitivity increases, positive predictive value also increases, but to a much lesser extent.

Predictive Value...

432

□ PPV:- The proportion of diseased individual out of positive test

$$\square PPV = \frac{\text{Diseased individual + ve by the test}}{\text{The whole positive test}} * 100$$

$$\square PPV = \frac{TP}{TP + FP} * 100 = [a / (a + b)] \times 100$$

□ NPV:- The probability of not having the disease when the test is negative.

$$\square NPV = \frac{\text{Non-disease individual -ve by the test}}{\text{The whole negative test}} * 100$$

$$\square NPV = \frac{TN}{TN + FN} * 100 = [d / (c + d)] \times 100$$

Predictive Value...

433

$$PPV = \frac{\text{sensitivity} \times \text{prevalence}}{(\text{sensitivity} \times \text{prevalence}) + (1 - \text{specificity}) \times (1 - \text{prevalence})}$$

$$NPV = \frac{\text{specificity} \times (1 - \text{prevalence})}{[(\text{specificity} \times (1 - \text{prevalence}))] + [(1 - \text{sensitivity}) \times \text{prevalence}]}$$

Reliability

434

- Reproducibility, repeatability, reliability all mean that the results of a test or measure are identical or closely similar each time it is conducted
- Because of variation in
 - Laboratory procedures,
 - Observers
 - Changing conditions of test subjects
 - **such as time, location , emotions** a test may not consistently yield the same result when repeated
- Different types of variation
 - **Intra-subject variation**
 - **Intra-observer variation**
 - **Inter-observer variation**

Exercise on Screening of diabetes

435

- In a hypothetical study, 1000 patients attending a hospital general outpatient department were tested for diabetes using the following two tests: fasting blood sugar (FBS) and glucose tolerance test (GTT). There were 100 patients who had a positive GTT, and they were classified as true cases of diabetes.
- There were also 140 patients with FBS at least 6 mmol/l (cut -off point to distinguish people with diabetes from those who do not have diabetes). Among these 140 patients, only 98 were true cases of diabetes (i.e. only 98 had a positive GTT as well)

Exercise on Screening of diabetes...

436

- Based on the above scenario:
- Create a 2 x 2 table, and
- Calculate the sensitivity, specificity and positive and negative predictive values of the FBS in this study population?

Screening Test Result	Disease Status			
		Positive	Negative	Total
	Positive	98	42	140
	Negative	2	858	860
	Total	100	900	1000

Yield

437

- Prevalence is the proportion of individuals with the diseases.
 - ▣ $\text{Prevalence} = p(D^+) = (a + c) / n$
- Yield is the proportion of cases detected by the screening program.
- $\text{Yield} = \frac{\text{persons with the disease detected by the test}}{\text{total screened}} \times 100$
- $\text{Yield} = a/n \times 100$

Yield...

438

- The **number of cases** of the condition detected by screening test in **relation to the total number of persons screened**

$$Yield = \frac{Tp}{Tn + Tp + Fn + Fp} \times 100\%$$

- Affected by sensitivity, prevalence of unrecognized cases, number of tests employed, frequency of screening and participation of people in screening and follow-up

$$Yield = \frac{\text{persons with the dx detected by the test}}{\text{Total screened}} \times 100\%$$

Biases in screening program

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- Problems exist when comparing survival of screened patients with that of unscreened patients.
- Observational studies and especially survival data are acutely sensitive to:
 - A. Patient self-selection (volunteer) bias
 - B. Lead -time bias
 - C. Length bias
 - D. Over diagnosis bias

Biases in screening...

440

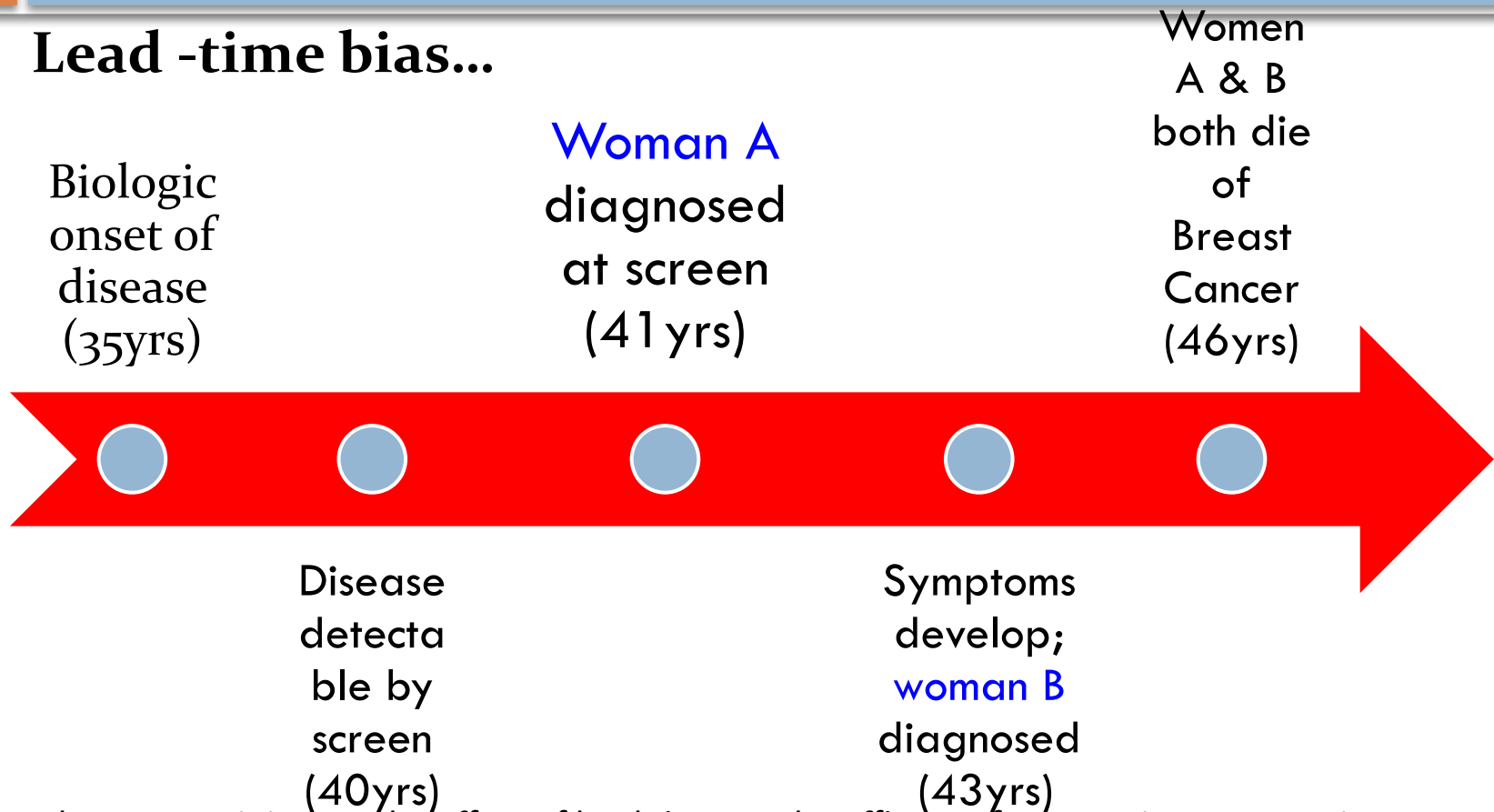
A. **Lead -time bias**

- Interval between the diagnosis of a disease at screening and when it would have been detected due to development of symptoms
- Apparent increased survival duration introduced by the lead time that results from screening.
- Screen-detected cases survive longer even without benefit of early treatment.
- Length depends on how fast the disease progresses and how soon screening is done.
- For unscreened patients, disease-specific survival is measured as the length of time from clinical diagnosis of disease to death from the disease.
- For screened patients, disease-specific survival is the length of time from disease detection by the screening test to death from the disease.
- Survival may erroneously appear to be increased among screen-detected cases.

Biases in screening...

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A. Lead -time bias...



■ The only way to minimize the effect of lead time on the efficacy of a screening program is to compare the [ASDR](#) in the screened and unscreened groups rather than the length of survival from diagnosis to death.

Biases in screening...

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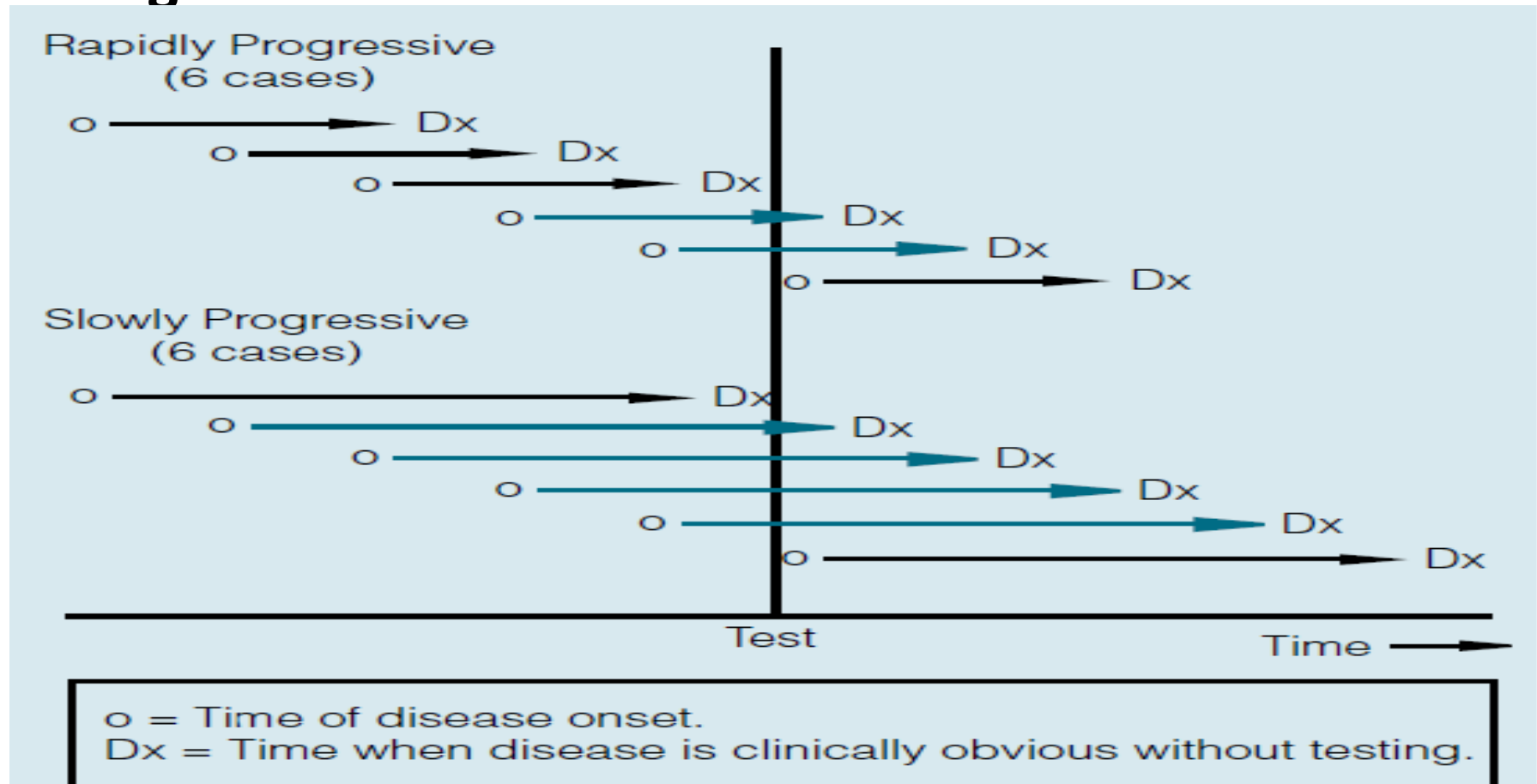
B. Length bias

- Length bias occurs because not all patients' disease progresses at the **same rate**.
- Cases of slowly progressing disease have **a longer detectable** preclinical phase than cases of quickly progressing disease.
- The slowly progressing disease is easier to detect &
 - ▣ They tend to be overrepresented in the screening cohort.
 - ▣ They have naturally longer survival regardless of the effectiveness of treatment but the longer survival is usually **wrongfully** attributed to early treatment.

Biases in screening...

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B. Length bias...





Thank You!!!